

内容由8所高校IGEM团队编写

2024 E.coli Nissle 1917 White Paper

editor-in-chief

PekingHSC NJTech-China

joint writing

WHU-China TJUSX-China GXU-China NJTech- China-A
USTC HUBU-4-CHN LZU- Medicine-China

Table of Contents

1.Explanation of the term EcN.....	2
2.Introduction to EcN and its history of use	3
3.Application scenarios of EcN.....	10
4.Overview of EcN Research Group.....	18
5.EcN modification related methods.....	21
6.iGEM Mid-Year EcN-Related Projects in the Past Five Years	29
7.EcN clinical drugs and progress currently under development	59
8.Analysis of EcN delivery strategy.....	60
9.Comparison of some bacterial therapeutic transformations with ECN	64
10.Policies related to bacterial therapy	75
11.Overview of key companies in the ECN industry market	79
12.Forecast of the development prospects of EcN in China.....	81
13.Side effects and biosafety prospects of EcN-related therapies	85
14.Introduction to the EcN project used by each team	88
15.Contributors.....	97
16.Legal Statement.....	98

1.Explanation of the term EcN

As one of the most important industrial production strains, *Escherichia coli* has a long history of research, a clear genetic background, and is easy to perform genetic manipulation. It has been used in scientific research and industrial production. However, most *E. coli* strains are conditional pathogens, and their applications are limited to a certain extent. The discovery of *E. coli* Nissle 1917 (EcN) has provided more possibilities for the study of *E. coli*^[1].

EcN, the full name of *Escherichia coli* Nissle 1917, is named after its inventor Alfred Nissle. EcN is a non-pathogenic *E. coli* that is sensitive to serum and easily cleared by serum. Studies have shown that it has probiotic properties such as lack of virulence factors, production of special adaptation factors, and strengthening of the natural barrier function of the intestine^[2]. It is widely used in the prevention of infectious diarrhea and immune regulation.

2.Introduction to EcN and its history of use

The discovery of EcN can be traced back to World War I, when German physician and bacteriologist Alfred Nissle isolated a strain of non-pathogenic *E. coli* from the feces of two soldiers who were not affected by the outbreak of dysentery (shigellosis) on the Dobruka front^[3]. This strain can survive in the intestines and inhibit the colonization of pathogens and maintain intestinal health. This strain was named after Dr. Alfred Nissle, and because it was discovered in 1917, it was named Nissle 1917. EcN is serum-sensitive and does not produce any enterotoxins or cytotoxins associated with pathogenic *E. coli* strains. It has the functions of inhibiting pathogenic microorganisms, biofilm formation, and regulating the immune system. It can be used not only for infectious diseases, but also for the treatment of gastrointestinal diseases such as diarrhea and ulcerative colitis^[4]. In the past few decades, the microbial characteristics and molecular genetic background of EcN have been explained in detail.

1. Microbial characteristics

1.1 The LPS in EcN has a special structure

EcN is a typical Gram-negative bacterium that contains lipopolysaccharide (LPS) components in its outer cell membrane structure. In terms of molecules, the LPS of EcN is different from that of all other types of *E. coli*. For example, the O6 polysaccharide side chain of the LPS in EcN is very short and contains only one repeating unit composed of oligosaccharide building blocks of the O6 antigen; modifications to the *E. coli* LPS were found in the oligosaccharide core fragment of the molecule. The special structure of LPS may explain why EcN exhibits immunomodulatory rather than immunotoxic effects^[5].

1.2 EcN is serum-sensitive

EcN is sensitive to serum. The so-called "capsule" structure exists in many extraintestinal pathogenic *E. coli*, which protects them from attack by nonspecific defense components in serum, allowing the bacteria to acquire resistance to serum, thereby prolonging their survival time in the blood and enhancing the toxicity of the pathogen^[6]. In contrast, EcN strains are rapidly killed when present in human serum or other mammalian serum (cow and pig serum) due to their own serum sensitivity^[7].

1.3 Fixation factors in EcN

EcN produces specific fixation factors. Pathogenic bacteria contain virulence factors encoded by special chromosomal DNA sequences (pathogenicity islands ^[8], referred to as PAIs), which increase the infection ability of the strain. Different fixation factors have also been detected in EcN, which enhance the competitiveness of EcN with other strains in the intestine, improve its survival in the intestine, and promote effective communication with the host. These factors are encoded by special chromosomal DNA sequences (genomic islands ^[9], referred to as GEIs).

1.4 Metabolic characteristics of EcN

As a typical *E. coli*, researchers conducted biochemical analysis of the unique metabolic capacity and fermentation characteristics of the EcN strain and conducted a series of biochemical tests. The results showed that in addition to the ability to metabolize arginine, EcN has typical metabolic characteristics of *E. coli*. Gas chromatography analysis of the spent culture supernatant ^[10] showed that under aerobic and anaerobic growth conditions, EcN produced short-chain fatty acids as the metabolic end products of carbohydrates.

1.5 Toxicological potential and biosafety of EcN

E. coli includes two types: enteropathogenic and extra-intestinal pathogenic, so the Nissle strain has received special attention in terms of toxicology and biosafety. In the selection principles of probiotics in the Enterobacteriaceae family, probiotic candidate bacteria should not exhibit pathogenic adhesion factors, nor form enterotoxins and cytotoxins or invade epithelial cells. They should not carry typical virulence factors of extra-intestinal pathogenic *E. coli*, such as serum resistance or hemolysin, nor should they exhibit urinary pathogenicity or cause immunotoxic effects. EcN strains do not produce any toxins associated with pathogenic *E. coli* strains, such as heat-labile enterotoxins (H-LT), heat-stable enterotoxins (H-ST), cytotoxins (such as cytotoxic necrosis factor, CNF 1) or Shiga toxins (SLT I, SLTII). Studies have shown that EcN does not have the genetic information to produce the above-mentioned toxins^[11].

2. Molecular genetic background

2.1 Genome structure of EcN

Non-pathogenic *E. coli* and enteropathogenic and extra-intestinal *E. coli* have different chromosomal regions. Five large genomic islands and some small genomic islands have been detected in EcN, carrying genomic islands encoding special fixation factors in EcN^[12]. An important reason for

the probiotic properties of EcN may be the lack of virulence factors, including α -hemolysin, p-fimbriae adhesin, and semi-rough lipopolysaccharide phenotype expressed in conjunction with fixation factors^[13].

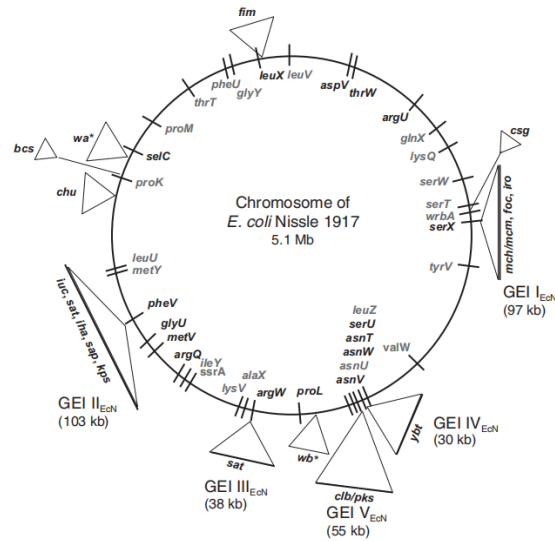


Figure 1. Functional genomic map of *E. coli* Nissle 1917 (EcN)^[12]

2.2 Plasmid characterization of EcN

EcN contains two small cryptic plasmids, named pMUT1 and pMUT2^[14]. Both are genetically stable and cannot be transferred to other *E. coli*, and have been completely sequenced^[15]. These two cryptic plasmids have not been found in other *E. coli* so far, and can be used as a method to identify EcN strains. Although the function of these stealth plasmids in EcN is still unclear, there is evidence that the stealth plasmids themselves may play a role in protecting the carrier bacteria from attacks by mobile genetic elements such as bacteriophages, thereby affecting the genetic stability of the strain^[16].

3. History of EcN use

In 1917, Alfred Nissle isolated EcN from the feces of a German soldier and found through experimental determination that it exhibited strong antagonistic activity against different pathogenic *Escherichia coli*. This was the first time that EcN entered the scientific field of research. Over the past 100 years, EcN strains have been used as the active ingredient of a licensed drug sold in Germany and several other countries^[17]. In recent decades, with the continuous deepening of EcN research, scientists have continuously discovered new probiotic activities in EcN, and the application of EcN in the field of probiotics has continued to expand, and it has also been used in many important applications in disease treatment.

3.1 Alfred Nissle and the discovery and use of EcN

After Nissle first discovered and isolated the EcN strain, he cultured the collected strains in the laboratory. He first cultured the strain on large agar plates and then placed the strain in gelatin capsules sealed with wax or paraffin for the treatment of individual patients. In 1917, he applied for and obtained a patent for the trademark Mutaflor from the then-existing Imperial Patent Office in Berlin. The word Mutaflor comes from the Latin words for variation and strain, demonstrating the ability of EcN to affect the intestinal flora through antagonism. In order to mass-produce the drug to meet the needs of more patients, Nissle first chose G. Pohl Company for the production of strain preparations, which had special experience in the production of gelatin capsules. In 1932, Nissle transferred the production license of the strain to Hageda Company in Berlin. In 1944, the production equipment was destroyed by bombs during the war. After the war, Hageda Company rebuilt the production equipment and resumed the production of Mutaflor. Nissle has been providing subcultures of EcN strains as starters for Mutaflor preparations until his death in November 1965. Later, the EcN strain was deposited in the German Collection of Microorganisms and Cell Cultures for industrial use and was named *E. coli* DSM 6601.

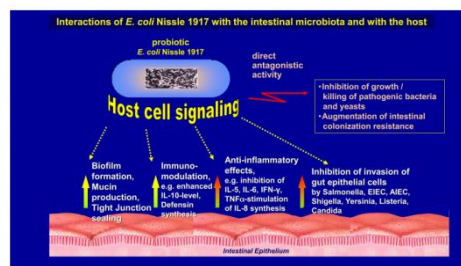


Figure 2. Interactions between *Escherichia coli* Nissle 1917 and intestinal flora and host^[17]

In the 1930s and 1940s, EcN was commercialized and popularized, especially in Germany and European countries, where EcN-related products were gradually commercialized. As the research on EcN continued to deepen, scientific research began to frequently discuss the benefits of EcN as an early example of probiotics for intestinal health and was recognized in clinical applications. In the late 20th century, researchers conducted in-depth research and developed standard clinical trials. In the 1970s and 1980s, controlled clinical trials on EcN began to increase, and researchers confirmed that it has a positive effect in digestive system diseases (such as IBD, constipation and diarrhea)^[17].

In the 21st century, the function and mechanism of EcN have received more attention, and researchers have explored its potential therapeutic effects in immune regulation, severe diarrhea and other gastrointestinal diseases. As one of the representatives of probiotics, EcN has been included in a variety of health foods and dietary supplements for the mass market. In their studies, researchers have also found that EcN has the potential to be useful in vaccine development, mainly in inducing

immune responses. When attenuated bacteria or viruses are used as vaccines, protective immune responses are initiated by presenting the necessary antigens to the immune system, and in theory, the use of genetically modified probiotics can enhance the above effects. EcN is also widely used in diagnosis and drug development. EcN has tumor-specific colonization properties in mice, so it has been used to diagnose solid tumors in clinical studies. In recent studies, researchers have attempted to engineer EcN to enhance its ability to induce immune responses and improve its utility in diagnostic and therapeutic applications ^[18].

4. Use of EcN in intestinal diseases

Since the 21st century, research on the function and mechanism of EcN has been continuously improved. On this basis, researchers have turned their attention to the immunomodulatory properties of EcN and applied it to related intestinal diseases. The main task of intestinal inflammation research is to develop new treatments that can accurately target the intestinal mucosa. Genetically modified bacteria can serve as a good carrier for local antigens to enter the intestine, but their biosafety is also crucial. Studies have shown that EcN has the potential to serve as a safe carrier for targeting recombinant proteins to the intestinal mucosa. The researchers tested EcN in a defined and sensitive immune system to explore whether it affects the migration, clonal expansion, and activation state of specific CD4⁺ T cells. The results showed that EcN did not affect the above processes, either in healthy mice or in animals with acute colitis. The excellent colonization properties make the EcN strain an ideal carrier candidate for the intestinal focused in situ synthesis of therapeutic molecules.

4.1 Use of EcN in tumor diseases

In recent years, the research on EcN in targeted cancer therapy has gradually become a hot topic, and it is expected to bring new hope to cancer treatment. Traditional tumor treatment methods such as chemotherapy and radiotherapy have improved the survival rate of patients to a certain extent, but there are still many problems, such as large toxic side effects, poor targeting, easy to develop drug resistance and high recurrence rate. Tumor-targeted bacteria can achieve continuous synthesis and release of anticancer drugs in the tumor environment through specific colonization in the solid tumor environment under the conditions of genetic engineering, thereby improving the selectivity of drugs for tumor tissues. Compared with chemotherapy, tumor-targeted bacteria can also reduce drug damage to tissues. The hypoxic and immunosuppressive tumor microenvironment creates conditions for the proliferation of some bacteria in it. At present, obligate anaerobes such as *Bifidobacterium infantis* and *Clostridium novyi* and facultative anaerobes such as *Salmonella typhimurium* and *Escherichia coli* have been studied for tumor targeted therapy. EcN has good tumor targeting ability. The serum-

sensitive lipopolysaccharide (LPS) present in its outer membrane ensures that it can complete "suicide" in the body and be cleared from normal organs [5].

In addition, EcN does not produce any enterotoxins and cytotoxins related to pathogenic strains, has a clear genetic background, and is convenient for genetic manipulation. Therefore, EcN has become an ideal living drug for tumor targeted therapy.

Genetically modifying EcN to directly express therapeutic factors is one of the important strategies for tumor targeted therapy. EcN directly expresses therapeutic factors such as cytotoxic drugs, prodrug converting enzymes, and angiogenesis inhibitory proteins to achieve targeted therapy for tumors, with good therapeutic effects. EcN can also assist chemotherapy, radiotherapy, immunotherapy, and phototherapy to treat tumors, injecting new vitality into traditional tumor treatments. The key challenge faced in tumor bacterial therapy is to improve the targeting, controllability, and safety of bacteria [19]. Studies have shown that the main strategy to improve the tumor targeting of EcN is to enhance EcN's recognition of the hypoxic and acidic microenvironment of tumors and promote its binding to tumor cell surface antigens. As a multifunctional probiotic, EcN has become an excellent chassis strain for tumor bacterial therapy. With the development of technology and in-depth research, EcN will become a powerful weapon for cancer treatment[20].

Reference:

- [1] 汤佳冰, 张颖, 叶佳微, et al. 大肠杆菌 Nissle 1917 的多组学分析及其产微菌素的功能验证. 微生物学报 [J]. 1-19.
- [2] 温杨, 万朝敏. 益生菌大肠埃希菌 EcN 研究进展. 中国实用内科杂志 [J]. 2012, 32(09): 714-6.
- [3] OZEN M, DINLEYICI E C. The history of probiotics: the untold story [J]. Beneficial Microbes, 2015, 6(2): 159-65.
- [4] KRUIS W. Maintaining remission of ulcerative colitis with the probiotic *Escherichia coli* Nissle 1917 is as effective as with standard mesalazine [J]. Gut, 2004, 53(11): 1617-23.
- [5] GROZDANOV L, ZÄHRINGER U, BLUM-OEHLER G, et al. A Single Nucleotide Exchange in the *wzy* Gene Is Responsible for the Semirough O6 Lipopolysaccharide Phenotype and Serum Sensitivity of *Escherichia coli* Strain Nissle 1917 [J]. Journal of Bacteriology, 2002, 184(21): 5912-25.

- [6] Taylor P W. Bactericidal and bacteriolytic activity of serum against gram-negative bacteria[J]. Microbiological reviews, 1983, 47(1): 46-83.
- [7] Hughes C, Phillips R, Roberts A P. Serum resistance among *Escherichia coli* strains causing urinary tract infection in relation to O type and the carriage of hemolysin, colicin, and antibiotic resistance determinants[J]. Infection and Immunity, 1982, 35(1): 270-275.
- [8] Hacker, J., & Kaper, J. B. (2000). Pathogenicity islands and the evolution of microbes. Annual Reviews in Microbiology, 54(1), 641-679.
- [9] Hacker, J., & Carniel, E. (2001). Ecological fitness, genomic islands and bacterial pathogenicity: A Darwinian view of the evolution of microbes. EMBO Reports, 2(5), 376–381.
- [10] Zijlstra J B, Beukema J, Wolthers B G, et al. Pretreatment methods prior to gaschromatographic analysis of volatile fatty acids from faecal samples[J]. Clinica Chimica Acta, 1977, 78(2): 243-250.
- [11] Blum, G., Hacker, J., & Marre, R. (1995). Properties of *Escherichia coli* strains of serotype O6. Infection, 23(4), 234-236.
- [12] SONNENBORN U, SCHULZE J. The non-pathogenic *Escherichia coli* strain Nissle 1917 – features of a versatile probiotic [J]. Microbial Ecology in Health and Disease, 2009, 21(3-4): 122-58.
- [13] DOBRINDT U. (Patho-)Genomics of *Escherichia coli* [J]. International Journal of Medical Microbiology, 2005, 295(6-7): 357-71.
- [14] YU X, LIN C, YU J, et al. Bioengineered *Escherichia coli* Nissle 1917 for tumour-targeting therapy [J]. Microbial Biotechnology, 2019, 13(3): 629-36.
- [15] Blum-Oehler, G., Oswald, S., Eiteljörge, K., Sonnenborn, U., Schulze, J., Kruis, W., & Hacker, J. (2003). Development of strain-specific PCR reactions for the detection of the probiotic *Escherichia coli* strain Nissle 1917 in fecal samples. Research in microbiology, 154(1), 59-66.
- [16] FELDGARDEN M, GOLDEN S, WILSON H, et al. Can phage defence maintain colicin plasmids in *Escherichia coli* [J]. Microbiology, 1995, 141(11): 2977-84.
- [17] SONNENBORN U, ROBERTSON L. *Escherichia coli* strain Nissle 1917—from bench to bedside and back: history of a special *Escherichia coli* strain with probiotic properties [J]. FEMS Microbiology Letters, 2016, 363(19).
- [18] OU B, YANG Y, THAM W L, et al. Genetic engineering of probiotic *Escherichia coli* Nissle 1917 for clinical application [J]. Applied Microbiology and Biotechnology, 2016, 100(20): 8693-9.
- [19] WEBER W, FUSSENEGGER M. Emerging biomedical applications of synthetic biology [J]. Nature Reviews Genetics, 2011, 13(1): 21-35.
- [20] 李雨桐, 崔天琦, 张海林, et al. 肿瘤靶向细菌 *Escherichia coli* Nissle 1917 在癌症治疗中的研究进展. 中国生物工程杂志 [J]. 2023, 43(06): 54-68.

3.Application scenarios of EcN

1. Detection and treatment of cancer

Gene circuits expressing various anticancer drugs are constructed in EcN to improve its ability to kill tumor cells; butyrate has biological activity in reducing carcinogenesis. In order to achieve targeted cancer therapy, Chiang C J et al. developed bacterial cancer therapy (BCT) with butyrate as a payload. Through metabolic engineering, *Escherichia coli* Nissle 1917 (EcN) was recombined to synthesize biobutyrate, named EcN-BUT. The strategy adopted includes constructing biobutyrate synthesis pathways and metabolic pathways to increase the production of biobutyrate at the expense of acetate. In tumor-bearing mice, injection of EcN-BUT showed tumor-specific colonization and significantly reduced tumor volume by 70% [1]. By constructing a synchronized cleavage circuit to achieve bacterial self-triggered release of drugs, or by precisely controlling the time, location and dosage of drug release through external signals (such as small molecules, light, heat, and ultrasound), the controllability of drug release can be improved; immune checkpoint inhibitors targeting programmed cell death protein-ligand 1 (PD-L1) and cytotoxic T lymphocyte-associated protein-4 (CTLA-4) have revolutionized the paradigm of cancer immunotherapy and achieved tumor regression in several cancers. By leveraging the convergence of immunotherapy and synthetic biology, a probiotic EcN can be designed to release PD-L1 and CTLA-4 antagonists in a localized and controllable manner. Specifically, it combines immunotherapeutic expression with an optimized cleavage mechanism, allowing the probiotic to carry the nanobody back to the core of the necrotic tumor, grow to a critical density, and lyse, effectively and continuously releasing therapeutic drugs within the tumor microenvironment (TME) [2].

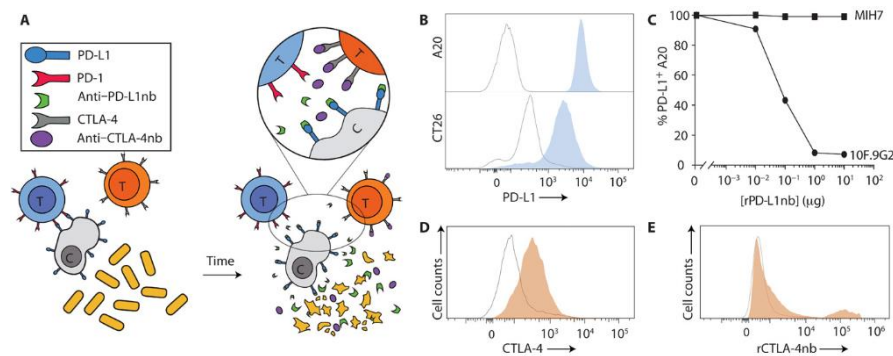


Figure 1. Design and characterization of a probiotic cancer treatment system for releasing functional PDL1- and CTLA-4 blocking nanobodies^[2]

Nguyen, DH, etc. enrich bacterial functions by expressing enzymes, antibodies (such as CD47 antibodies, PD-L1 antibodies, CTLA-4 antibodies), cytokines, etc., to assist traditional cancer therapies, such as improving the tumor hypoxic microenvironment to improve the effectiveness of radiotherapy properties and produce immune activators to improve the effect of immunotherapy [3]. Mohamad H. Abedi et al. designed a therapeutic bacterium controlled through focused ultrasound, a form of energy that can be applied noninvasively to specific anatomical sites, such as solid tumors. This control is provided by a temperature-driven genetic state switch that produces long-lasting therapeutic output after brief application of focused ultrasound hyperthermia. Through a combination of rational design and high-throughput screening, we optimized the switch circuit in engineered cells and Linking its activity to the release of immune checkpoint inhibitors. In clinically relevant cancer models, ultrasound-activated therapeutic microorganisms successfully opened in situ and induced significant inhibition of tumor growth. This technology provides a key tool for the spatiotemporal targeting of bacterial therapeutics that is effective in a variety of biological and clinical situations [4].

STING (stimulator of interferon genes) agonists are emerging as a new approach for cancer therapy because of their ability to enhance intratumoral immune responses. As a potential means of delivering STING agonists directly to tumors, Leventhal and colleagues developed SYNB1891, an EcN variant that produces high levels of cyclic AMP via a hypoxia-inducible promoter. SYNB1891, like SYNB1618, is trophic for thymine and diaminopimelic acid. Phagocytosis of SYNB1891 by phagocytes triggers the secretion of interferons, which can be used to treat tumors [7].

SLIC strains constitutively expressing anti-PD-L1 Nb (nanobody), anti-CTLA-4 Nb, and granulocyte-macrophage colony-stimulating factor (GM-CSF) (using a QS switch) have the ability to effectively promote antitumor immunity and tumor growth [7].

EcN is engineered to increase the availability of the phytochemical sulfathionin, an antitumor compound derived from glucosinolates in cruciferous vegetables. Ho and colleagues engineered Eda-11-HlpA to constitutively secrete the horseradish myrosinase enzyme fused to the YebF secretion sequence. The strain was also designed to express HlpA on its outer surface, which promotes interaction with heparan sulfate proteoglycans on the surface of colorectal tumors. To display HlpA on the surface of EcN, it was fused to the ice nuclease protein. When administered with a cruciferous vegetable diet, mice receiving the resulting strain, Eda-11-HlpA, exhibited smaller and fewer colorectal tumors in a chemically induced colorectal cancer (CRC) model [7].

L-arginine availability is reduced in tumors, which limits T cell responses, and Canale and colleagues developed L-Arg EcN, an EcN variant optimized for ammonia conversion to L-arginine, which promotes T cell activation and inhibits tumor growth [7].

The article [8] designed EcN to produce a small molecule salicylic acid. Oral administration of this strain resulted in increased levels of salicylic acid in the urine of adenoma-bearing mice compared

with healthy controls, and salicylic acid has therapeutic effects on rectal cancer. The basic approach was to clone the genes *mbti*, *irp9*, *menF*, and *entC* into a plasmid encoding *pchB*, which was then cloned into *EcNATTΔclbA-lux* (*EcNATT*), which includes genomically integrated *aroG*, *tktA*, and *talB* genes. These genes are involved in the shikimate and pentose phosphatase pathways, through which salicylic acid is synthesized [8]. Subsequently, the article [8] designed an *EcN* that delivers nanobodies that block PD-L1 and CTLA-4 targets and the cytokine GM-CSF, and it was also found to have an anti-tumor effect (oral administration of this strain reduced adenoma burden by approximately 50%) [8].

Systemic administration of *EcN* to tumor-bearing mice revealed its preferential accumulation in tumors compared with clearance in organs such as the spleen and liver. Here, Stritzker, Jochen et al. compared the tumor-specific colonization efficiency of several different microbial strains. All strains efficiently colonized and replicated in tumors, with each strain producing more than 1×10^8 CFU per gram of tumor tissue. Colonization of the spleen and liver was significantly reduced when using *EcN*. compared to *S. typhimurium*, and the non-pathogenic strain did not colonize these organs at all. Further studies with *E. coli* Nissle1917 showed that no significant differences were observed in colonization and expansion when using immunocompetent and immunocompromised animals, and Stritzker, Jochen et al. were able to show that *E. coli* Nissle1917 replicated at the edges of both viable and necrotic tumor tissue. Stritzker, Jochen et al. also demonstrated exogenously applied L-arabinose-dependent gene activation in colonized tumors in live mice. These findings will pave the way for bacteria-mediated controlled protein delivery to solid tumors [11]. *Escherichia coli* Nissle 1917 (*EcN*) is able to penetrate the barrier and proliferate at the interface between the viable and necrotic areas of tumors. By genetically engineering *EcN* to create nanoscale microcells, Zhang, YL targeted the delivery of chemotherapeutic drugs to hypoxic areas of tumors for cancer treatment. [12]

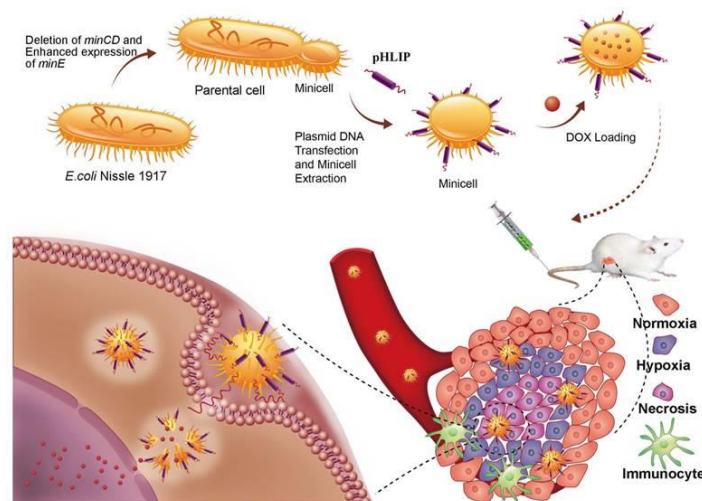


Figure 2. Schematic diagram depicting the construction of miniature cell pHILIP for targeted delivery of chemotherapeutic drugs to hypoxic regions of solid tumors to kill cancer cells. [12]

2. Treatment of gastrointestinal dysfunction and other gastrointestinal-related diseases.

EcN has been engineered to synthesize 3-hydroxybutyric acid (R)-3-hydroxybutyl ester (3HB). After colonizing the stomach and intestines of mice for 14 days, the 3HB and short-chain fatty acid (SCFA) levels in the intestines were increased by 8.7 times and 3.1 times, respectively, compared to the wild-type EcN. The persistent presence of 3HB in the mouse intestines promotes the growth of beneficial bacteria and can be used to treat colitis [5].

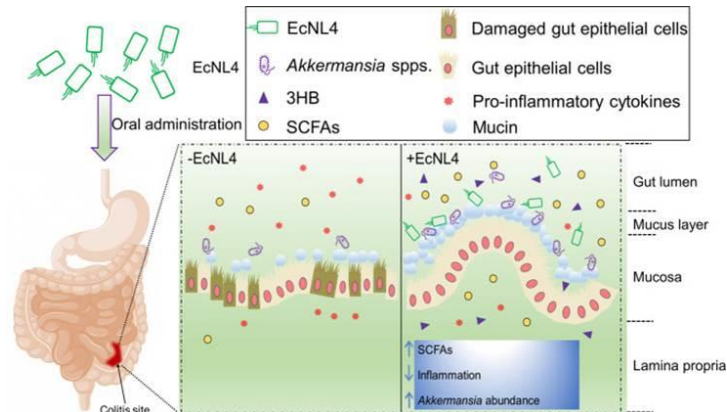


Figure3. EcNL4 is administered orally and colonizes the inflammatory bowel disease site, improving IBD by increasing SCFA concentrations, reducing inflammation, and enriching beneficial bacteria^[5] (such as *Akkermansia* spp.)

Escherichia coli Nissle 1917 (EcN) has been genetically engineered to overexpress catalase and superoxide dismutase (EcN-pE) for the treatment of intestinal inflammation. To enhance the bioavailability of EcN-pE in the gastrointestinal tract, the effective biomaterials chitosan and alginate were used to coat EcN-pE via a layer-by-layer electrostatic self-assembly strategy. In mouse IBD models induced by different chemical drugs, the chitosan/alginate-coated EcN-pE (EcN-pE(C/A)2) effectively alleviates inflammation and repairs the intestinal epithelial barrier [6].

Praveschotinunt and colleagues developed the PATCH (Probiotic Associated Therapeutic Curli Hybrid) system, a platform for displaying high concentrations of proteins on the surface of EcN. This is achieved by fusing the self-assembling protein (CsgA) subunit to the target protein. Once secreted, the CsgA-fused protein forms Curli, nanofibers that extend from the bacterial surface. PBP8 CsgA-TFF3 is a variant of EcN that expresses CsgA fused with TFF3, a TFF3 is a factor that promotes mucosal healing^[7].

To enhance the ability of EcN in the intestine to convert toxic metabolites (including ammonia and phenylalanine) into non-toxic forms, Kurtz and colleagues designed EcN to effectively direct intestinal ammonia into the L-arginine (L-arg) biosynthesis pathway. The resulting strain lacks ArgR

(a negative regulator of the pathway) and expresses a feedback-resistant variant of ArgA (an L-arg biosynthesis enzyme). Modified ArgA is expressed under an anaerobic inducible promoter, allowing the bacteria to treat hyperammonemia [7].

Lynch JP and colleagues developed EcN variants of SYN1618 and SYN1934 to promote phenylalanine metabolism. In anoxic intestinal environments, both express phenylalanine transporter (PheP) and phenylalanine aminotransferase (PAL), the former promoting amino acid uptake by bacteria and the latter an enzyme that converts phenylalanine into harmless metabolite trans-cinnamic acid ester, both of which are expressed under the control of an anoxia-inducible promoter. The two strains also express L-amino acid deaminase (LAAD), a protein that metabolizes phenylalanine into harmless phenylalanine in the presence of oxygen through arabinose-inducible promoter in aerobic conditions. These bacteria can be used to treat phenylketonuria (PKU) [7].

The article^[9] constructed an engineered EcN bacterial strain for tyrosine degradation, and HpaBC is a natural hydroxylase that has previously been used for the synthesis of hydroxytyrosol from bio-based L-tyrosine. It was selected for converting tyrosine to levodopa. To maximize tyrosine degradation rate, the tyrosine:H⁽⁺⁾ symporter (encoded by the tyrP gene) was chosen to work together with HpaBC, thus increasing the intracellular availability of tyrosine [9].

β -glucans (GN) are common prebiotics widely found in food and plants, which can be metabolized by the gut microbiota into short-chain fatty acids and exhibit rich physiological activities. Besides biological activities, GN is also used for the directed delivery of natural products. The basic principle of GN carrier is to take advantage of the stability of glomerulonephritis in the upper gastrointestinal tract to prevent the degradation and destruction of natural biologically active substances, and deliver them to the target area at high concentrations to maximize therapeutic effects, which is similar to the principle of nanocarriers that have been developed and applied for effective disease-targeted therapy. At the same time, since GN can be specifically recognized by Dectin-1 on M cells, it can be delivered to the lesion site via a complex macrophage-mediated transport mechanism, overcoming the biological barriers encountered by most nanocarriers in oral delivery. Moreover, GN and its derivatives have significant biodegradability and bioactivity in the intestinal environment, so they are also used as a class of functional materials or surface modifiers in microbial engineering. Inspired by this, the article envisions that GN can act as a "shield" to protect probiotics from the adverse pressure in the stomach and intestinal tract, followed by enzyme-assisted degradation by the gut microbiota to release probiotics and produce SCFAs for synergistic functions [10].

The method described in [10] involves anchoring Fe³⁺ ions via electrostatic adsorption onto the surface of EcN, forming a Fe-TA network layer with gallic acid and Fe through cross-linking, and finally depositing a layer of mGN onto the Fe-TA layer due to the strong attraction between the Fe-

TA network and the mGN via hydrogen bonds. This results in the formation of a modified "shield" based on prebiotics (Fe-TA@mGN) ^[10].

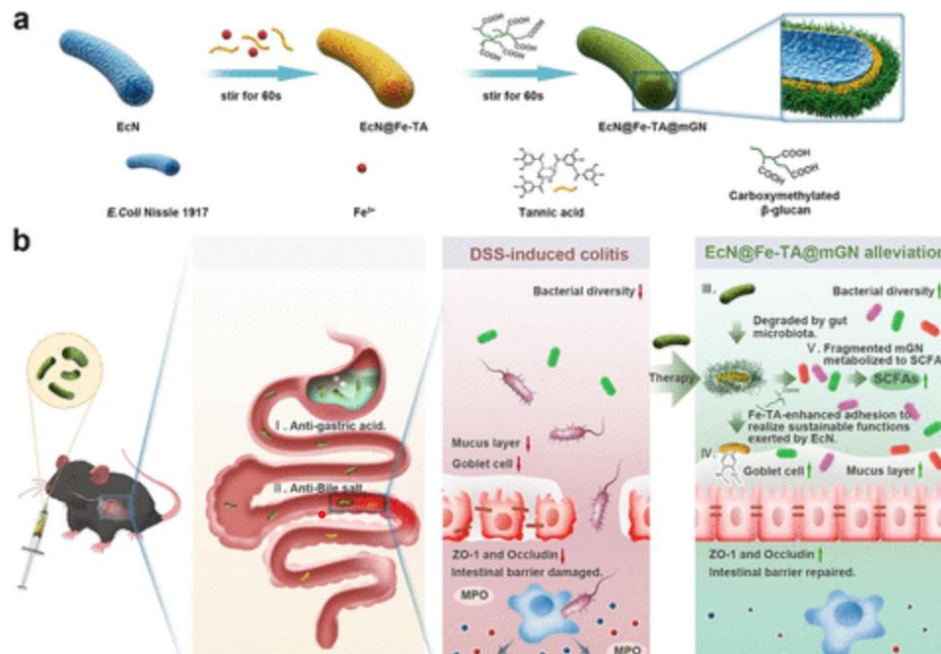


Figure 4. Schematic diagram of the preparation of EcN@Fe-TA@mGN and its alleviation process of DSS-induced colitis in mice.

Suspend the bacteria in a solution and mix it with TA and Fe for 60 seconds to form a Fe-TA network layer on the EcN. Then, encapsulate them with mGN by simply stirring. (b) The probiotics, equipped with the Fe-TA@mGN "shield", exhibit excellent resistance to gastric acid and bile salts and remain intact in the upper gastrointestinal tract. Once the EcN@Fe-TA@mGN reaches the colon, the mGN layer is degraded and metabolized into SCFAs by the gut microbiota, which work together with the EcN to alleviate colitis ^[10].

3. Diagnosis and treatment of spleen-related diseases

The *E. coli* Nissle 1917 (EcN) bacterium described in ^[9] was designed to express genes involved in tyrosine metabolism under the anoxic conditions of intestinal microenvironment (EcN-HT). It consumes tyrosine and produces L-DOPA (levodopa). Oral administration of EcN-HT enhances tyrosine degradation, reduces the accumulation of toxic metabolites, and prevents fatal liver damage ^[9] (preventing fatal liver damage in type 1 tyrosinemia).

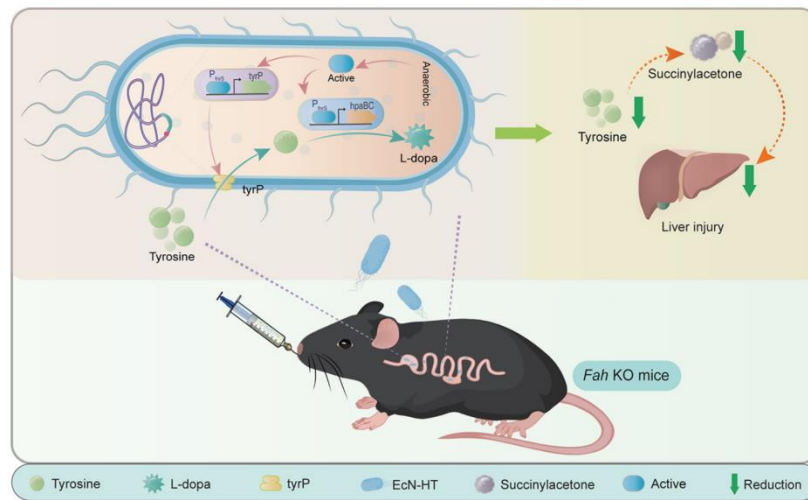


Figure 5. Schematic diagram of the action of EcN-HT

Reference:

- [1] Chiang, C. J., & Hong, Y. H. (2021). In situ delivery of biobutyrate by probiotic *Escherichia coli* for cancer therapy. *Scientific reports*, 11(1), 18172. <https://doi.org/10.1038/s41598-021-97457-3>
- [2] Gurbatri C R, Lia I, Vincent R, et al. Engineered probiotics for local tumor delivery of checkpoint blockade nanobodies. *Science Translational Medicine*, 2020, 12(530) : eaax0876.
- [3] Nguyen, DH., Chong, A., Hong, Y. *et al.* Bioengineering of bacteria for cancer immunotherapy. *Nat Commun* 14, 3553 (2023). <https://doi.org/10.1038/s41467-023-39224-8>
- [4] Abedi MH, Yao MS, Mittelstein DR, Bar-Zion A, Swift MB, Lee-Gosselin A, Barturen-Larrea P, Buss MT, Shapiro MG. Ultrasound-controllable engineered bacteria for cancer immunotherapy. *Nat Commun*. 2022 Mar 24;13(1):1585. doi: 10.1038/s41467-022-29065-2. PMID: 35332124; PMCID: PMC8948203.
- [5] Yan X, Liu XY, Zhang D, Zhang YD, Li ZH, Liu X, Wu F, Chen GQ. Construction of a sustainable 3-hydroxybutyrate-producing probiotic *Escherichia coli* for treatment of colitis. *Cell Mol Immunol*. 2021 Oct;18(10):2344-2357. doi: 10.1038/s41423-021-00760-2. Epub 2021 Sep 3. PMID: 34480146; PMCID: PMC8484604.
- [6] Zhou J, Li M, Chen Q, Li X, Chen L, Dong Z, Zhu W, Yang Y, Liu Z, Chen Q. Programmable probiotics modulate inflammation and gut microbiota for inflammatory bowel disease treatment after effective oral delivery. *Nat Commun*. 2022 Jun 14;13(1):3432. doi: 10.1038/s41467-022-31171-0. PMID: 35701435; PMCID: PMC9198027.

- [7]Lynch JP, Goers L, Lesser CF. Emerging strategies for engineering *Escherichia coli* Nissle 1917-based therapeutics. *Trends Pharmacol Sci.* 2022 Sep;43(9):772-786. doi: 10.1016/j.tips.2022.02.002. Epub 2022 Feb 26. PMID: 35232591 ; PMCID: PMC9378478
- [8]Gurbatri CR, Radford GA, Vrbanac L, Im J, Thomas EM, Coker C, Taylor SR, Jang Y, Sivan A, Rhee K, Saleh AA, Chien T, Zandkarimi F, Lia I, Lannagan TRM, Wang T, Wright JA, Kobayashi H, Ng JQ, Lawrence M, Sammour T, Thomas M, Lewis M, Papanicolas L, Perry J, Fitzsimmons T, Kaazan P, Lim A, Stavropoulos AM, Gouskos DA, Marker J, Ostroff C, Rogers G, Arpaia N, Worthley DL, Woods SL, Danino T. Engineering tumor-colonizing *E. coli* Nissle 1917 for detection and treatment of colorectal neoplasia. *Nat Commun.* 2024 Jan 20;15(1):646. doi: 10.1038/s41467-024-44776-4. PMID: 38245513; PMCID: PMC10799955.
- [9]Gu P, Xie L, Chen T, Yang Q, Zhang X, Liu R, Guo J, Wei R, Li D, Jiang Y, Chen Y, Gong W, Chen P. An engineered *Escherichia coli* Nissle strain prevents lethal liver injury in a mouse model of tyrosinemia type 1. *J Hepatol.* 2024 Mar;80(3):454-466. doi: 10.1016/j.jhep.2023.10.037. Epub 2023 Nov 10. PMID: 37952766.
- [10]Xie A, Ji H, Liu Z, Wan Y, Zhang X, Xiong H, Nie SP, Wan H. Modified Prebiotic-Based "Shield" Armed Probiotics with Enhanced Resistance of Gastrointestinal Stresses and Prolonged Intestinal Retention for Synergistic Alleviation of Colitis. *ACS Nano.* 2023 Aug 8;17(15):14775-14791. doi: 10.1021/acsnano.3c02914. Epub 2023 Jul 21. PMID: 37477584.
- [11] Stritzker J, Weibel S, Hill PJ, Oelschlaeger TA, Goebel W, Szalay AA. Tumor-specific colonization, tissue distribution, and gene induction by probiotic *Escherichia coli* Nissle 1917 in live mice. *Int J Med Microbiol.* 2007 Jun;297(3):151-62. doi: 10.1016/j.ijmm.2007.01.008. Epub 2007 Apr 19. PMID: 17448724.
- [12]Zhang, Y., Ji, W., He, L., Chen, Y., Ding, X., Sun, Y., Hu, S., Yang, H., Huang, W., Zhang, Y., Liu, F., Xia, L. (2018). *E. coli* Nissle 1917-Derived Minicells for Targeted Delivery of Chemotherapeutic Drug to Hypoxic Regions for Cancer Therapy. *Theranostics*, 8(6), 1690-1705. <https://doi.org/10.7150/thno.21575>

4. Overview of EcN Research Group

1 Hu Qinglian Research Group—Polyphenol Nanoparticles Modified Probiotics for Targeted Therapy of Microenvironment Reconstruction

Inflammatory bowel diseases (IBDs) refer to multifaceted disorders of the intestinal microenvironment and microbiota homeostasis. Given the wide range of bioactivities and high compatibility of polyphenols, there is considerable interest in developing a polyphenol-based collaborative platform to reshape the IBD microenvironment and regulate the microbiota. Here, we demonstrated the synergistic assembly of nanostructured polyphenols to modify probiotics and simultaneously deliver drugs for IBD treatment. Inspired by the unique structure of tannic acid (TA), we prepared nanostructured pBDT-TA using self-polymerized aromatic dithiols (BDT) and TA, which exhibited good antioxidant and anti-inflammatory abilities *in vitro*. Therefore, we layered pBDT-TA and sodium alginate (SA) on the surface of *Escherichia coli* Nissle 1917 to construct the collaborative platform EcN@SA-pBDT-TA. The modified probiotics showed enhanced resistance to oxidative and inflammatory stress, leading to better colonic accumulation and retention in IBD model mice. In addition, EcN@SA-pBDT-TA could alleviate dextran sulfate sodium (DSS)-induced colitis by controlling inflammatory responses, repairing the intestinal barrier, and regulating the intestinal microbiota. Importantly, EcN@SA-pbdt-TA-mediated IBD drug delivery could improve the therapeutic efficacy of DSS model mice. Considering the availability and functionality of polyphenols and prebiotics, we expect that nanostructured polyphenol-modified probiotics provide a solution for developing a collaborative platform for IBD treatment.

2 Yenifer Olivo-Martinez Group - Regulation of serotonin-related genes in an intestinal epithelial cell inflammation model by probiotic 1 β 1917 extracellular vesicles

Inflammatory bowel disease (IBD) is a chronic inflammatory disease involving dysregulated immune responses and imbalanced intestinal microbiota in genetically susceptible individuals. Current IBD treatments often have significant side effects and limited success, prompting the search for new therapeutic strategies. Microbiome-based approaches aim to restore the balance of the gut microbiota to achieve anti-inflammatory and mucosal healing properties. Extracellular vesicles (EVs) derived from beneficial gut microbes are emerging as potential postbiotic agents. Serotonin plays a crucial role in intestinal homeostasis, and its dysregulation has been associated with the severity of IBD. Our study utilized an IL-1 β -induced Caco-2 cell model of inflammation to investigate the effects of EVs from the probiotic Nissle 1917 (EcN) and commensal *Escherichia coli* on intestinal serotonin

metabolism under inflammatory conditions. We found strain-specific effects. Specifically, EcNev up-regulated SERT expression by downregulating miR-24, miR-200a, TLR4, and NOD1, thereby reducing free serotonin levels. Furthermore, EcNev reversed IL-1 β -induced changes in tight junction proteins and oxidative stress markers. These findings highlight the potential of postbiotic interventions as a therapeutic approach for IBD and related pathologies, with EcNev showing promise in modulating serotonin metabolism and preserving gut barrier integrity. This study is the first to demonstrate the regulatory effects of probiotic-derived EVs on miR-24 and miR-200a.

3 Shi Jun's group - Outer membrane vesicles derived from probiotic *Escherichia coli* Nissle1917 regulate the intestinal microbiome and host enterohepatic metabolism in obese and diabetic mice

Obesity and diabetes are common chronic metabolic disorders that can lead to imbalances in intestinal flora and intestinal-liver metabolism. Several studies have shown that probiotics, including *Escherichia coli* Nissle 1917 (EcN), promote microbial balance and metabolic health. However, how EcN outer membrane vesicles (EcN-OMVs) affect intestinal flora and how they affect metabolic disorders in obesity and diabetes has not been studied. **Methods** In this study, we evaluated the effects of EcN-OMVs on high-fat diet (HFD)-induced obesity and HFD + streptozotocin (STZ)-induced diabetes. **Results** EcN-OMVs reduced body weight, lowered blood glucose, and increased plasma insulin in obese mice. Similarly, EcN-OMVs treatment can change the ratio of Firmicutes/Bacteroidetes in the intestine, increase the intestinal flora that produces short-chain fatty acids (SCFAs), and affect the intestinal SCFA content. In addition, the intestinal metabolites ornithine and fumaric acid, liver ω -6 unsaturated fatty acids and SCFAs were significantly increased after EcN-omv. In summary, this study suggests that EcN-OMVs may act as a post-biological factor to regulate intestinal liver metabolism and improve the pathophysiology of obesity and diabetes.

4 O'Neill's group-*Escherichia coli* Nissle 1917 can inhibit the biofilm formation of *Pseudomonas aeruginosa* and reduce its toxicity

In order to find ways to alleviate bacterial virulence, we tested the activity of cell-free supernatants (CFS) from 25 human commensal bacteria and related bacteria against *Pseudomonas aeruginosa*. Among them, *E. coli* Nissle 1917 CFS significantly inhibited biofilm formation and dispersed existing *Pseudomonas* biofilms without inhibiting the growth of planktonic bacteria. Exposure to *E. coli* Nissle CFS resulted in a reduction of eDNA in biofilms as observed by confocal microscopy. *E. aeruginosa* larval virulence assays also showed significant protection 24 h before *P. aeruginosa* challenge. No inhibitory effects were observed against other *E. coli* strains against *P. aeruginosa*. Accord-

ing to proteomic analysis, *E. coli* Nissle CFS downregulated expression of several *P. aeruginosa* proteins involved in motility (flagellar secretion chaperone FliSB, b-type flagellar protein Flip, type IV pilus assembly enzyme PilB) and quorum sensing (acyl homoserine lactone synthase Las and HTH-type quorum sensing regulator rhlR), which are implicated in biofilm formation. Physicochemical characterization of the putative antibiofilm compound(s) indicated the involvement of thermolabile protein factors with molecular size exceeding 30 kDa.

5. EcN modification related methods

Escherichia coli Nissle 1917 (EcN) is a non-pathogenic *Escherichia coli* isolated from feces by Alfred Nissle in 1917. It does not express P-fimbriae adhesin, does not synthesize semi-rough lipopolysaccharide, and does not secrete α -hemolysin, that is, it does not produce known virulence factors and is biosafe. The use of EcN to treat intestinal diseases and conduct biofermentation production has unique advantages.

EcN has two cryptic plasmids, pMUT1 and pMUT2, which are genetically stable and non-transferable and are only found in EcN. pMUT1 carries a ColE1-type replication system, and pMUT2 contains a ColE2-like replication system and other replication systems, but no other open reading frames with known functions were found in these two plasmids [1]. These two cryptic plasmids may be related to resistance to phage infection. Blum-Oehler et al. constructed a plasmid-free variant of *Escherichia coli* DSM 6601 (EcN), laying the foundation for the use of cryptic plasmids for gene expression and modification.

The genetic modification of EcN is mainly focused on disease treatment, cell factories, CRISPR tool development and gene circuits, and the genetic modification operations related to EcN are also relatively complete.

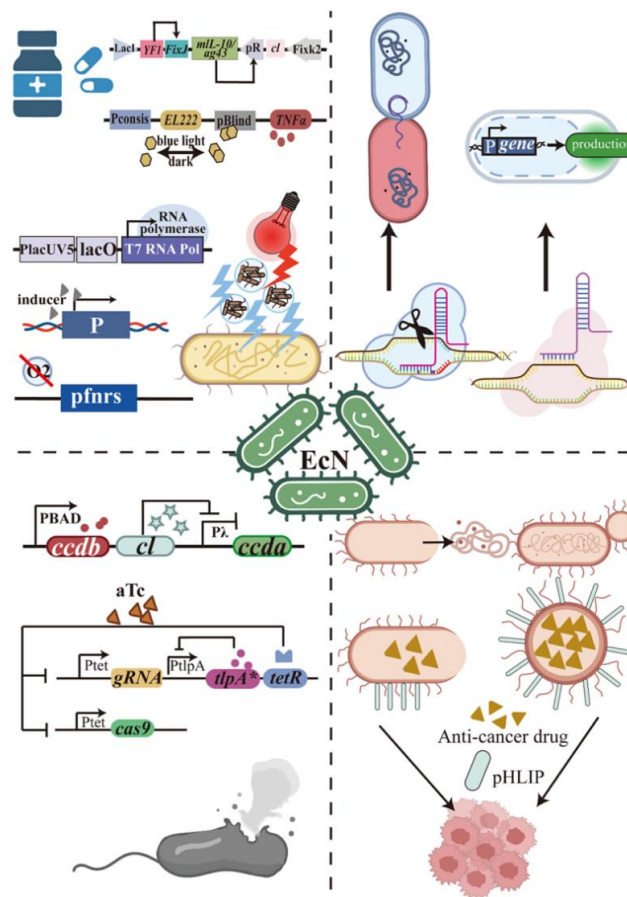


Figure 1. EcN-related gene modification and application^[21]

1. Disease treatment

EcN, as an intestinal probiotic, prevents pathogen infection by strengthening the intestinal mucosal barrier and stimulating the immune system. In recent years, genetically engineered EcN has been further applied to the treatment of inflammatory bowel disease, colorectal cancer and metabolic diseases.

In the treatment of IBD, Wang et al. relieved colitis and promoted intestinal antioxidant capacity by secreting the immunomodulatory protein Sj16 in EcN through the α -hemolysin secretion system^[2], and further increased Treg cells and reduced Th17 cells to alleviate immune response through butyrate^[3]. Praveschotinunt et al. developed a probiotic PATCH treatment system that relieves inflammation by displaying a fusion of the self-assembling protein subunit CsgA and TFF3 on the surface of EcN. TFF3 is a factor that promotes mucosal healing. This gene circuit is expressed through an arabinose-dependent promoter, and this EcN variant is named PBP8 CsgA-tff3^[4].

Yu et al. transferred the gene encoding the lipase ABC transporter of *Erwinia* into the EcN chromosome, which recognized and secreted transforming growth factor β (TGF- β 1) to promote mucosal healing^[5].

Cui et al. designed blue light-responsive expression of Ag43, an adhesin that promotes biofilm formation and enables bacteria to form "bio-glue", thereby increasing the survival rate of bacteria in the gastrointestinal tract^[6].

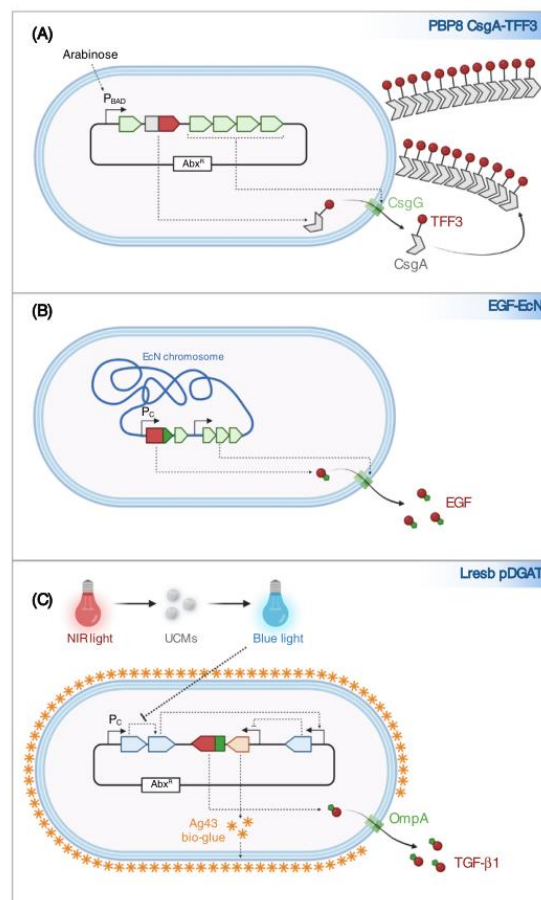


Figure 2. EcN modification is used to promote mucosal healing^[21]

At the same time, engineered EcN can also be used as an anticancer agent to treat cancer. The main ideas currently include improving the targeting and lethality of EcN and giving it diagnostic capabilities. Leventhal developed the EcN variant SYNB1891, which expresses *Listeria diadenylate cyclase* through the *Pfnrs* hypoxia-inducible promoter to produce high levels of cyclic AMP. The uptake of SYNB1891 by phagocytes can trigger the secretion of interferon. In mouse studies, the strain was directly injected into mouse cold tumors, which promoted their clearance and induced the development of immune memory, thereby preventing tumor recurrence^[7].

Table 1. EcN tumor therapy related research ^[22]

Function	Principle	Implement
Improve the targeting of the EcN	Microenvironment in response to low tumor pH	EcN-ca-dox release in the low pH tumor environment Dox
	Binding of specific targeting proteins	pHLIP packages the EcN target to the tumor environment HlpA binds to the tumor marker HSPG
	Expression of tumor therapeutic agents	EcN produces the neoplastic killer protein asulin
Increasing the tumor lethality of EcN	Blocking the nutritional source of the tumors	The antiangiogenic protein Tum-5 synergizes with the tumor killer protein p53
	Promote drug entry into the tumor	Apoptotin was coexpressed with cellular Tat
	To restore the T-cell immune activity	PD-L1 inhibitor bind to tumor PD-L1
Ening diagnostic power to EcN tumors	In response to the glucose concentration	Trz 1 senses the glucose concentration and displays different fluorescence intensities
	The urine color was analyzed	LacZ decomposes LuGal expression after tumor colonization and urine color changes

Li et al. linked the tumor-killing drug doxorubicin and the acid-labile linker of cis-aconitic anhydride to the surface of EcN based on the low pH characteristics of the tumor microenvironment to form EcN-ca-Dox targeted drugs, so that doxorubicin is released only in the tumor environment [8]. Similarly, Zhang et al. expressed the low pH-responsive insertion peptide pHLIP in EcN and anchored it on the surface of EcN cells to enhance the targeting effect [9].

Gurbatri et al. constructed PD-L1 inhibitors in EcN, using the quorum sensing effect to lyse EcN bacteria, release PD-L1 inhibitory proteins to bind to PD-L1 on the tumor surface, and activate T cells to continuously kill tumor cells [10].

Given that oral cancer is difficult to diagnose in the early stages, Hamilton et al. constructed an extracellular lactate-responsive fluorescent reporter gene-engineered EcN based on the lactate marker, including a lactate-binding response transcription factor LldR and a green fluorescent protein expressed by the LLD promoter. When lactate is absent, the LldR protein inhibits the promoter controlled by lld and blocks GFP expression [11]. It is very promising to transform EcN through synthetic biology technology and apply it to the treatment of diseases. Using EcN as a chassis cell to assemble biochips, etc., can further improve EcN's response and killing ability to tumors, and use its natural anti-inflammatory, targeting and tumor-suppressing properties to become a new powerful weapon for inflammation and tumor treatment.

2. EcN cell factory

EcN can reach a cell density equivalent to that of commonly used microbial matrix strains in the culture medium, making it an ideal high-density fermentation chassis organism.

Hu et al. metabolically modified EcN to enable it to produce β -alanine, and the yield reached 11.9 g/L after batch fermentation^[12]. At the same time, Datta et al. achieved the production of 3 g/L heparin polysaccharide through high-density fermentation and glucose as a carbon source^[13]. EcN is also a good microbial chassis for the production of bacterial cellulose. The natural bacterial cellulose producer *Xylacetobacter* grows slowly and is not suitable for industrial production. Sajadi et al. constructed a bacterial cellulose metabolic pathway in EcN and achieved a fermentation yield of 1.9 g/L in a shake flask^[14]. ω -3FA, including eicosapentaenoic acid and docosahexaenoic acid, are dietary components involved in fetal development, immunity and Alzheimer's disease. By expressing the EPA and DHA gene clusters in EcN, the shake flask production reached 31.36 mg/L^[15].

Table 2. EcN cell factory related research ^[21]

Metabolin	Function	Implement	Output
-----------	----------	-----------	--------

β -ALA	As a pharmaceutical intermediate	Expression of engineered aspartyl transferase increases the supply of fumarate Strengthening the pathway of fumarate to L-ASP Promote the conversion of oxaloacetate to L-ASP	11.9 g / L
L-Arg (Levostin)	Take the NH ₃ and convert it to the L-Arg	The negative regulator gene was deleted and the feed-back resistance enzyme gene was inserted	-
L-Arg (Levostin)	Increase in the number of T cells Synergism between L-Arg and PD-L1 inhibitors	Deleting the arginine repressor gene	-
Butyric acid	Improve anti-inflammatory efficacy	Heterologous da BCD and BUT genes	297.3±34.25 mg /L
Butyrate	Cell cycle arrest in the G1 phase induces the mitochondrial apoptosis pathway	Integrative butanyl-CoA from the glucose pathway converts butanyl-CoA to the butyrate gene	20 mm
5-ALA	As a component of the PDT	Overexpressing the genes encoding the 5-ALA components Inhibition of the downstream pathway, LA	300 mg /L
GLP-1	Neuropeptide expression regulating energy intake and consumption	Overexpression of GLP-1	-
TFF	Mucosal healing	The synthetic curli operon-encoding plasmid was constructed	-

3. EcN gene editing tools and gene circuit construction

The development of EcN-related gene editing tools and the construction of gene circuits will further facilitate the transformation and utilization of EcN. Fiege et al. integrated T7 RNA polymerase into the EcN genome and constructed the EcN::T7 strain. The engineered EcN strain can recognize the T7 promoter and achieve large-scale expression of heme-dependent proteins ^[16]. Studies have found that leaky expression of the T7 promoter is more severe in EcN than in BL21, and replacing the lacI promoter with a constitutive promoter can significantly increase the stringency of the T7 expression system in EcN ^[17]. The near-infrared light response system can also be applied to EcN. Due to the poor tissue transmittance of blue light, the near-infrared induction system EcN (ROEN) has been constructed, which contains a chimeric light sensor Cph8 that is inactivated under red light 635nm. This inhibits the expression of cI, thereby initiating the expression of cleavage genes controlled by p λ ^[18].

EcN-related CRISPR toolboxes are also gradually being developed and used. Studies have used related technologies to remove the cryptic plasmid in EcN and edit its genome, thereby improving its cell factory production capacity.

Pan et al. constructed a suicide system composed of L-arabinose-induced promoter, lambda promoter, lambda promoter inhibitor and toxin-antitoxin system. The toxin protein coding gene (ccdb) and cI are controlled by the pBAD promoter, and the antitoxin protein The encoding gene (ccda) is controlled by the p λ promoter. When arabinose is supplemented, cI is highly expressed and inhibits ccda transcription, thereby inducing EcN death ^[19].

More gene circuits can be developed by combining with the quorum sensing system. Hwang et al. constructed a suicide system that responds to the quorum sensing signal (N-acyl homoserine lactone). When *Pseudomonas aeruginosa* grows to a threshold cell density, the increased N-Homoserine lactone triggers the expression of cleavage genes of engineered EcN, which releases S5 and DspB of *septicarum*, thereby inhibiting *Pseudomonas aeruginosa* ^[20].

Reference:

- [1] Decanio, M. S., Landick, R. & Haft, R. J., The non-pathogenic *Escherichia coli* strain W secretes SslE via the virulence-associated type II secretion system beta. BMC MICROBIOL 13 130 (2013).

- [2] Wang, L. et al., rSj16 Protects against DSS-Induced Colitis by Inhibiting the PPAR-alpha Signaling Pathway. *THERANOSTICS* 7 3446 (2017).
- [3] Wang, L. et al., An engineered probiotic secreting Sj16 ameliorates colitis via Ruminococcaceae/butyrate/retinoic acid axis. *Bioeng Transl Med* 6 e10219 (2021).
- [4] Praveschotinunt, P. et al., Engineered E. coli Nissle 1917 for the delivery of matrix-tethered therapeutic domains to the gut. *NAT COMMUN* 10 5580 (2019).
- [5] Yu, M., Kim, J., Ahn, J. H. & Moon, Y., Nononcogenic restoration of the intestinal barrier by E. coli -delivered human EGF. *JCI INSIGHT* 4 (2019).
- [6] Cui, M. et al., NIR light-responsive bacteria with live bio-glue coatings for precise colonization in the gut. *CELL REP* 36 (2021).
- [7] Leventhal, D. S. et al., Immunotherapy with engineered bacteria by targeting the STING pathway for anti-tumor immunity. *NAT COMMUN* 11 2739 (2020).
- [8] Xie, S. et al., Doxorubicin-conjugated Escherichia coli Nissle 1917 swimmers to achieve tumor targeting and responsive drug release. *J CONTROL RELEASE* 268 390 (2017).
- [9] Zhang, Y. et al., E. coli Nissle 1917-Derived Minicells for Targeted Delivery of Chemotherapeutic Drug to Hypoxic Regions for Cancer Therapy. *THERANOSTICS* 8 1690 (2018).
- [10] Gurbatri, C. R. et al., Engineered probiotics for local tumor delivery of checkpoint blockade nanobodies. *SCI TRANSL MED* 12 (2020).
- [11] Hamilton, S., Shea, D., Ibsen, S. & Brasino, M., On-chip dielectrophoretic recovery and detection of a lactate sensing probiotic from model human saliva. *6.ELECTROPHORESIS* 44 442 (2023).
- [12] Hu, S. et al., Development of probiotic E. coli Nissle 1917 for 尾-alanine production by using protein and metabolic engineering. *APPL MICROBIOL BIOT* (2023).
- [13] Datta, P., Yan, L., Awofiranye, A., Dordick, J. S. & Linhardt, R. J., Heparosan Chain Characterization: Sequential Depolymerization of E. Coli K5 Heparosan by a Bacterial Eliminas Heparin Lyase III and a Bacterial Hydrolase Heparanase Bp to Prepare Defined Oligomers. *BIOTECHNOL J* 16 (2021).
- [14] Sajadi, E. et al., Increased cellulose production by heterologous expression of bcsA and B genes from Gluconacetobacter xylinus in E. coli Nissle 1917. *BIOPROC BIOSYST ENG* 42 2023 (2019).
- [15] Amiri-Jami, M., Abdelhamid, A. G., Hazaa, M., Kakuda, Y. & Griffiths, M. W., Recombinant production of omega-3 fatty acids by probiotic Escherichia coli Nissle 1917. *FEMS MICROBIOL LETT* 362 (2015).
- [16] Fiege, K. & Frankenberg-Dinkel, N., Construction of a new T7 promoter compatible Escherichia coli Nissle 1917 strain for recombinant production of heme-dependent proteins. *MICROB CELL FACT* 19 (2020).

- [17] Effendi, S. S. W. & Ng, I., Reprogramming T7RNA Polymerase in *Escherichia coli* Nissle 1917 under Specific Lac Operon for Efficient p -Coumaric Acid Production. *ACS SYNTH BIOL* 11 3471 (2022).
- [18] Fernandez-Rodriguez, J., Moser, F., Song, M. & Voigt, C. A., Engineering RGB color vision into *Escherichia coli*. *NAT CHEM BIOL* 13 706 (2017).
- [19] Pan, H. et al., Engineered NIR light-responsive bacteria as anti-tumor agent for targeted and precise cancer therapy. *CHEM ENG J* 426 (2021).
- [20] Hwang, I. Y. et al., Engineered probiotic *Escherichia coli* can eliminate and prevent *Pseudomonas aeruginosa* gut infection in animal models. *NAT COMMUN* 8 (2017).
- [21] Yu, M., Hu, S., Tang, B., Yang, H. & Sun, D., Engineering *Escherichia coli* Nissle 1917 as a microbial chassis for therapeutic and industrial applications. *BIOTECHNOL ADV* 67 (2023).
- [22] 王彦雯, 叶静怡 & 王鹏超, 合成生物学在 *E.coli* Nissle 1917 靶向治疗癌症中的应用. *中国生物化学与分子生物学报* 37 20 (2021).

6.iGEM Mid-Year EcN-Related Projects in the Past Five Years

2019

3081 Peking

<https://2019.igem.org/Team:Peking>

The host cell, *E. coli*, also exhibits significant changes in its general physiology over time, which can directly or indirectly affect the "desired" functions of the synthetic biological part. Many challenges can be addressed if we can better control the growth rate of the cells. However, previous methods have many shortcomings for growth rate control. Therefore, we developed a novel system to achieve precise growth rate control by targeting the DNA replication origin using dCas9.

Solution: Growth control toolbox directly targeting genome replication

In *E. coli*, genome replication starts from a single site, *oriC*. The formation of DnaA protein filaments on the DnaA box within *oriC* accurately regulates the opening of the replication bubble and the subsequent helicase loading. Here, we managed to block DnaA binding (mainly based on the competition of CRISPR/dCas9 with the DnaA box array).

So their method: a new method for prokaryotic genome replication interference (CRISPRri), similar to transcriptional inhibition by CRISPR interference [5]. The CRISPRri system can well control the growth of *E. coli* Nissle 1917 strain.

Considering that the *oriC* sequence of *E. coli* Nissle is the same as the *oriC* sequence in Top10, we hope to prove that the function of the CRISPRri system in the new *E. coli* Nissle is the same as that in Top10.

They transformed the CRISPRri system into *E. coli* Nissle 1917. Through the microfluidic imaging system, we observed that under 0.25% arabinose treatment, the bacterial proliferation of the experimental group (whose sgRNA targeted the M box in the *oriC* region) was significantly slower than that of the control group (polyA). Quantitative results showed that the average cell doubling time of the experimental group (M⁺) was greater than 2 hours, while the control group (polyA) only needed about 40 minutes to complete cell division. This shows that the CRISPRri system can well control the growth of EcN.

3036 BNU-China

<https://2019.igem.org/Team:BNU-China>

Obesity leads to many diseases and has become a serious problem worldwide. They proposed to develop a synthetic lean-promoting intestinal microorganism (SLIM) that can colonize the human intestine and promote the catabolism of fat (both assimilated and unassimilated fat) through two separate pathways. One of them continuously enhances the β -oxidation of excessive high fatty acids

consumed by humans, while the other leads to excessive production of acetic acid (which acts as a signal to promote the consumption of human white adipose tissue). The latter is placed under the control of a sensing module so that it is activated only at the optimal time, thus preventing interference with digestion caused by acidification.

They chose *Escherichia coli* Nissle 1917 as the chassis of the final product. Because *E. coli* nissle1917 is a probiotic strain derived from the human intestine, it is not easy to trigger an immune response. In addition, to prevent the engineered bacteria from being excluded, they enhanced their proliferation ability to make them more competitive. BNU-China 2018 has demonstrated that strains can gain significant proliferation advantages by overexpressing glucose dehydrogenase (GDH), a key enzyme in the pentose phosphate pathway. The same enzyme can be used to promote microbial colonization.

On the other hand, they also do not want the engineered bacteria to disturb the natural microbiome. Therefore, we included a quorum sensing system orthogonal to the natural microbiome in our design. Due to the limitations of the quorum sensing mechanism, once a certain population threshold is reached, the proliferation rate of the engineered bacteria will slow down, ensuring that it does not interfere with the intestinal microbiome.

3245 Fudan

<https://2019.igem.org/Team:Fudan>

Lactose intolerance is a common disease in my country, which is insignificant for adults but critical for infants. We used *E. coli* Nissle 1917 as our chassis and added several modules to improve its viability and lactose digestion ability so that it can stably solve lactose intolerance.

The first is the acid-resistant module. In order to make the bacteria survive when passing through the highly acidic environment of the stomach, we improved the acid resistance system by knocking out the key gene *hns* in the acid resistance system 2 (AR 2) of *E. coli* Nissle 1917. Therefore, the acid resistance of *E. coli* Nissle 1917 was significantly improved, and it did not suffer a huge loss when passing through gastric acid.

The second is the antimicrobial peptide module. We used microcin B17 (Mcc B17) to moderately inhibit the reproduction of other flora, thereby improving the competitiveness of *E. coli* Nissle 1917 by secreting antimicrobial peptides. Despite the fierce competition with the intestinal flora, it can reproduce to a certain number.

The third is the secretion module. In order to make lactose decompose more efficiently, we chose the latter between increasing lactase activity and expression. The fusion of *luxpR* and the high-strength promoter J23100 significantly increased the expression of the *lacZ* gene.

We used the Quorum Sensing System (QS System) to control our independent modules. When our engineered bacteria reach the stomach, they can survive thanks to the acid-resistant module. When they are in the intestine, the antimicrobial peptide module works, so our bacteria have an advantage and multiply rapidly. When the population reaches a certain level to trigger the QS system, the antimicrobial peptide module is turned off and lactase secretion begins. In this way, we reduce the burden on the bacteria, allowing them to multiply before secreting lactase. And when the number is not enough, they can start multiplying again, repeating the previous cycle, keeping their population stable.

2967 NEU_CHINA

https://2019.igem.org/Team:NEU_CHINA/homepage.html

We treat IBD by using intestinal flora transplantation. *E.coli* Nissle 1917 (EcN), a probiotic in the human intestine, serves as a chassis cell. We tried to transform our plasmid into EcN and characterized EcN.

1. Considering the sensitivity of EcN, we optimized the NO sensor of last year's team and successfully reduced the leakage rate of the sensor by additional regulatory protein binding sequences. In addition, we constructed another new NO sensor and characterized it for better selection.

2. We found a viable system *Pseudomonas syringae* and designed a tunable gain amplifier. We built a leaky model of a fixed gain amplifier to further explore the tunable amplifier.

3. We characterized the therapeutic compounds IL-10 and myosinase using immunoblotting. To confirm the activity of these compounds, we tested the activity of the immune cytokine IL-10.

4. With the toxin-antitoxin system - mazEF, we significantly improved cell viability and precisely controlled the expression of antibacterial toxins at different temperatures.

All these works and optimizations contributed to the creation of EcN, which can precisely colonize the inflammatory areas of the gastrointestinal tract; successfully express and secrete anti-inflammatory proteins, and function in the intestine with little biohazard and minimal side effects.

We used *E.coli* Nissle 1917 (EcN) as a competent cell to express inflammation-sensing proteins. EcN was chosen not only because it is an *E. coli* strain (which can grow and proliferate rapidly, easily obtain components, and can be easily stimulated to transform into competent cells to absorb foreign DNA), but also because it is a clinically confirmed harmless probiotic, lacks virulence factors, is used to promote intestinal health, and is confirmed as a therapeutically safe strain. In addition, EcN has several unique functions, including microcin synthesis and a different iron absorption system, which provide EcN with advantages in intestinal colonization. In addition, recombinant EcN has no harmful effects on migration, clonal expansion, and immune cell activation in immune-competent hosts. Therefore, *E. coli* Nissle 1917 may be an ideal bioengineered strain for in situ synthesis of gastrointestinal-related inflammation-sensing proteins and anti-inflammatory molecules.

5313 NEFU-China

The team used *E. coli* Nissle 1917 to design a novel controllable system to release anti-tumor drugs in response to changes in uric acid levels.

The project is based on in vivo bacterial therapy, selecting a bacterial carrier that can survive in human tissues without causing an unbearable inflammatory response. The system is mainly divided into four core design modules: tumor microenvironment sensor module, uric acid regulation module, targeting and self-regulation module, and plasmid protection module.

To ensure that the treatment system works only when it reaches tumor cells, the team selected hypoxia and high lactate levels, characteristics of the tumor microenvironment, as detection indicators for the sensor module, and activated the expression and secretion of the mutant form of tumor necrosis factor- α (mTNF- α) by sensing these two parameters, thereby enhancing the activity of killing tumor cells. To prevent these bacteria from adversely affecting normal tissues, the team used cytosine deaminase (CD) as part of the targeting and self-regulation module, which can convert 5-fluorocytosine (5-FC) into 5-fluorouracil (5-FU) to kill bacteria. The team also designed *E. coli* Nissle1917 to express *ftnA-M*, a mutant form of the ferritin gene, which can effectively accumulate iron ions in the bacteria, which will enable the bacteria to locate at the tumor site when exposed to a magnetic field to enhance tumor cell killing. When the uric acid level at the tumor site is low, the bacteria will release anti-tumor drugs. However, when uric acid reaches a certain level (threshold), the bacteria slow down the release of anti-tumor drugs and instead express uricase to reduce uric acid levels. This achieves the goal of killing tumors while avoiding the occurrence of acute uric acid nephropathy, thereby treating cancer patients gently and safely. To prevent the plasmid from being lost after injection into mice, the team knocked out the *alr* and *dadX* genes in the *E. coli* genome. Once the coding sequence of the *alr* or *dadX* gene is inserted into the vector, the bacteria can only survive with this vector. Therefore, the bacteria will not lose this vector.

The team used *E. coli* Nissle1917 to deliver and release anti-cancer agents while preventing the accumulation of uric acid during cancer treatment, which will lead to the elimination of cancer cells and the prevention of kidney disease.

2997 NCKU

The team used engineered Nissle1917 to reduce the accumulation of p-cresol in humans to treat chronic kidney disease (CKD).

The project engineered *E. coli* Nissle1917 to have two specific functions. First, by diverting the original p-cresol production pathway, the precursor tyrosine of p-cresol is converted into a substance that is beneficial to the body. The team engineered *E. coli* Nissle to express TyrP and TAL to convert

excess tyrosine in the intestine into p-coumaric acid, which is beneficial to humans. By reducing the total amount of tyrosine (the precursor of p-cresol), the amount of p-cresol produced by other intestinal flora can be reduced. Second, by producing bacteriocins (CBM-B) from commonly used probiotics, the number of other *Clostridium* strains (the main p-cresol producers) can be reduced. The team designed three constructs, one of which is a fusion of full-length CBM-B with an *E. coli* secretory protein (yebF), and two of which are the C-terminal domains of CBM-B and yebF (which have bactericidal activity against certain *Clostridium* strains), with two different linkers: a GS linker (three glycine serine repeats: GSGSGS) and a TB linker (thrombin cleavage site: LVPRGS).

For biosafety reasons, the team implemented an inducible kill switch by deleting the *can* gene, thus ensuring that *E. coli* Nissle1917 can only survive in the human intestine. With the *can* gene knocked out, *E. coli* cannot convert CO₂ into bicarbonate (a metabolic substrate) quickly enough before CO₂ diffuses out and causes cell death. Inside the intestine, where *E. coli* lives, CO₂ concentrations are high enough to allow spontaneous conversion of CO₂ into bicarbonate ions. However, when *E. coli* leaves the human body, CO₂ concentrations decrease, causing its death. The team also made it a kill switch by deleting DapA (4-hydroxy-tetrahydrodiaminopimelate synthase, an enzyme necessary for cell wall synthesis), so that the bacteria will have to rely on exogenous diaminopimelate (DAP) for cell wall synthesis and growth, as the bacteria will die without a supply of DAP.

Pseudomonas has the ability to sense p-cresol. The team centered on *Pseudomonas fluorescens* (PC24), which has a gene cluster consisting of a positive inducible operon, *pchR*, which enables it to sense p-cresol and promote transcription of a downstream region to degrade p-cresol. *pchR* encodes an activator protein that binds to an inducer (p-Cresol) and causes a conformational change in the protein. Thus, the activator-p-cresol complex will bind to the p-cresol sensing region located downstream of *pchR* and initiate transcription. The team replaced the downstream gene region with GFP so it can sense p-cresol and emit light.

3096 Tuebingen

The team developed a probiotic based on *E. coli* Nissle 1917, which secretes Exendin-4, an incretin mimetic that responds to glucose. The therapeutic agent is the probiotic secreting the incretin mimetic that is placed in the patient's intestine via an orally administered capsule. This makes treatment easier, during which no medication needs to be taken regularly.

The system works via a carbon catabolite repression system that turns on transcription of *tetR*. Upstream of Exendin-4 is a TetR repressible promoter that ensures that Exendin-4 is only transcribed in the presence of glucose. In addition, Exendin-4 is coupled to an N-terminal secretion tag and a C-terminal cell penetrating peptide (CPP). The secretion tag ensures that Exendin-4 is secreted into the

intestine, while the CPP ensures that Exendin-4 is absorbed into the blood, thereby increasing overall bioavailability.

To ensure its safe use as a transgenic organism, the team integrated a CRISPR/Cas3-based Kill-Switch using our probiotic. Once the stop switch is activated, the Cas3 nuclease degrades the bacterial DNA, thus preventing the spread of the GMO into the environment. Once the probiotic leaves its designated area, the inhibition of the stop switch is abolished and the CRISPR/Cas3 system is activated. Therefore, the stop switch is a device that can be used in a variety of therapies by exchanging drugs and/or conditions.

3282 Lund

The team genetically modified the probiotic - *Escherichia coli* Nissle 1917 (EcN) to increase its ability to absorb and accumulate lead and arsenic.

The team was able to increase the arsenic and lead accumulation ability of EcN by taking some genes from *Candida metallidiae* CH34 and inserting them into the probiotic chassis EcN. The bacteria will then be ingested as a probiotic and, thanks to EcN's innate abilities, it adapts to the gut microbiome. There, it will accumulate arsenic and lead in your place. Soon, the metal-filled bacteria will be excreted, thus reducing the amount of toxic metals in the body.

PbrT, a leader transporter, and PbrD, a putative leader binding protein, were used to enhance lead accumulation. Overexpression of ArsR, a regulatory protein in the ars operon, in transgenic *E. coli* increased arsenic accumulation 5-60-fold, so this protein was included as an accumulation protein.

2020

3121 IISER_Bhopal

The team used modified engineered probiotics *E. coli* Nissle 1917 to express therapeutic proteins in situ in the intestine, promote the transdifferentiation of intestinal crypt cells, and then produce insulin in the body to achieve the therapeutic effect on diabetes.

The team's modification of EcN is based on an innovative ternary system: sensor (such as NO or thiosulfate) - secretory peptide (CsgA) - therapeutic protein. When the intestinal environment changes, such as inflammation, the sensor triggers EcN, leading to the secretion of therapeutic proteins, promoting the repair and functional recovery of intestinal cells. To enhance the secretion capacity of EcN, the team specially fused CsgA secretory peptide to ensure the effective release of therapeutic proteins.

Due to the possible presence of adherent invasive *Escherichia coli* (AIEC) in the intestines of diabetic patients, the team introduced the production capacity of Microcins. Microcins, as a group of

secretory peptides that can inhibit the growth of AIEC, work by binding to ATPase on the AIEC cell membrane. To achieve this function, EcN was endowed with a mature *E. coli* CA46 micromycin expression system, enabling it to inhibit AIEC in the intestine and help restore the balance of intestinal flora.

To solve the harsh conditions of the digestive tract environment faced by EcN during transportation, the team used PCS (polydopamine co-deposited with chitosan) technology. PCS can form a protective film on the surface of EcN, protecting EcN from damage by the digestive tract environment through chemical self-assembly.

Finally, by encapsulating the engineered EcN into capsules, the innovative product iBETA can be obtained, which provides a non-invasive treatment method that allows patients to receive treatment through oral capsules. This method is expected to reduce the dependence of diabetic patients on external insulin injections, improve the convenience and compliance of treatment, and reduce the economic burden of diabetes treatment worldwide.

3661 CPU_CHINA

The team used the engineered probiotic *E. coli* Nissle 1917 (EcN) as the core and genetically engineered it to give it two main functions: one is to secrete specific bacteriocins to reduce the abnormally high density of *Enterococcus faecalis* in the intestine and restore the balance of the intestinal flora; the other is to secrete the anti-inflammatory cytokine IL-22 to alleviate hepatitis, thereby developing an innovative treatment for alcoholic liver disease (ALD).

The team's EcN design uses a ternary system similar to that used to treat inflammatory bowel disease (IBD): sensor (such as alcohol metabolites) - secreted peptide (CsgA) - therapeutic protein. When biomarkers of ALD are detected, EcN is triggered to secrete therapeutic proteins into the intestine to promote intestinal health and anti-inflammation. At the same time, the team specifically selected the CsgA secretory peptide to enhance EcN's ability to secrete therapeutic proteins.

Drawing on IBD treatment strategies, the team introduced cutting-edge research and discovered the important role of *Enterococcus faecalis* (*E. faecalis*) in the development of ALD. In subsequent designs, the team used EcN to specifically reduce the density of *E. faecalis* by secreting bacteriocins JM79, $\text{plw}\alpha$, and $\text{plw}\beta$, and controlled its expression through a quorum sensing system, thereby enhancing the specificity of treatment.

At the same time, the team used the alcohol-induced promoter palcA to control the expression of IL-22 to ensure that it is activated only when alcohol is present, thereby exerting an anti-inflammatory effect in the liver. This design is similar to the use of sensors to detect inflammation and trigger the secretion of therapeutic proteins in IBD treatment.

In addition, in exchanges with the Alcoholic Anonymous community, the team learned about the limitations of existing alcohol removal drugs and optimized the design accordingly, focusing on controlling the balance of intestinal flora and providing anti-inflammatory support through EcN, rather than simply decomposing acetaldehyde.

3475 Tongji_China

The team used engineered probiotics *E. coli* Nissle 1917 to produce microcins and therapeutic proteins.

The team used an innovative ternary system design: inflammation-sensing sensors (such as nitric oxide NO or thiosulfate) - auxiliary secretion peptides (CsgA) - therapeutic proteins. When inflammation occurs, the sensor can quickly detect inflammatory markers, trigger the secretion of therapeutic proteins, and with the assistance of CsgA, effectively deliver these proteins to the intestine to promote the healing of damaged areas.

Targeting the common adherent-invasive *Escherichia coli* (AIEC) in IBD, by utilizing Microcins, a secreted peptide produced by a specific strain of *E. coli*, it can specifically bind to the ATPase of AIEC and inhibit its growth without affecting the EcN strain that produces them. In order to enhance this function of EcN, the team integrated the mature *E. coli* CA46 microcin expression system into EcN, enabling the production of functional Microcins, effectively inhibiting AIEC in the intestines of IBD patients and promoting the recovery of intestinal flora.

As for the stability of engineered bacteria in the digestive tract, the team used PCS technology (polydopamine co-deposited with chitosan). Through chemical self-assembly, this material forms a protective film on the surface of EcN cells to protect the bacteria from being destroyed by digestive enzymes, while improving its stability and persistence in the intestine.

3482 UNI Australia

The team used the engineered probiotic *E. coli* Nissle 1917 (EcN) as the core, and genetically engineered it to produce the anticancer peptide azurin to achieve the therapeutic effect on colorectal cancer.

The team's treatment plan is based on a ternary system: inflammation-sensing sensors (such as nitric oxide NO or thiosulfate) - secretory peptides (CsgA) - anticancer peptides (azurin). When colorectal cancer cells release inflammatory signals, the engineered EcN can respond quickly and secrete the anticancer peptide azurin into the intestine with the help of the CsgA secretory peptide, directly acting on tumor cells and effectively inhibiting their growth, while promoting the healing of damaged intestinal tissue.

In order to ensure the stability and therapeutic efficiency of EcN in the intestine, the team used PCS technology (polydopamine co-deposited with chitosan) to form a protective film on the surface of EcN cells. This film not only protects EcN from damage by the digestive tract environment, but also enhances its persistence in the intestine, ensuring the continuity and effectiveness of treatment.

In order to solve the problem of positioning the engineered bacteria in the intestine, the team knocked out the *FliC* gene of EcN, and used Lambda Red recombination and I-SceI cutting technology to effectively eliminate the active movement ability of EcN and improve its positioning accuracy in the intestine.

In terms of biosafety, the team designed two killing switch mechanisms to ensure that EcN can be safely removed from the patient's body after completing the treatment task. The internal killing switch quickly eliminates EcN remaining in the gastrointestinal tract through arabinose-induced E protein expression; the external termination switch uses the temperature-sensitive RBS and RelE-RelB system, which is triggered when the ambient temperature is lower than the human body temperature, effectively killing EcN that may leak into the environment.

3121 IISER_Bhopal

The team used modified engineered probiotics *E. coli* Nissle 1917 to express therapeutic proteins in situ in the intestine, promote the transdifferentiation of intestinal crypt cells, and then produce insulin in the body to achieve the therapeutic effect of diabetes.

The team's transformation of EcN is based on an innovative ternary system: sensor (such as NO or thiosulfate) - secretory peptide (CsgA) - therapeutic protein. When the intestinal environment changes, such as inflammation, the sensor triggers EcN, resulting in the secretion of therapeutic proteins, promoting the repair and functional recovery of intestinal cells. To enhance the secretion capacity of EcN, the team specially fused the CsgA secretory peptide to ensure the effective release of therapeutic proteins.

Due to the possible presence of adherent invasive *Escherichia coli* (AIEC) in the intestines of diabetic patients, the team introduced the production capacity of Microcins. Microcins, as a group of secretory peptides that can inhibit the growth of AIEC, work by binding to ATPase on the AIEC cell membrane. To achieve this function, EcN was endowed with a mature *Escherichia coli* CA46 microcin expression system, enabling it to inhibit AIEC in the intestine and help restore the balance of intestinal flora.

To solve the harsh conditions of the digestive tract environment faced by EcN during delivery, the team used PCS (polydopamine co-deposited with chitosan) technology. PCS can form a protective film on the surface of EcN, and protect EcN from damage by the digestive tract environment through chemical self-assembly.

Finally, by encapsulating the engineered EcN into capsules, the innovative product iBETA can be obtained, which provides a non-invasive treatment method that allows patients to receive treatment through oral capsules. This method is expected to reduce the dependence of diabetic patients on external insulin injections, improve the convenience and compliance of treatment, and reduce the economic burden of diabetes treatment worldwide.

3661 CPU_CHINA

The team took the engineered probiotic *E. coli* Nissle 1917 (EcN) as the core and genetically engineered it to give it two main functions: one is to secrete specific bacteriocins to reduce the abnormally high density of *Enterococcus faecalis* in the intestine and restore the balance of intestinal flora; the other is to secrete the anti-inflammatory cytokine IL-22 to reduce hepatitis, thereby developing an innovative treatment for alcoholic liver disease (ALD).

The team's EcN design uses a ternary system similar to that used to treat inflammatory bowel disease (IBD): sensor (such as alcohol metabolites) - secretory peptide (CsgA) - therapeutic protein. When ALD biomarkers are detected, EcN is triggered to secrete therapeutic proteins into the intestine to promote intestinal health and anti-inflammation. At the same time, the team specifically selected the CsgA secretory peptide to enhance EcN's ability to secrete therapeutic proteins.

Drawing on IBD treatment strategies, the team introduced cutting-edge research and discovered the important role of *Enterococcus faecalis* (*E. faecalis*) in the development of ALD. In subsequent designs, the team used EcN to specifically reduce the density of *E. faecalis* by secreting bacteriocins JM79, *plw* α , and *plw* β , and controlled its expression through a quorum sensing system, thereby enhancing the specificity of treatment.

At the same time, the team used the alcohol-inducible promoter *palcA* to control the expression of IL-22, ensuring that it is activated only in the presence of alcohol, thereby exerting an anti-inflammatory effect in the liver. This design is similar to the use of sensors to detect inflammation and trigger the secretion of therapeutic proteins in IBD treatment.

In addition, in communication with the Alcoholic Anonymous community, the team learned about the limitations of existing alcohol removal drugs and optimized the design accordingly, focusing on controlling the balance of intestinal flora and providing anti-inflammatory support through EcN, rather than simply decomposing acetaldehyde.

3475 Tongji_China

The team used engineered probiotics *E. coli* Nissle 1917 to produce micronutrients and therapeutic proteins.

The team used an innovative ternary system design: inflammation sensing sensors (such as nitric oxide NO or thiosulfate) - auxiliary secretion peptides (CsgA) - therapeutic proteins. When inflammation occurs, the sensor can quickly detect inflammatory markers, trigger the secretion of therapeutic proteins, and with the assistance of CsgA, these proteins are effectively delivered to the intestine to promote the healing of damaged areas.

Targeting the common adherent invasive *Escherichia coli* (AIEC) in IBD, the team used Microcins, a secreted peptide produced by a specific strain of *E. coli*, to specifically bind to the ATPase of AIEC and inhibit its growth without affecting the EcN strain that produces them. In order to enhance this function of EcN, the team integrated the mature *E. coli* CA46 microcin expression system into EcN, enabling the production of functional Microcins, effectively inhibiting AIEC in the intestines of IBD patients and promoting the recovery of intestinal flora.

As for the stability of engineered bacteria in the digestive tract, the team used PCS technology (polydopamine co-deposited with chitosan). Through chemical self-assembly, this material forms a protective film on the surface of EcN cells to protect the bacteria from damage by digestive enzymes, while improving its stability and durability in the intestine.

3482 UNI Australia

The team used the engineered probiotic *E. coli* Nissle 1917 (EcN) as the core, and genetically engineered it to produce the anticancer peptide azurin to achieve the therapeutic effect on colorectal cancer.

The team's treatment plan is based on a ternary system: inflammation-sensing sensors (such as nitric oxide NO or thiosulfate) - secretory peptides (CsgA) - anticancer peptides (azurin). When colorectal cancer cells release inflammatory signals, the engineered EcN can respond quickly and secrete the anticancer peptide azurin into the intestine with the help of the CsgA secretory peptide, directly acting on tumor cells and effectively inhibiting their growth, while promoting the healing of damaged intestinal tissue.

In order to ensure the stability and therapeutic efficiency of EcN in the intestine, the team used PCS technology (polydopamine co-deposited with chitosan) to form a protective film on the surface of EcN cells. This membrane not only protects EcN from damage by the digestive tract environment, but also enhances its persistence in the intestine, ensuring the continuity and effectiveness of treatment.

In response to the problem of positioning of engineered bacteria in the intestine, the team knocked out the *FliC* gene of EcN, and used Lambda Red recombination and *I-SceI* cutting technology to effectively eliminate the active movement ability of EcN and improve its positioning accuracy in the intestine.

In terms of biosafety, the team designed two kill switch mechanisms to ensure that EcN can be safely removed from the patient's body after completing the treatment task. The internal kill switch quickly eliminates EcN remaining in the gastrointestinal tract through arabinose-induced E protein expression; the external termination switch uses the temperature-sensitive RBS and RelE-RelB system, which is triggered when the ambient temperature is lower than the human body temperature, effectively killing EcN that may leak into the environment.

3562 WHU-China

The team designed and modified the probiotic Nissle 1917 to bind to pathogens and promote the immune system. As one of the most suitable chassis cells, Nissle1917 has a rationally designed gene module targeting the Gram-negative quorum sensing system.

Probiotics are considered to be the next generation of chassis in the biomedical field, and have been shown by animal models to protect hosts from respiratory infections through nasal or oral methods. It is also speculated that probiotics can secrete chemicals to stimulate and activate the immune system.

The team used *Pseudomonas aeruginosa* as a pathogen, and the design of probiotics included four main aspects, namely quenching module, sensing module, safety measures and cell-free quorum sensing.

The quenching module is constructed based on quorum quenching enzymes. Involving AHL acylase and AHL lactonase, it can degrade the main communication tool of Gram-negative bacteria-N-acyl homoserine lactones (AHLs) (affecting the quorum sensing network of *Pseudomonas aeruginosa*). As the concentration of AHLs decreases, Gram-negative bacteria cannot express virulence. To degrade AHLs with different acyl chain lengths, enzyme combinations can be selected with the help of quorum sensing kinetic models.

The sensing module is built based on immune cell chemotaxis. The goal is that at the same infection level (or natural chemotaxis), probiotics are more sensitive to the effects of immune cells than pathogens. The homologous transcription factor PqsR will be overexpressed in the host through a strong constitutive promoter to capture PQS (a major contributor to virulence, its autoinducer *Pseudomonas* quinolone signal (PQS) is a characteristic molecule of the genus *Pseudomonas*). A phage-derived system is used to amplify the signal to achieve robust expression of downstream recombinant chemokines, which include the sec tag DsbA for excretion. The chemotaxis of several specific pathogen-killing immune cells is used to eradicate *Pseudomonas aeruginosa*.

The team designed two measures to ensure the safety of probiotics. The first safety measure is to develop a flexible mechanism to control the secretion of chemokines, which involves the expression of TEV protease to solve the possible leakiness of the upstream pqs promoter and regulate the

secretion of chemokines by cutting off the sec tag to prevent cytokine storm. A model was designed to describe this process in mathematical language. The second safety measure is to add a toxin-anti-toxin system in important modules to prevent horizontal gene transfer. The loss of the transformation plasmid will lead to the suicide of the engineered probiotics. The toxin-antitoxin system of MazE and MazF was finally used to test its efficiency.

2021

3771 NCKU_Tainan

The team's goal is to alleviate chronic stress-induced depression (CSID) and designed a probiotic that can catalyze the production of taurine in response to the increase of stress biomarkers in the intestine. Taurine is a semi-essential amino acid that can act as a neuroprotectant. The chassis cell selection used *Escherichia coli* Nissle 1917 is based on biosafety, a non-pathogenic *E. coli* strain. The resistance genes were removed to eliminate concerns about the spread of antibiotic resistance.

The team designed two sensing systems for *E. coli* Nissle 1917, which are induced by the stress biomarkers interferon- γ (IFN- γ) and reactive oxygen species (ROS). (Depression is highly correlated with inflammatory response and oxidative stress, and the detection of CSID is difficult, so ROS and IFN- γ are set as two potential biomarkers for CSID.) When the concentrations of these two biomarkers increase, the enzymes required for taurine synthesis are expressed and coordinately convert L-cysteine into taurine (which has both anti-inflammatory effects and the role of scavenging ROS).

Oxidative stress sensing system: Using the soxRS regulator, which is the most effective and sensitive promoter, *E. coli* mainly responds to superoxide and can act as a transcription factor to stimulate the expression of various antioxidant genes. Small changes in oxidative stress in the human intestine are essential for the production of taurine synthase. In *E. coli*, SoxR mainly responds to superoxide. The soxS gene is a component of the soxRS regulon. Oxidized SoxR can induce soxS expression by distorting the soxS promoter. The soxR promoter and the coding sequence of SoxR were added in front of the soxS promoter in opposite directions, the same as it is located in the chromosome of *E. coli*. With the increase of SoxR production, the soxS promoter can promote higher expression of CSAD.

Interferon- γ (IFN- γ) sensing system: An OmpA/OprF chimeric protein was constructed. The OmpA/OprF chimeric protein is used to detect IFN- γ in the human intestine. It consists of the OmpA protein from *E. coli* (responsible for maintaining the stability of the bacterial membrane) and a small part of the OprF protein from *Pseudomonas aeruginosa* (with the ability to bind to IFN- γ). The two structures are homologous, so the extracellular loop of each protein can be constructed.

In addition, the team designed bubbles containing engineered *E. coli* Nissle 1917, which is intended to alleviate CSID by producing taurine when sensing high concentrations of IFN- γ and oxidative stress. Its main component is sodium alginate, which is edible and can resist degradation in low pH environments, such as gastric acid. In environments with higher pH, such as the small intestine, Menbles is able to release our engineered bacteria into the surrounding environment and begin to produce taurine.

3733 HZAU-China

The team's goal is to detect pet IBD early. It is divided into detection and reporting, anti-inflammatory, health care, health care, and suicide modules. Nissle 1917 serves as its chassis cell.

The detection and reporting module plays the role of sensing abnormal intestinal conditions and reporting to the outside. It is based on two TCSs (NarXL and ThsSR) and an enzyme (ScGS). Among them, NarXL is an endogenous two-component system of *E. coli*, and ThsSR is a two-component system from marine *Shewanella*. They can sense the concentration of nitrate and thiosulfate, respectively, to affect the downstream circuit. When the two TCS are activated, the geosmin synthase from *Streptomyces* A3 (ScGS) will be expressed and release the odor substance geosmin. Since the concentration of nitrate and thiosulfate in the intestine will increase significantly when IBD occurs. Therefore, nitrate and thiosulfate are selected as biomarkers for the occurrence of IBD. When the concentration of both increases at the same time, it is believed that IBD has occurred in the intestine, and the pet owner is notified with the geosmin report.

The "anti-inflammatory" module will gradually begin to relieve inflammation. When IBD breaks out, lipopolysaccharide (LPS) as an antigenic determinant of bacteria triggers Toll-like receptor 4 (TLR4) to cause the outbreak of IBD. LL-37, one of the antimicrobial peptides (AMPs), can block this pathway by neutralizing LPS. This module can relieve intestinal inflammation by activating the promoter to produce the small peptide LL37-LTA, which can kill some bacteria that may cause inflammation. After the pet owner detects the smell, he will take the pet to the hospital for further treatment.

The health care module is always activated. When IBD occurs, the azurin produced by this module can prevent further deterioration of inflammation, and can also stabilize the p53 protein, a protein expressed by tumor suppressor genes, thereby producing a certain anti-cancer effect. Long-term inflammation may lead to intestinal cancer. Azurin has anti-cancer effects and can stabilize p53. *Escherichia coli* Nissle 1917 has a certain anti-cancer effect on cancer cells, and using it as a chassis for expressing Azurin can further enhance anti-cancer.

In order to prevent the engineered bacteria from leaking into the environment, a "suicide" system was added to the engineered bacteria. If the engineered bacteria enter the external environment, the

"suicide" system will be activated. Considering that the intestinal temperature of dogs is usually higher than room temperature, it was decided to let the bacteria kill themselves at low temperatures, and RNA thermometers and HepT toxins (and MntA antitoxins) were used to build this module.

3838 SZU-China

Treatment of IBD

Choice of chassis cells: *Escherichia coli* Nissle 1917 was selected as the main carrier cell, which belongs to the category of probiotics. It is non-pathogenic and can colonize. It can enhance the intestinal immune regulation ability and the protective ability of the intestinal epithelial mucus barrier to a certain extent. It can also express microbotin, which can inhibit the growth of other microorganisms such as *Salmonella typhimurium* and enterohemorrhagic *Escherichia coli* in the inflamed intestine. It itself has the potential to treat IBD. Because it belongs to *Escherichia coli*, it has more experimental operability than *Lactococcus lactis*.

Project design: cocktail therapy. SOD: Improve the state of peroxidative stress Excessive reactive oxygen metabolites play a role in mediating intestinal damage in IBD. Transform the plasmid containing the exocrine expression of superoxide dismutase into engineered bacteria. TES: Beneficial to the human intestine. The proportion of butyrate-producing bacteria is reduced in the intestines of IBD patients. Thioesterases were selected from *Bacteroides polymorpha*, and expression vector plasmids containing their genes were transferred into EcN. BSH: Intervention of abnormal bile acid metabolism Impaired microbial enzyme activity in IBD patients can lead to impaired bile acid metabolism, characterized by the inability to hydrolyze, convert, and desulfurize bile salts. It is hoped that the genes encoding well-characterized BSH enzymes can be heterologously expressed by engineered bacteria and allowed to colonize the intestine and provide high levels of enzyme activity locally to supplement the insufficient abundance of BSH genes caused by IBD. LL37: Targeted decomposition of endotoxins. In UC patients, the intestinal flora is unbalanced, the mucosal barrier function and permeability are altered, and Gram-negative bacteria are translocated and enter the lymphatic and blood systems. LL-37 is an antimicrobial peptide secreted by human immune cells that inhibits harmful bacteria such as *Helicobacter pylori* and *Staphylococcus aureus*, and has strong anti-endotoxin properties.

It is hoped that engineered bacteria can directly synthesize LL37 as a therapeutic element and use exocrine vectors for exocrine expression as a supplementary substance to participate in the immune regulation and targeted inhibition of LPS. The plasmid containing the exocrine expression of LL37 was transformed into engineered bacteria.

3875 BUCT

Treatment of Obesity

The team hopes that by taking it orally, this probiotic will enter your intestines, break down fat into glycerol and fatty acids, help regulate dietary balance, absorb oils and reduce the risk of diseases such as high blood pressure. Then, use the small molecule precursors they produce to synthesize new substances-gamma-aminobutyric acid (GABA) and 5-hydroxytryptophan (5-HTP). Our *E. coli* can not only relieve weight gain, but also produce GABA and 5-HTP, which can help you reduce anxiety, prevent depression, promote the formation of melatonin, improve sleep quality, and even suppress appetite.

The first part is the deletion of *fadR* in *EcN* and the overexpression of *fadL* & *fadD* to increase the consumption of fatty acids. The second part is to obtain better GABA production by mutating the wild-type *gadB* into a mutant form, and the mutant *GadB* protein is active in a wide pH range. In *E. coli*, the chorion is the branch point for the synthesis of aromatic amino acids. In the third part, the team inserted four related genes that promote the four-step conversion of this substance to 5-HTP. Tetrahydromonascins (MH4) was used as a coenzyme for P4H, and the expression of PCD was also added to create an artificial MH4 cycle and ensure the synthesis of 5-HTP. Finally, considering the impact of biosafety, we chose the toxin-antitoxin system, which envisions that the engineered bacteria survive in a low-oxygen, lactose-containing environment and commit suicide under aerobic conditions. We placed an anoxic promoter *phyb* upstream of the antitoxin gene to ensure that *E. coli* died when it had no resistance to the toxin under aerobic conditions.

3784 BNUZ-China

Treatment of chronic kidney disease (CKD)

Using *E. coli* Nissle 1917 as the chassis, the team developed a new type of intestinal colonization engineered bacteria, which knocked out the gene *TnaA* of the original *E. coli* to metabolize tryptophan to produce indole, and overexpressed the engineered bacterial glucose dehydrogenase to enhance its competitiveness, thereby achieving the purpose of replacing intestinal *E. coli*, thereby eliminating the production of indole from the source. At the same time, the gene cluster that catalyzes tryptophan to produce indolepropionic acid is expressed in the engineered bacteria, and indolepropionic acid can enhance the intestinal mucosal barrier. In addition, the engineered bacteria will promote the growth of other probiotics by overexpressing β -galactosidase to catalyze lactose to form oligosaccharides. Through the above means, the intestinal toxin indole sulfate can be eliminated from the source, while promoting the proliferation of intestinal probiotics, repairing the intestinal mucosal barrier, and ultimately achieving the purpose of treating chronic kidney disease.

3903 St_Andrews

A probiotic, sustainable sunscreen that does not harm the marine environment

EcN was selected to overexpress shinorine so that this bacterium can provide users with UV protection when applied to the surface of human skin. The use of Nissle 1917 *Escherichia coli* cells is crucial to our project because this bacterial strain does not cause disease, is approved by the FDA, and is a probiotic, which means that applying this synthetic engineered organism to human skin will theoretically be a completely safe process.

Shinescreen (Reef Safe Sunscreen) will utilize shinorine, a non-toxic UV-A protective molecule that is already naturally produced by marine organisms such as red algae and cyanobacteria. Due to synthetic engineering, an FDA-approved probiotic strain (i.e. Nissle 1917 *E. coli*) will overproduce shinorine: in the final product, customers will be able to apply these engineered bacteria to the skin surface for UV protection.

All genes involved in the shinorine production pathway (DHQS, O-MT, ATPG, and NRPS) were cloned into plasmids, these plasmids were transformed into EcN-expressing cells, and finally over-expression of the four enzymes involved in shinorine production was confirmed. Our team then had to confirm that EcN was able to synthetically produce sufficient amounts of shinorine to guarantee UV-A protection.

3924 Tsinghua

The team used engineered probiotic *E. coli* Nissle 1917 to produce microtoxins and therapeutic proteins, targeting the composition of the gut flora and the mucus healing process, respectively.

To heal the intestine damaged by IBD, the team used in situ expression of therapeutic proteins from *E. coli* Nissle 1917 (EcN). The design is based on a ternary system: sensor (NO or thiosulfate) - secretory peptide (CsgA) - therapeutic protein. When intestinal inflammation occurs, NO/thiosulfate will be presented and detected by the engineered Nissle 1917, resulting in the secretion of therapeutic proteins into the intestine to promote wound healing. Bacteria like *E. coli* Nissle 1917 usually do not have the ability to actively secrete engineered proteins. The team fused a special secretory peptide (CsgA) to help the secretion of therapeutic proteins.

Adhesion-invasive *Escherichia coli* (AIEC) can adhere to and invade intestinal epithelial cells and macrophages, which will cause an inflammatory response in the intestine. Microcins are a group of secreted peptides produced by certain *E. coli* strains that can bind to ATPases on the plasma membrane of other *E. coli* strains (including AIEC), thereby inhibiting their growth. Due to the presence of autoimmune genes, the strains that produce microcins themselves will not be affected. The chassis bacterium *E. coli* Nissle 1917 has an endogenous micromycin expression system that produces MccH47, but it will only turn on and express mature micromycin under certain nutrient depletion conditions. Therefore, the team introduced the mature *E. coli* CA46 micromycin expression system

into *E. coli* Nissle 1917 to produce functional micromycin, enabling it to inhibit AIEC in the intestines of IBD patients and help them restore the composition of intestinal flora.

In addition, a major problem in the delivery of engineered bacteria lies in the digestive tract environment. The team chose PCS (polydopamine co-deposited with chitosan) because it works well in *EcN*. Polydopamine and chitosan can be deposited on the cell surface through chemical self-assembly to form a membrane on the cell surface, thereby protecting our engineered *EcN* from the effects of the digestive tract environment.

3925 XHD-Wuhan-Pro-China

The team chose Nissle 1917 as the chassis cell and designed probiotics containing the ADH gene for producing ADH and the ALD gene for producing ALDH to solve the problem from the root. Compared with existing products, the team fundamentally solved the problem by enabling probiotics to synthesize ADH and ALDH in the stomach and intestines to reduce alcohol damage to the liver. In addition, to ensure that the energy provided to the reaction is sufficient and the activity of the enzyme remains high, the team added *nadE* and *nox* genes to maintain the balance between the coenzymes NAD⁺ and NADH.

3932 UI_Indonesia

The team designed three systems, two of which are engineered: biofilm dispersion and *H. pylori* eradication, and the other is naturally presented: chemotaxis.

The team used AI-2 as a chemical attractant for *EcN*, which is the only known quorum sensing signal secreted by *H. pylori* and a chemical attractant signal for *E. coli*.

The team used protease- λ to disperse the biofilm of *H. pylori*, which is mainly composed of proteins. The synthesis of protease- λ is controlled by the arabinose-dependent pBAD promoter, while the secretion is accomplished by autolysis of the E7 cleavage protein, which is triggered by ammonium sensing through the *glnAp2* promoter.

The team designed a system to eradicate *H. pylori* through a proven and effective antimicrobial peptide (AMP), PGLa-AM1. Since PGLa-AM1 is also toxic to *E. coli* as a host, the team constructed AMP in an inactive form, namely fused AMP (FAMP), by linking AMP to SUMO protein. In the time-controlled system, SUMO-AMP is then cleaved by Ulp protease and PGLa-AM1 is activated. Subsequently, the holin-antiholin system is also activated and cleaves *E. coli* to secrete our activated PGLa-AM1. 、

3984 Think_Edu_China

The use of bio-enzyme catalysts has the advantages of low energy consumption, strong stability, and simple operation steps under different temperature and pressure environmental conditions. The

whole-cell biocatalyst EcN-IL can be a favorable solution to the above-mentioned antibiotic abuse problem. Using cell surface display technology, the Laccase gene *lacc6* derived from *Pleurotus ostreatus* can be displayed on the surface of the probiotic *Escherichia coli* Nissle 1917 (EcN) and form a membrane mechanism to decompose antibiotic residues.

The team is committed to studying the degradation of the antibiotic sulfadiazine. The laccase gene *Lacc6* derived from *Pleurotus ostreatus* can be displayed on the probiotic *Escherichia coli* Nissle 1917 cells using cell surface technology and named EcN-IL to degrade antibiotics. Subsequently, the whole-cell catalyst was prepared as a freeze-dried powder strain and applied to animal feces.

The team used ice nucleoprotein for cell surface display technology. Ice nucleation protein (INP) is a secreted surface protein that is widely regarded as a carrier protein in bacterial cell surface display systems and is used to display different types of exogenous proteins.

3986 LZU-HS-China

The team used *Escherichia coli* Nissle 1917 as the host strain of the cell biocatalyst. Through synthetic biology methods, the team used engineered bacteria to reduce selenium in the intestine, improve the bioavailability of selenium, and explore more efficient supplementation methods.

Different forms of selenium supplements have different bioavailability. Nano-selenium (SeNPs) has lower toxicity than inorganic selenium (such as selenite), and also has antioxidant and immunomodulatory functions. Due to its small particle size, it is easier to be absorbed by the human body. In order to reduce sodium selenite in the intestine to nano-selenium and increase the selenium content in the body, the team designed a new cell biocatalyst EcN-IS. Using the microbial cell surface display system, the selenium reductase *SerV01* was displayed on the surface of EcN cells. This system enables the fusion protein to be directly expressed on the host cell surface by combining specific exogenous functional proteins with anchor proteins. In the fusion protein construction part, the team selected ice nucleoprotein (INP-N) as the carrier protein and fused its N-terminal sequence with the peptide sequence of bioreductase *SerV01* to form a fusion polypeptide INP-*SerV01* that can be anchored on the outer membrane of EcN cells. In order to improve the stability and utilization of passenger proteins, the team achieved overexpression of homologous or heterologous proteins through high-copy plasmid transformation, reducing costs and improving protein expression efficiency.

To optimize the expression system, the pSB1A3 plasmid in the iGEM data resource library was used in the project, and a strong promoter and a long gene sequence were inserted as the spacer region to ensure high-quality expression of subsequent proteins.

In addition, the surface-displayed fusion protein can be recycled as a catalyst, which is more cost-effective compared with the high recycling cost of immobilized enzymes.

The team focuses on cardiovascular diseases, especially deep vein thrombosis (DVT), a serious health problem that may be caused by thrombosis. They proposed an innovative solution, including the development of a home-assisted test kit to detect D-dimer concentrations in saliva, and the design of an *E. coli* Nissle 1917 live biotherapeutic product (LBP) capable of producing an improved nattokinase, which can control the production of nattokinase through a remote light-induced system.

The team designed the *E. coli* Nissle 1917 live biotherapeutic product (LBP), first cloned the gene (*aprN*) of nattokinase from *Bacillus subtilis*, then connected *aprN* with pET-21a (+) to produce a cloning vector, used *E. coli* Nissle 1917 to produce nattokinase, and tried to increase its activity through mutation.

In order to help bacteria attach to the small intestinal wall and prolong the residence time of LBP in the intestine, the team developed an adhesion protein. The system consists of the transmembrane protein OmpA located in the outer membrane, GS Linker, and the gene encoding the adhesion pili *fimH*.

To protect EcN from the irritating gastrointestinal environment and deliver it to the small intestine, the team designed an oral delivery system. EcN was first freeze-dried to ensure long-term storage, and then loaded into acid-resistant HPMC capsules by loading freeze-dried EcN, which can help oral probiotics face obstacles such as low gastric pH, bile salts, host immune system and competition.

The team also used optogenetics to remotely control the gene expression of nattokinase in EcN by near-infrared light. And designed a kill switch based on the MazE-MazF toxin-antitoxin system to achieve controlled clearance of EcN through temperature changes and chemical induction.

2022

Unhealthy diet affects the levels of low-density lipoprotein (LDL) and triglycerides in the body, which will lead to a series of cardiovascular diseases (CVD) such as thrombosis. The team plans to develop an edible probiotic *Escherichia coli* Nissle 1917 to reduce the possibility of thrombosis from two aspects: inhibiting the production of TMA and degrading existing TMA. At the same time, the engineered bacteria also release many preventive substances that are beneficial to vascular protection.

The project mainly designed three functional modules and one suicide module. The three functional modules are degradation, inhibition and relief. The three functional modules are connected by a simple theophylline small molecule. Theophylline, as a sensor substance, can specifically turn on the switch of the engineered bacteria in the patient's body. This avoids the expression of degradation and inhibition modules when only the relief module needs to be expressed. The team provides two main functional cycle designs for different scenarios:

When patients take probiotics containing theophylline, the expression of trimethylamine, dimethylamine and formaldehyde dehydrogenase gene clusters will be induced.

If the patient only takes probiotics, the probiotics will only express short-chain fatty acids that are beneficial to vascular protection and intestinal mucosal repair.

In the suicide module, the team designed a heat-inhibitory RNA. This is followed by a thermometer with a toxin protein. The former senses the ambient temperature and plays a role in controlling the switch to close and open, which affects the subsequent path, while the latter can express the toxin protein HepT, as an RNase, to degrade mRNA at the transcriptional level and eventually lead to cell death. Through such a design, the suicide of engineered bacteria in vitro is ensured to minimize environmental pollution.

4156 lzu-china

The team used *Escherichia coli* Nissle 1917 as a host strain and gene-edited probiotics to target colorectal cancer (CRC).

In order to construct a bacterial biosensor system that can distinguish unique organ environments, the team used oxygen, pH, and lactate as unique indicators of colorectal cancer tumor tissues, and used the unique microenvironmental characteristics of colorectal cancer tissues for identification. Three operons were designed to specifically identify hypoxia, low pH, and lactate, including: hypoxia-sensing promoter (pPepT), regulated by transcription activator (FNR); L-lactate sensing system (lldPRD operon), pLldP promoter drives the expression of the gene of interest, LldR repressor protein inhibits the expression of the reporter gene when lactate is not bound; pH-sensitive promoter (pCadC), regulated by membrane-bound activator protein (CadC), more active in acidic media.

However, the current clinical application of biosensor systems faces many challenges, such as limited signal processing capabilities, inability to integrate multiple biomarkers for accurate diagnosis, and response time incompatible with diagnoses that require rapid results. In order to enable living cells to perform complex signal processing operations, a biosensor system was designed using serine integrase-based amplifying genetic switches and Boolean logic gates. The team designed a tumor cell adhesion module, using HlpA protein to bind to HSPG on the tumor surface to improve the ability of engineered bacteria to penetrate colorectal tumor cells.

The treatment part uses hemolysin E, CCL21 and CDD-iRGD (a fusion protein Bit1 of the cell death domain and tumor perforin) to activate the host's immune response. A phage lysis gene (phiX174 E) was introduced, which causes bacterial lysis and death when expressed, releasing therapeutic factors stored in the strain.

For safety reasons, the team introduced an arabinose-induced kill switch as an additional protective barrier. Due to unsatisfactory results of the kill switch experiment, microcapsules composed of sodium alginate and chitosan were also made to ensure the safety and effectiveness of the treatment.

4205 NJMU-China

The team built the first and second systems for mitochondrial relief.

In the second system, the team aimed to eliminate heavy metal ions such as Hg^{2+} and Pb^{2+} in the intestines of autistic patients. *E. coli* 1917 was selected as the chassis and pET-28a (+) as the vector.

The team constructed the pET28a-MT plasmid at the DNA level, transformed it into *E. coli* Nissle 1917, and extracted the plasmid to prove the successful transformation. First, enzyme digestion verification was performed by adding restriction endonucleases SphI and XhoI to obtain double enzyme digestion and run agarose gel. Secondly, plasmid PCR verification was performed by designing primers to amplify the target fragment and run agarose gel. After enzyme digestion, the sample ran out of a band close to 2000 bp from the extracted plasmid, which is consistent with the target band (2399 bp). After plasmid PCR, the extracted plasmid sample also ran out of a band close to 2000 bp, which is consistent with the target band (2399 bp). This further indicates that the plasmid transformation was successful.

Subsequently, the experimental group conducted electroporation of *E. coli* Nissle 1917, prepared competent cells and electroporation. It was used to transform *E. coli* Nissle1917 to facilitate the construction of the required plasmids for mitochondrial relief.

4387 Uzurich

The project of the University of Zurich team is aimed at the treatment of inflammatory bowel disease (IBD). Using *E. coli* Nissle 1917 as a base bacteria, they designed a new genetic circuit to sense nitric oxide and produce and secrete anti-TNF- α nanobodies to reduce inflammation.

The ECN is designed to sense nitric oxide levels in the intestine. When high levels of nitric oxide are detected, the ECN will begin to produce and secrete anti-TNF- α nanobodies, which are biological molecules targeting inflammatory responses.

The team also developed a model to simulate the diffusion and action of ECN in the intestine to evaluate the effectiveness of treatment.

4244 Wageningen-Ur

Developing a live diagnostic tool called "Colorectal" for the detection of colorectal cancer.

E. coli Nissle 1917 was engineered in this project to bind to tumor cells and detect two different cancer biomarkers: lactate and matrix metalloproteinase 9 (MMP-9).

When lactate levels are significantly elevated, the ECN produces a signal protein, and the presence of MMP-9 activates the color of this signal protein. The colored signal protein is only observed in the feces when both biomarkers are present.

The team also implemented a biosafety circuit to limit the ECN to work only in the colon and control its distribution through temperature and mucin dependence, while providing a mechanism to remove the ECN when needed.

4199 Nwu-China-A

The team selected *E. coli* Nissle 1917 as the chassis. By culturing, culturing and observing the color of bacteria in the feces, the diagnosis of chronic lower gastrointestinal bleeding can be achieved. Two recombinant plasmids, pET28a (+) -ChuA-HrtR and pSB1C3-HtrO-cjBlue, were transformed into the non-pathogenic edible probiotic Nissle 1917 for the manufacture of biosensors for detecting intestinal microbleeding. When the probiotic is administered to patients with microintestinal bleeding, it transports heme from the intestine into the cells, activating the expression of the reporter gene in the bacteria, thereby enabling the detection of intestinal microbleeding.

The team's sensor needs to enter the human intestine to work, so *E. coli* Nissle 1917 was chosen as the expression strain instead of *E. coli* BL21. Since there is no T7 RNA polymerase in *E. coli* Nissle 1917, the team plans to use the J23119 promoter instead of the T7 promoter.

Considering that the suicide switch requires blue light activation, which is not available in ordinary households. By consulting the literature, the team found another simple suicide switch, which is to knock out a specific gene in *E. coli*.

4183-HS-China

The team proposed a syn-bio approach to address lactose intolerance, developing and utilizing the *E. coli* strain *E. coli* Nissle 1917, which can be safely transplanted into the intestines and secrete lactase from the inside to break down any lactose that is causing trouble to the digestive system. This not only eliminates physical discomfort, but also unlocks the nutritional value of milk for people who were once unable to digest milk.

The treatment designed by the team is divided into three steps. First, find a suitable chassis strain and adjust its genetics to produce lactase. Then, the chassis organism is safely transplanted into the patient's small intestine. Finally, ensure that the strain survives and operates effectively in the intestinal environment for a long time. Therefore, the chassis strain must be able to produce the enzymes

needed to catalyze the breakdown of lactose in the patient's abdomen. First, use the *E. coli* strain Nissle 1917 as the chassis organism. *E. coli* has always been the most popular chassis strain with high productivity and operability. Our goal is to use this strain of *E. coli* to produce the enzyme LacZ, a beta-galactosidase, in the small intestine, which can effectively break down lactose into glucose and galactose, both of which can be degraded by the human digestive tract. The enzyme produced will be transported to the extracellular environment via a signal peptide. This ensures that the bacteria themselves do not ingest lactose to produce gas, effectively preventing all forms of bloating, discomfort or diarrhea. Secondly, the plan is to use enteric-coated pills to contain the freeze-dried bacteria to be ingested. When swallowed, the enteric-coated pills will not dissolve in the stomach like ordinary pills, but will only begin to break down in the small intestine. This helps the bacteria bypass the chemical barrier of stomach acid and ensure the strain is safely transported to the intestine. Third, CAP (capsular polysaccharide surrounding the *E. coli* strain Nissle 1917) is used to create a protective film to ensure that the bacteria are safely implanted in the intestinal environment without being damaged. The gene for CAP is encoded in the bacteria and it will be produced spontaneously. CAP helps the bacteria adapt to a wider range of temperatures and pH values, blocks harmful chemicals and toxins, and helps them adhere to the intestinal wall.

4331-LZU-HS-China-B

The team chose *Escherichia coli* Nissle 1917 as our chassis organism to solve various diseases caused by alcohol. The mechanism of action of EcN is that it can colonize in the human intestine, prevent pathogens from attacking the intestinal mucosa, and has a protective and repairing effect on the intestinal mucosal barrier. EcN also participates in the immune regulation of the host organism, balances the secretion of immune factors, and enhances the host's immune ability, thereby alleviating and treating inflammation. By increasing the production of nad^+ by increasing the NOX enzyme, the efficiency of alcohol dehydrogenase and acetaldehyde dehydrogenase also increases with the increase in the number of nad^+ factors. Ultimately, the results we hope to achieve are: increasing the conversion rate of ethanol to acetaldehyde and reducing the harm of ethanol and acetaldehyde to humans.

The design of the project initially followed the following steps. When people drink alcohol, alcohol passes through the cell membrane, where NAD^+ promotes ethanol dehydrogenase to become acetaldehyde, and then NAD^+ promotes aldehyde dehydrogenase to convert acetaldehyde into acetic acid, and then these substances become acetyl coenzyme (acetyl-CoA), and finally, they join the TCA cycle. The NAD^+ we mentioned earlier is the conversion of NADH through NOX, which helps the conversion between ethanol and acetate. In these cells, alcohol dehydrogenase can prevent people from flushing when drinking or some larger body reactions, while aldehyde dehydrogenase can help people sober up faster.

Decided to better verify their ideas, they made two engineered strains: pSB-AA and pSB-AN. For pSB-AA, they only improved its degradation system; while pSB-AN modified genes in both the acceleration system and the degradation system. To achieve this effect, they finally chose the alcohol dehydrogenase gene, acetaldehyde dehydrogenase gene, NAD⁺ synthase gene, and NADH oxidase gene. The reason is that they control ADH, ALDH, NAD⁺ and NADH respectively, which play a vital role in both the acceleration and degradation systems.

4300-LZU-HS-Pro-A

In the team's project, *E. coli* Nissle 1917 was genetically modified so that the bacteria could produce GABA and 5-HTP, thereby achieving the purpose of treating and relieving depression. We plan to inject this transgenic strain into the human intestine to alleviate depression. We also designed a cell apoptosis system to kill the bacteria once they come out of the human body and prevent transgenic contamination.

Since scientists have demonstrated the existence of the gut-brain axis, they have studied the neurological and psychiatric effects of gut microbiota on the brain. Many studies have shown that there is a significant relationship between depression and gut microbiota. 5-HTP and gamma-aminobutyric acid (GABA) produced by gut microorganisms can effectively help patients with depression recover from depression. Based on the results of this study, we designed a 5-HTP and gamma-aminobutyric acid production synthetic biological strain based on *E. coli* Nissle1917.

Glutamate decarboxylase can promote bacteria to produce GABA, and human tryptophan hydroxylase can promote bacteria to produce 5-HTP. We engineered *E. coli* Nissle1917 to add genes that overexpress these two substances to produce large amounts of GABA and 5-HTP. By delivering these bacteria to the intestines of patients with depression, we can help them relieve or get rid of depression. We also need to solve other problems. In order to protect the environment and life from genetically modified pollution, we consider designing a suicide system to prevent strain escape. Under normal circumstances, when arabinose is present, the arabinose promoter starts the expression of the following three enzymes to decompose arabinose. The plan is to insert an antitoxin gene after the production of these three enzymes, and then express a toxin gene. Arabinose is present in the small intestine, and arabinose is not broken down by the small intestine. So when *E. coli* is in the small intestine, it can receive arabinose in the small intestine, express antitoxin genes in the back, and offset the toxin genes in the back, and the bacteria can survive normally. When *E. coli* leaves the small intestine, the arabinose operon without arabinose regulation does not express the following antitoxin genes, only the toxin genes. The bacteria will eventually kill themselves.

In addition, after consumers take our probiotics, the probiotics will produce GABA and 5-HTP. When needed, when we drink tea, the tea will activate the lysis system, and after the bacteria are lysed, more GABA and 5-HTP will be released into the intestine.

4410-Worldshaper-HZBIOX

The main idea of our project is to use the non-pathogenic bacteria *E. coli* Nissle 1917 to produce L-arginine to promote colorectal cancer immunotherapy. Since the ArgR gene was knocked out and the ArgJ gene was inserted to overcome the arginine inhibition mechanism. To ensure biosafety, we inserted the hemolysin gene into our designed EcN so that it would have done their work in the bacteria. The team found that in the genetically engineered bacteria ENC1917, there are two genes that have a profound effect on the production of arginine, they are the ArgR gene and the ArgJ gene. Our project aims to provide a safe, effective and practical method to introduce and maintain high levels of L-arginine in solid tumors, especially in colorectal tumors, to promote cancer immunotherapy. The main idea of this project is to utilize the non-pathogenic bacteria EcN, which can enter the digestive tract orally and colonize the tumor site in the digestive tract. EcN in this project is designed to be able to continuously produce large amounts of L-arginine in tumors by using ammonia, a metabolite of cancer cells, as a substrate. After treatment, a regulatory mechanism is introduced to eliminate EcN in the system to ensure safety.

Our project consists of two main parts: EcN engineering for arginine synthesis and preliminary verification of the anti-tumor effect of high-concentration arginine on colorectal cancer cells.

In order to locally utilize the anti-tumor effect of high-concentration arginine synergistic immunotherapy on solid tumors such as colorectal cancer, our project uses probiotic EcN as a therapeutic carrier to produce large amounts of L-arginine in tumors. We engineered the bacteria by adding the argJ gene and deleting the argR gene to overcome the arginine inhibition of the arginine synthesis pathway. To ensure biosafety, we inserted the hemolytic protein gene into our engineered EcN to lyse the bacteria after the bacteria have done their work. Probiotic EcN has tumor targeting effect, can specifically accumulate in tumor tissue, and play the role of targeted transporter.

Our engineered probiotics can be quickly colonized in the intestine by oral administration, have a good adhesion rate to porcine intestinal epithelial cells, can stay in the tumor tissue of the intestine for a long time, have the potential to treat digestive system tumors, and have a positive effect. In addition, our probiotics can also be delivered to the target by injection in experimental animals, so this direction can be developed in future clinical applications.

4762 NWU-CHINA-A

Super Probiotics: Intestinal Knight engineered bacteria beat IBD

Project Overview: This project mainly treats inflammatory bowel disease (IBD) by genetically engineering *Escherichia coli* Nissle 1917 (EcN). In this study, the researchers tried to express three beneficial protein genes - catalase (CAT), superoxide dismutase (SOD) and Elafin - in EcN at the same time to enhance its ability to fight IBD.

Reasons for using ECN as the expression strain: EcN can colonize the gastrointestinal tract (GI) and exhibits better growth advantages than other bacteria in the intestine, inhibiting pathogenic Enterobacteriaceae. More importantly, EcN is the active ingredient of the pharmaceutical preparation Mutaflor. Mutaflor is a microbial drug currently approved in some European countries for human medication to treat various diseases and dysfunctions of the intestine. Given its long-term and safe delivery record for human medication, by engineering the strain into a living drug, industrial platform and drug delivery carrier. Therefore, EcN was selected to express the target gene.

Experimental design: First iteration: The researchers used the pET-28a (+) vector to express the CAT and SOD genes separately to verify the expression and activity of these genes in *E. coli* BL21. Second iteration: To achieve simultaneous expression of CAT and SOD, the researchers modified the vector by adding the SOD gene sequence downstream of the CAT gene and changing the promoter from T7 to T5 to accommodate expression in EcN, which lacks T7 RNA polymerase. Third iteration: To further enhance the function of EcN, researchers introduced the Elafin gene. To avoid the problem of plasmid incompatibility, pCDFDuet-1 from different sources was selected as the expression vector of Elafin. Fourth iteration: The researchers connected the Elafin gene sequence to the vector that already contained the CAT and SOD genes, and constructed a vector that can express these three proteins at the same time. Fifth iteration: Considering food safety, the researchers transformed the constructed vector into the probiotic EcN in order to use this modified probiotic to treat IBD.

2023

4965 RDFZ-CHINA

A Double-stage Patch Set for Facial Acnes

Project Overview: Studies have shown that *P. acnes* is highly associated with the formation of acne because it releases extracellular enzyme products that degrade facial sebum into LCFA, leading to inflammation. Therefore, we tried to solve the problem of inflammation by simultaneously eliminating acne and the LCFA they produce. The team engineered *Escherichia coli* Nissle 1917 (EcN), a non-pathogenic *E. coli* strain, to absorb and degrade LCFAs from the face.

Reasons for using ECN in the chassis microbiome: ECN has probiotic properties and non-pathogenic properties, possesses multiple adaptive factors, and can effectively inhibit the growth of opportunistic pathogens, thus ensuring the safety of the product. In addition, the gene expression of EcN

is under extensive transcriptional and translational control, and its metabolic circuits are well understood.

Experimental Design: Design 1: Removal and degradation of free fatty acids Acne releases lipases that degrade facial sebum into LCFAs, which leads to inflammatory lesions. Therefore, we decided to design a bacterial metabolic pathway with FadD and FadL with a T7 promoter, a strong RBS, and a double terminator, using the fadL and fadD genes to remove and decompose LCFAs. The cooperative function of these two genes is believed to complete the metabolic function of LCFAs. Design 2: Prevent scar formation Another EcN engineered strain was designed with the ability to produce epidermal growth factor (EGF) for the second patch. Design 3: Add phoA secretion tag phoA is a signal peptide that guides newly synthesized proteins to the secretory pathway.

Although *E. coli* Nissle 1917 can slowly secrete hEGF growth factor, the secretion rate is relatively low, which is not effective enough for treating acne. Therefore, it was decided to add a phoA secretion tag to the hEGF gene to promote the secretion of hEGF into the periplasmic space of *E. coli*. phoA is a signal peptide that guides newly synthesized proteins to the secretory pathway. hEGF is the gene responsible for the synthesis of human epidermal growth factor hEGF. A plasmid was constructed with the fused phoA-hEGF gene, transformed into *E. coli*, and its production and secretion were verified using ELISA and Western Blot.

5024 BZK

Gout "Firefighter" Probiotics

Project Overview: Uric acid is a waste product of purine metabolism in food, and most of it is excreted through the liver. However, if the body cannot effectively excrete uric acid, uric acid levels will rise, forming uric acid crystals and deposited in the joints and surrounding soft tissues, causing pain and inflammatory responses, leading to joint pain and swelling. The project aims to use the probiotic Nissle 1917 to synthesize enzymes (UAO, allantoinase and allantocase) that can digest uric acid to treat gout. Specific plasmids pGex-4T-1-UAO (BBa_K5024007) and pGex-4T-1-UAA (BBa_K5024008) were constructed and transformed into the target probiotic Nissle 1917.

Reasons for using ECN in chassis microorganisms: ECN has probiotic properties and non-pathogenic properties, and possesses multiple adaptive factors.

Experimental design: Plasmid construction: The research team first designed and constructed three different plasmids, each containing the gene for a key enzyme. These plasmids are based on the pET28a vector, and the genes of urate oxidase (UAO), allantoinase and allantocystase are inserted respectively. **Replacing the expression vector:** The initially constructed pET28a plasmid is designed to express these enzymes in *Escherichia coli*. In order to adapt to the ECN expression system, the target gene needs to be transferred to a vector that is more suitable for ECN. Therefore, the research

team chose the pGEX-4T-1 vector, which is a vector suitable for expressing foreign proteins in a variety of bacteria, with different resistance markers and inducible expression systems.

4872 Canton-HS

Project Overview: This project is dedicated to developing a dual vaccine that can provide immune protection against both norovirus and rotavirus. Two different plasmids were constructed to express the two antigen genes GII.4-VP1 and GII.17-VP1 of norovirus, respectively, and these plasmids were transformed into *Escherichia coli* BL21 (DE3) strains for expression and purification. Based on this basic work, vaccines that can simultaneously target norovirus and rotavirus were further developed. A fusion protein that combines the GII.17-VP1 antigen of norovirus and the RV-VP7 antigen of rotavirus was designed and expressed, and the plasmid encoding this fusion protein was transformed into *Escherichia coli* BL21 (DE3) and Nissle 1917 strains for protein expression and purification.

Disadvantages: The expression of RV-GII.17-VP1 in *E. coli* Nissle 1917 is low, and the protein expression method needs to be improved and optimized to achieve large-scale expression of RV-GII.17-VP1.

4955-Japan-United

Crocin synthesis via EcN for depression treatment

Antidepressant components of saffron (crocetin, crocin, crocin) were produced using *E. coli*. These components were integrated into cookies as a new means of controlling depression.

Design of a new metabolic pathway (enzyme group A) that biosynthesizes 4-hydroxy-2,6,6-trimethyl-1-cyclohexene-1-carboxaldehyde (HTCC), a precursor of crocin, in *E. coli*, and a new metabolic pathway (enzyme group B) that biosynthesizes crocin from HTCC in vitro. Enzyme groups A and B were introduced into the BL21(DE3) strain to biosynthesize crocin. Detecting soil information via EcN.

4916-WLC-Milwaukee

Design of a biological soil sensor based on EcN to collect information needed to balance soil nutrient levels and maximize crop growth.

The engineered *Escherichia coli* (*E. coli*) uses only bioavailable phosphate and senses phosphate concentration by controlling the expression of β -lactamase (*bla* gene). Finally, a spectrophotometer is used for colorimetric analysis to accurately determine the phosphate concentration in soil samples.

4980- Jilin-China

The engineered non-pathogenic *E. coli* Nissle 1917 is used as a drug delivery system. It is able to grow in the immunosuppressive environment of tumors and deliver anticancer drugs to provide targeted treatment.

Nissle 1917 was selected as the chassis organism. Based on the anticancer protein melittin, a fusion peptide was designed to achieve targeted and effective treatment. The TCI42 and pR, pL promoters of λ phage were used, which can turn off the target gene expression at room temperature and turn on the expression above 42°C. The Axe-Txe system is also used to ensure the correct transfer of plasmids into bacteria and control cell growth.

7. EcN clinical drugs and progress currently under development

Use: Mutaflor is used to treat ulcerative colitis (especially in remission) and constipation. It restores intestinal balance by regulating intestinal microbiota and helps relieve symptoms.

Status: It has been on the market as a drug or dietary supplement for many years in countries such as Germany, Canada, and Australia.

2. Symbioflor® 2

Use: Mainly used to treat chronic constipation and irritable bowel syndrome (IBS). It relieves symptoms by enhancing intestinal barrier function and regulating immune response.

Status: Available in some European countries, usually as a supportive treatment for intestinal health.

3. EcN-TNF α inhibitor

Use: By engineering EcN to enable it to secrete tumor necrosis factor α (TNF α) inhibitors, it is used to treat chronic inflammatory diseases such as inflammatory bowel disease (IBD). This drug is still in the preclinical research stage and has not yet been commercialized.

4. Tum 5-p53 (experimental drug)

Purpose: This is an experimental anticancer drug that genetically modifies *E. coli* Nissle 1917 to express the tumor suppressor protein p53 and the anti-angiogenic protein Tum-5 in the hypoxic area of the tumor. It is designed to directly attack tumor cells and inhibit their growth and angiogenesis.

5. Engineered *E. coli* for Nanobody Delivery

Purpose: Using engineered *E. coli* Nissle 1917 as a carrier, therapeutic nanobodies are delivered to the intestine for the treatment of chronic diseases such as inflammatory bowel disease (IBD). This therapy is currently in the research stage and has not yet entered the clinic.

8. Analysis of EcN delivery strategy

In vivo delivery system of EcN 1917 engineered bacteria

The use of engineered probiotics is suitable for detecting and producing key biomolecules and responding to a range of diseases, and has great value in the prevention, diagnosis and treatment of diseases. For example, inflammatory bowel disease, diabetes, multiple sclerosis, autism and malignant tumors. For example, anticancer drug delivery elements designed using bacterial tumor targeting behavior. In addition, some engineered bacteria have the function of stimulating and activating the adaptive immune system and are used to assist immunotherapy. However, due to the xenobiotic biological characteristics of bacteria (such as rapid proliferation and colonization), the in vivo application of engineered bacteria is often severely limited, such as the invasion of normal tissues by bacteria themselves or immunogenic metabolites. For example, genetically engineered ECN1917 has been shown to be effective in intestinal microbiota transplantation therapy, but its implementation is largely limited due to risks such as intestinal epithelial barrier damage. Considering the inherent shortcomings of engineered bacteria, designing a dedicated bacterial delivery system to enhance the accuracy of diagnosis and treatment is a feasible approach and reduce potential safety issues.

1. Nanocoating

Surface decoration of engineered bacteria is a universal method to regulate their biological behavior. Many functional molecules with self-assembly and film-forming properties have been attached to the surface of genetically engineered bacteria, and change the behavior of bacteria in vivo and enhance their biosafety. For example, ECN1917 coated with erythrocyte membrane can escape capture by macrophages to prolong its retention time in tissues.

There are many other functional molecules that can be used for engineering decoration of bacterial surfaces to achieve specific biological effects. Here are some specific examples:

1 Display of antigen epitopes: The surface of engineered bacteria can display specific antigen epitopes for the development of vaccines or diagnostic tools. This method can use bacteria as carriers to effectively display antigen epitopes on their surface and stimulate the immune response of the host immune system. For example, researchers cloned the gene of influenza virus (such as HA protein) into the expression vector of *Escherichia coli* and used the surface of *Escherichia coli* to display the antigen epitope of influenza virus. This method has been used to develop recombinant vaccines to enhance immune response.

2 Drug release system: Decorating molecules with drug release function on the surface of bacteria can achieve targeted therapy. These decorative molecules can respond to specific physiological

conditions or external stimuli, release drugs when needed, improve therapeutic effects and reduce side effects. For example, polylactic acid (PLA) and polyvinyl alcohol (PVA). Polylactic acid is a biodegradable polymer. Researchers can modify PLA to the surface of bacteria so that it can release encapsulated drugs, such as antibiotics or anticancer drugs, under specific conditions. This method can be used for targeted therapy to reduce the impact on healthy cells. PVA is a water-soluble polymer. Studies have shown that after PVA is modified to the surface of bacteria, it can release drugs at a specific pH value to achieve the effect of targeted release.

3 Bioadsorption materials: attach biological materials with adsorption capacity to the surface of bacteria for environmental governance or pollutant removal. These engineered bacteria can adsorb specific chemicals or metal ions to remove pollutants in water or soil. Researchers can clone the genes of metal binding proteins (such as cadmium binding proteins and lead binding proteins) into bacteria and express them on the surface of bacteria. These proteins can specifically bind to and remove heavy metal ions in water. In addition, poly-L-lysine is a cationic polymer that can adsorb anionic pollutants. By modifying the bacterial surface with polylysine, researchers were able to improve the bacteria's ability to adsorb anionic pollutants in water.

By designing a nanocoating with specific functions on the bacterial surface, the activation of bacteria can also be programmed on demand without relying on promoters or recombinant protein strategies. Inhibition can trigger the bacteria in the nanocoating to reactivate on demand. The intestinal microbiota restricted in the nanocoating can respond to the gastrointestinal environment and restore its function in the intestine through pH-triggered dissolution of the nanocoating. To achieve the safety of bacteria after oral administration until the bacteria enter the intestine, while restoring their therapeutic effects in the intestine. For example, Pingping Wang et al. used calcium ions outside the bacterial membrane to guide the self-assembly of the enteric polymer Eudragit L100-55 on the surface of engineered ECN1917 and achieve pH-dependent drug release.

2. Nanoparticles

In order to achieve richer biological functions and controllability, it is an effective method to couple precisely designed nanoparticles to the surface of engineered bacterial membranes, such as polymer particles, polymer tubes, and magnetic nanoelements. This strategy can not only enhance the function of bacteria, but also expand its application potential in biomedicine, environmental governance, and synthetic biology. Here are some specific examples.

Researchers can use polymer microtubules as "transport vehicles" for bacteria, and use them for directional movement of bacteria and drug release. This method can control the release rate and location of drugs under specific physiological environments or stimuli. Modifying the surface of bacteria with polystyrene nanotubes can improve the ability of bacteria to adsorb specific molecules.

This structure can be used to capture pollutants or biomarkers in the environment, thereby achieving environmental monitoring or disease diagnosis. For example, the modified bacteria can capture specific toxic substances in water and detect their concentrations through fluorescent signals.

The application of magnetic nanoelements in bacterial surface decoration provides new possibilities for drug delivery and targeted therapy. These magnetic nanoparticles are usually made of magnetic materials (such as iron oxide, cobalt or nickel) and can achieve directional movement, aggregation or drug release under the action of an external magnetic field. For example, magnetic iron oxide, cobalt-based magnetic nanoparticles and magnetic polymer nanoparticles.

Magnetic iron oxide nanoparticles (such as Fe_3O_4) are widely used in bacterial surface decoration. Studies have shown that bacteria modified with these nanoparticles can target tumor cells and release anticancer drugs under the guidance of an external magnetic field. For example, bacteria modified with magnetic iron oxide nanoparticles can release chemotherapy drugs, such as cisplatin, in the tumor microenvironment, increasing local drug concentrations and reducing systemic side effects. Similar to ferroferric oxide, cobalt-based magnetic nanoparticles are also used to decorate the surface of bacteria to enhance their targeting and biocompatibility in vivo. These nanoparticles can be guided to specific locations under the action of an external magnetic field, thereby achieving precise control of drug release.

Magnetic polymers have more advantages in the transformation of intelligent engineering bacteria because their particle size and surface structure are more controllable. These composite materials are not only magnetic, but also can improve the drug loading capacity. For example, researchers have developed a magnetic polymer nanoparticle that can be used as a carrier to deliver antibiotics to the site of bacterial infection after modification on the surface of bacteria. This method effectively enhances the local concentration of antibiotics and reduces systemic side effects. For example, Mukrime Birgul Akolpoglu et al. orthogonally coupled magnetic nanoparticles and nanoliposomes loaded with photothermal agents and chemotherapy molecules on ECN1917 to construct a magnetically guided biorobot, which can navigate in the biological matrix and colonize in solid tumors under the guidance of an external magnetic field, and release drug molecules on demand under the stimulation of near-infrared light, greatly improving the intelligence of the engineered bacteria.

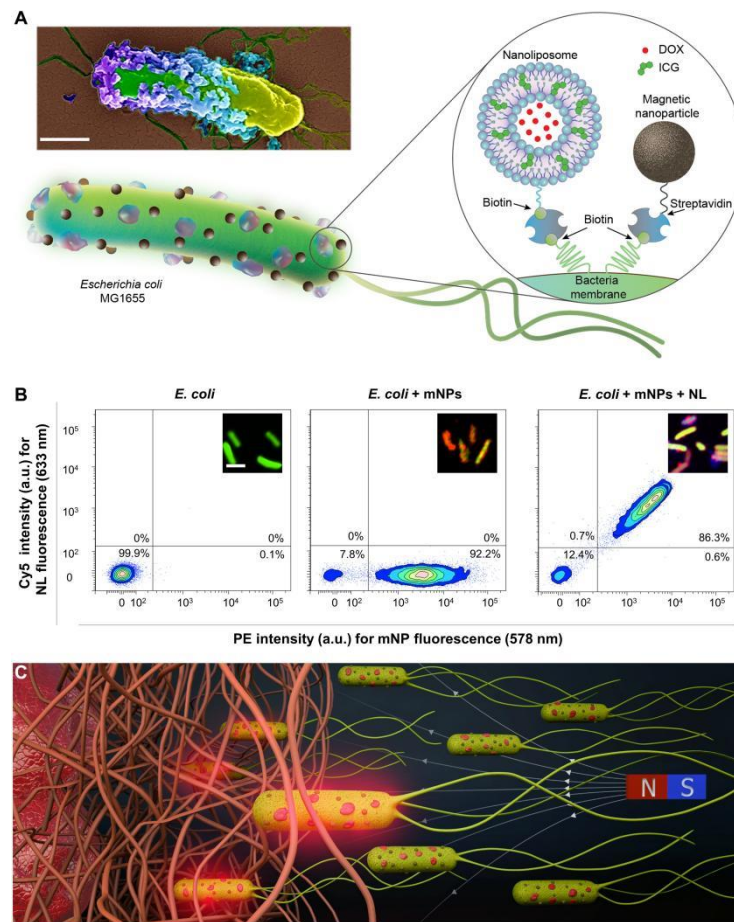


Figure 1. Magnetic-guided engineered bacteria delivery system designed by Mukrime Birgul Akolpoglu et al.

9. Comparison of some bacterial therapeutic transformations with ECN

1. Salmonella

1.1 Introduction

Salmonella is named after the American bacteriologist D. E. Salmon, who first isolated the bacterium from the pig intestine in 1884. ^[1] Salmonella is a Gram-negative, motile, hydrogen sulfide-producing bacterium belonging to the family Enterobacteriaceae. Salmonella is a mesophilic bacterium with an optimal growth temperature of 35-43°C. Their growth rate is greatly reduced at temperatures below 15°C, and most Salmonella are prevented from growing at temperatures below 7°C. Salmonella is a facultative anaerobe that can survive in low-oxygen environments.

Salmonella is an acid-labile facultative intracellular microorganism that can infect or colonize humans, causing a number of clinical infections with different clinical features, such as gastroenteritis, enteric fever, bacteremia, and chronic carrier state, and causing cross-infection between humans and animals. Many animals are known carriers of Salmonella. One of these animal vectors is chicken, which, when eaten undercooked, can lead to the transmission of Salmonella Enteritidis, which can cause inflammatory diarrhea. Salmonella can also cause severe focal infections in immunocompromised patients. Overall, there are more than 2,500 serotypes of Salmonella worldwide. Enteric fever is caused by Salmonella Typhi and Salmonella Paratyphi, while other Salmonella strains are known as non-typhoidal strains. ^[2]

Salmonella has tetrathionate respiration, which is closely related to its role in intestinal infections, especially in the context of inflammatory bowel disease (IBD). In IBD, intestinal inflammation provides an environment that is favorable for the growth of Salmonella. Salmonella is able to invade intestinal epithelial cells through its virulence factors and survive in mucosal macrophages, causing acute intestinal inflammation. Reactive oxygen species generated by the inflammatory response react with endogenous luminal sulfur compounds (such as thiosulfate) to form tetrathionate, a new respiratory electron acceptor. Salmonella uses this characteristic to grow in the inflamed intestine, thereby gaining an advantage in competition with other microorganisms in the intestine that rely on anaerobic fermentation. ^[3]

1.2 Application Examples

Through synthetic biology technology, scientists can edit the genes of *Salmonella*, knock out its pathogenic genes or introduce therapeutic genes, making it a potential biotherapeutic tool. For example, the attenuated *Salmonella* VNP20009 reduces its toxicity by knocking out the *purI* and *msbB* genes, improving tumor targeting and tumor treatment effects.

Salmonella VNP20009 is a genetically attenuated strain of *Salmonella typhimurium* obtained by knocking out the *purI* and *msbB* genes. This strain was evaluated in a phase I clinical trial for the treatment of non-responsive metastatic melanoma or renal cell carcinoma in humans and showed good safety and antibiotic sensitivity. *Salmonella* can convert chemical energy into mechanical energy through flagella, and its motility is more than 10 times that of *Escherichia coli*.^[4] Under normal circumstances, VNP20009 can move forward through the flagella bundle. When encountering an obstacle, the flagella bundle opens, allowing VNP20009 to roll and reposition to avoid the obstacle.^[5] This property makes VNP20009 an advantage as a carrier for antigen delivery, with higher efficiency compared to traditional passive nanocarriers. In addition, flagella are composed of flagellin, which has been shown to be a highly effective and clinically safe adjuvant that can increase the immunogenicity of antigens by binding to Toll-like receptors.^[6]

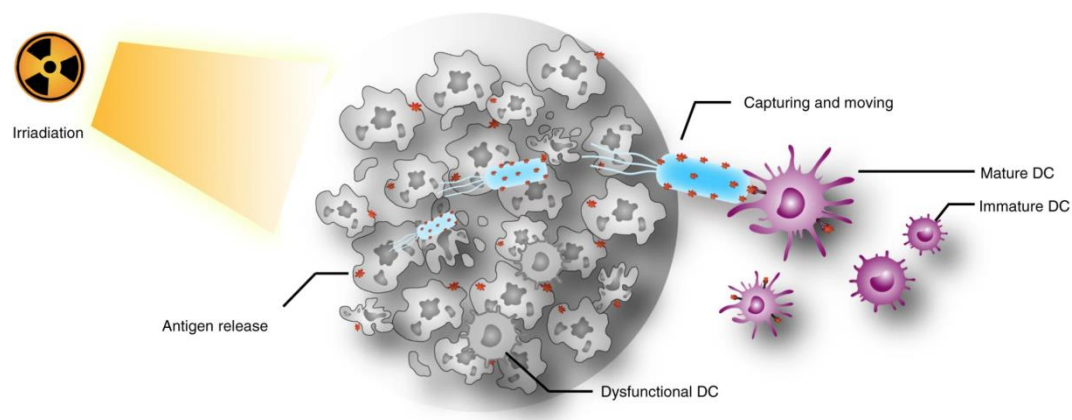


Figure 1. Intratumoral injection of a genetically attenuated *Salmonella* strain coated with antigen-adsorbing cationic polymer nanoparticles after radiotherapy resulted in the accumulation of tumor antigens in the tumor periphery. This enhanced the crosstalk between antigen and dendritic cells and led to a substantial increase in activated ovalbumin-specific dendritic cells that exerted antitumor effects *in vitro* and systemically and prolonged survival in multiple tumor models in mice, including metastatic and recurrent models.^[7]

1.3 Comparison of advantages and disadvantages with EcN

Salmonella can convert chemical energy into mechanical energy through flagella, and its motility is more than 10 times that of *E. coli*.^[4] At the same time, studies on tumor-specific colonization, tissue distribution and gene induction in living mice have shown that the colonization rate of spleen and liver is significantly reduced when using *E. coli* strains compared with *Salmonella typhimurium*.^[8]

However, due to its tumor-specific targeting and facultative intracellular parasitism, *Salmonella* is more used for tumor treatment, while the non-invasive EcN has a wider range of applications, such as being used as a delivery carrier for vaccines, cytokines and other substances. EcN shows strong antagonistic activity against different pathogenic *E. coli*. In addition, the application of *Salmonella* VNP20009 also faces clinical difficulties. After intravenous injection of VNP20009, the tumor response rate is low and the dose-dependent toxicity is problematic.

2. Listeria

2.1 Introduction

Listeria is a bacillus-shaped, Gram-positive, facultative anaerobic bacterium that is widespread in nature. *Listeria* comprises a diverse genus with four clades and 15 distinct species. Of these 15 distinct species, only *Listeria monocytogenes* and *Listeria ivanovii* are known to cause disease in humans and animals. Listeriosis is a very dangerous disease that is usually acquired through the consumption of food contaminated with *Listeria monocytogenes*.^[9] *Listeria monocytogenes* (Lm) is well known for infecting humans and producing a variety of symptoms, including gastroenteritis, meningitis, and encephalitis. In general, the human immune system mounts effective innate and adaptive immune responses that are able to control Lm infections. As a result, serious infections with Lm are rare and are usually limited to the elderly, pregnant women, or immunocompromised patients.^[10]

2.2 Application Examples

Lm has many properties that make it an attractive vehicle for cancer immunotherapy. Lm infection elicits only a modest humoral response that fails to block reinfection^[11], allowing repeated administration of Lm-based vectors as needed to boost T cell responses in patients. Furthermore, Lm strongly induces both innate and adaptive immune responses compared to DNA or peptide vaccines. Indeed, the ability of bacteria to activate innate immune mechanisms and elicit antitumor responses was first appreciated more than 100 years ago when William Coley observed spontaneous cancer

regression in patients with sarcomas who acquired bacterial skin infections^[12]. In addition to innate immune mechanisms, Lm is a potent stimulator of cytotoxic lymphocytes and cell-mediated immunity compared to other bacterial vectors such as *Salmonella*^[11]. More than 50 years ago, George Mackaness further demonstrated that mice exposed to sublethal doses of Lm developed long-lived, antibody-independent immune responses that protected against future challenge with lethal doses of Lm^[13]. These and other observations ultimately led to the exploration of Lm as a vaccine vector with the goal of inducing similar cell-mediated immune responses to foreign antigens. Paterson and colleagues used Lm-expressing β -galactosidase antigens to induce cytotoxic lymphocyte responses that killed β -galactosidase-expressing cancer cells^[14]. Since this seminal paper was published 25 years ago, a number of Lm vaccines have been developed for the treatment of various cancers.

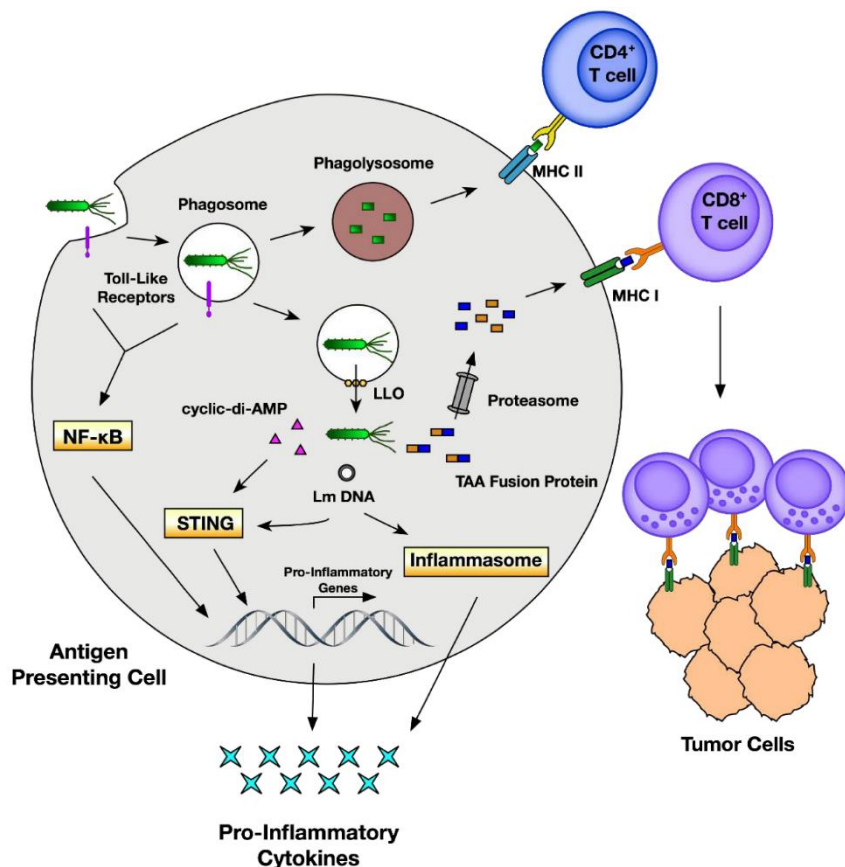


Figure 2. Innate and adaptive immune responses to recombinant *Listeria*.

Listeria monocytogenes (Lm) is internalized into phagosomes by antigen-presenting cells. During entry, Lm is sensed by toll-like receptors, leading to the activation of NF κ -B and the synthesis of pro-inflammatory genes. The phagosome may then fuse with the lysosome to form a phagolysosome, where Lm can be killed, resulting in the loading of its antigens onto MHC class II to activate CD4 helper T cells. Alternatively, Lm can express the pore-forming toxin listeriolysin O (LLO) to perforate the phagosome and enter the cytosol. Once in the cytosol, recombinant Lm secretes tumor-associated antigens (TAAs) as fusion proteins with Lm antigens into the cytosol, where they can be degraded by

the proteasome and loaded onto MHC class I to activate TAA-specific CD8 cytotoxic T lymphocytes. In addition, cytosolic Lm triggers further induction of pro-inflammatory pathways through the secretion of the cyclic dinucleotide cyclic-di-AMP and the detection of Lm DNA. Secreted cyclic-di-AMP directly stimulates the STING pathway and negatively regulates the NF- κ B inhibitor RECON (not shown), while the presence of genomic Lm DNA can lead to activation of the STING and inflammasome pathways, both of which contribute to the transcription of pro-inflammatory genes and cytokines. ^[15]

Because wild-type *Listeria* is pathogenic and unsuitable for clinical use, considerable effort has been expended to improve the safety of Lm. An ideal Lm strain would minimize pathogenicity while maximizing immunogenicity to target antigens. To this end, a variety of strategies have been developed to attenuate Lm and develop attenuated *Listeria* strains for vaccination. One is the deletion of virulence genes. A widely adopted strategy for attenuating Lm involves knocking out genes responsible for Lm tropism and cell-to-cell transmission. For example, the Lm Δ actA/ Δ inlB strain, which is knocked out of two virulence factor genes, also known as live attenuated double deletion (LADD), exhibits some anti-tumor responses ^[16] and is the basis of many vaccines in clinical trials. The second is to replace virulence or metabolic genes with free genes. The third is to develop killed but metabolically active (KBMA) strains. ^[15]

TAA fusion with *Listeria* antigens can enhance antitumor responses. Although Lm can be engineered to express TAA alone to generate therapeutic immune responses, almost all Lm vaccines express TAA as a chimeric protein fused to native Lm antigens. Initial studies of Lm vaccines fused TAAs to highly secreted Lm proteins as a method to increase the delivery of TAAs to the host cell cytosol ^[17]. However, it is now well recognized that certain Lm proteins can also act as adjuvants when fused to TAAs, improving therapeutic efficacy. The two most common fusion partners of TAA include variants of LLO and ActA. ^[15]

2.3 Comparison of advantages and disadvantages with EcN

As mentioned above, Lm has many properties that other strains (such as EcN) do not have, making it an attractive carrier for cancer immunotherapy, such as strong induction of innate and adaptive immune responses and being an effective stimulator of cytotoxic lymphocytes and cell-mediated immunity. Compared with *Salmonella*, Lm has unique intracellular parasitic processing and antigen presentation capabilities, and is therefore considered a good tumor vaccine carrier. ^[18]

However, Lm vaccines are still in their infancy and still require further clinical testing. At the same time, based on the invasiveness of Lm, Lm is more often used for immune-related treatments and modifications, while EcN has a relatively wider range of applications. Lm was originally a path-

ogen, so development should focus on attenuation. As a probiotic extracted from feces, EcN, as mentioned above, does not produce any toxins associated with pathogenic *Escherichia coli* strains, such as heat-labile enterotoxin (H-LT), heat-stable enterotoxin (H-ST), cytotoxins (such as cytotoxic necrosis factor, CNF 1) or Shiga toxins (SLT I, SLTII). Studies have shown that EcN does not have the genetic information to produce the above toxins, does not require attenuation, and is more convenient to use.

3. Microalgae

3.1 Introduction

In the strictest definition, microalgae are unicellular eukaryotic microorganisms that photosynthesize and typically have an aquatic lifestyle. Although cyanobacteria are prokaryotes and therefore not true algae, they share similar physiology and ecology with eukaryotic microalgae and share many biotechnological applications. ^[19]

Although oxygenic photosynthesis—the process by which algae convert inorganic carbon (CO₂) into biomass—is an important feature of microalgae, it is not necessarily a common denominator, as the ability to photosynthesize has been lost multiple times in several microalgal lineages. These species rely on heterotrophy (i.e., the use of organic carbon as an energy source) rather than photoautotrophic growth and are therefore able to grow in complete darkness. Other species combine these two strategies and are often referred to as mixotrophs. ^[19]

3.2 Examples of Applications

Because of their high exponential growth rates, microalgae achieve much higher areal biomass productivity than terrestrial crops, and microalgal biomass can be used almost entirely as it is very low in fiber and rich in protein, lipids, and/or other carbohydrates. Despite interest in large-scale production of microalgae for biofuels or bulk proteins, microalgae are still unable to compete with agricultural commodity crops. Microalgae are expensive to produce compared to traditional crops. ^[19]

Because microalgae lineages diverged long before terrestrial plants, they have high metabolic diversity. In fact, since the 1970s, several companies around the world have commercially produced microalgae (e.g., *Arthrospira* and *Chlorella*), not for biofuels or bulk protein, but for high-value applications such as nutritional supplements. For example, several microalgae accumulate high concentrations of carotenoids. Many microalgae contain high amounts of omega-3 fatty acids, such as eicosapentaenoic acid and docosahexaenoic acid, which are important for cardiovascular health and central nervous system development. Other examples of unique and promising metabolites that are

currently untapped include toxins produced by bloom-forming phytoplankton, some of which are so potent that they can be used to control tumor growth in targeted chemotherapy. Other substances, including various polysaccharides and plant hormones, can be used as plant biostimulants to optimize crop nutrient use efficiency and improve crop tolerance to environmental stress. Microalgae are also an emerging feedstock for bioplastic production. ^[19]

The unique metabolic pathways of microalgae can also be exploited by incorporating them into terrestrial crops. For example, genes of microalgal origin can be expressed in terrestrial crops to produce oils rich in omega-3 fatty acids. Vice versa, microalgae can be used as a platform for the synthesis of metabolites naturally occurring in other organisms. For example, microalgae are expected to be used in the manufacture and (oral) delivery of various biopharmaceuticals. ^[19]

Microalgae have attracted considerable attention in recent years for biomedical applications. Microalgae can be cultivated in large quantities, and many species have been commercialized as nutritional and food supplements, demonstrating their practicality and high biosafety as oral drug formulations ^[20,21]. In addition, many studies have shown that microalgae have great potential for targeted drug delivery in vitro and in vivo, as they can be effectively loaded with drug molecules through their active surfaces ^[22,23,24]. Given this paradigm, the spiral cyanobacterium *Spirulina platensis* (SP) can be used as a possible drug carrier for oral drug delivery. The helical structure of SP may make it not only more easily captured by intestinal villi, but also adhere to the intestinal wall, thereby prolonging the retention time of drugs in the intestine. In addition, the intrinsic fluorescent chlorophyll produced by SP allows non-invasive fluorescence imaging without the need for chemical modification ^[25,26], making the multifunctional SP particularly suitable for therapeutic and diagnostic applications. In this sense, the use of microalgae biomass-based drug delivery systems may provide innovative means to reduce costs, minimize toxicity, and improve the therapeutic efficacy of oral administration.

A microalgae-based oral drug delivery system (SP@Curcumin) was constructed for the treatment of various intestinal diseases. Curcumin, a drug approved by the US Food and Drug Administration with multiple pharmacological functions such as anti-inflammatory and anti-cancer, was loaded into SP. This study demonstrated that SP@Curcumin can pass through the stomach while its structure remains unchanged. SP@Curcumin can then be captured by intestinal villi and gradually degrade and release curcumin, thereby achieving ideal drug distribution in the intestine without causing adverse reactions. In traditional colon cancer radiotherapy, SP@Curcumin inhibits tumor progression by combining chemotherapy and radiotherapy, showing a synergistic therapeutic effect. At the same time, SP@Curcumin can also protect normal intestinal tissue during radiotherapy by eliminating reactive oxygen species (ROS) produced in normal cells and reducing ROS-induced DNA damage. In addition to its application in cancer treatment, the study also demonstrated the anti-inflammatory

ability of SP@Curcumin in the intestine, which can reduce the levels of proinflammatory cytokines and alleviate inflammation-related symptoms in colitis mice.

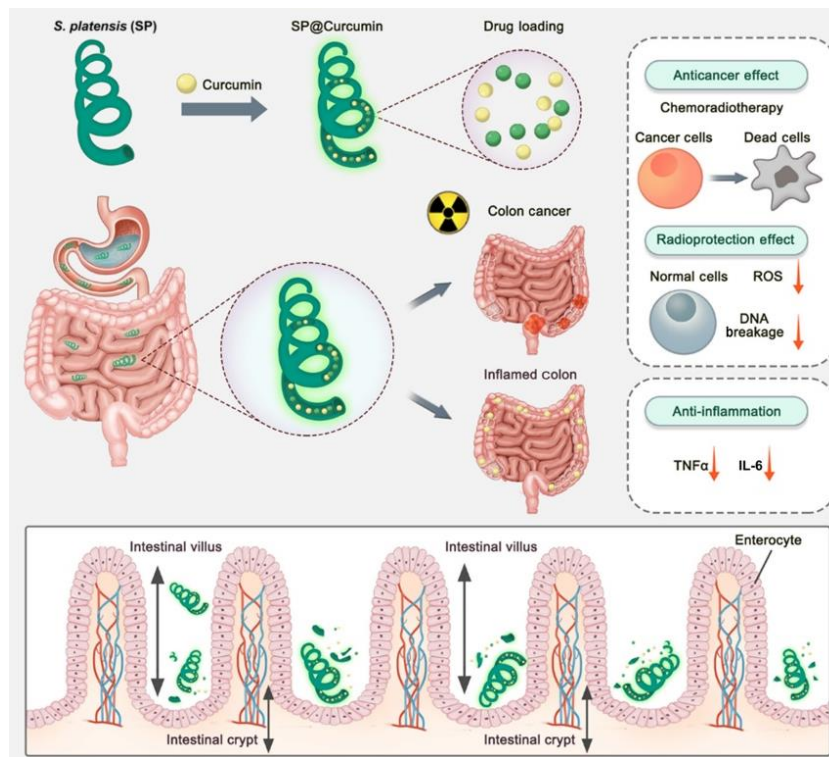


Figure 3. Schematic diagram of SP@Curcumin-mediated chemoradiotherapy for colon cancer Including radioprotection against tumor cells and normal cells, as well as anti-inflammatory effects on intestinal inflammation^[27]

In addition, biodegradable microalgae carriers can also be used for targeted delivery and imaging-guided treatment of breast cancer lung metastasis. High delivery efficiency, long-term drug release, and low systemic toxicity are effective weapons for drug delivery systems to win the battle against metastatic breast cancer. Spirulina (*S. platensis*) can be used as a natural carrier to construct a drug delivery system for targeted delivery and fluorescence imaging-guided chemotherapy for breast cancer lung metastasis. The chemotherapeutic drug doxorubicin (DOX) can be loaded into spirulina in just one step to prepare DOX-loaded SP (SP@DOX), which has ultra-high drug loading efficiency and pH-responsive drug sustained release. The rich chlorophyll gives SP@DOX excellent fluorescence imaging capabilities, which can be used for non-invasive tracking and real-time monitoring in vivo. In addition, the micron-sized and spiral-shaped SP carrier enables the prepared SP@DOX to passively target the lungs, thereby significantly improving the therapeutic effect on 4T1 breast cancer lung metastasis. Finally, the undelivered carrier can be biodegraded via renal clearance without significant toxicity. SP@DOX is a novel biohybrid strategy for targeted drug delivery and effective treatment of cancer metastasis.^[24]

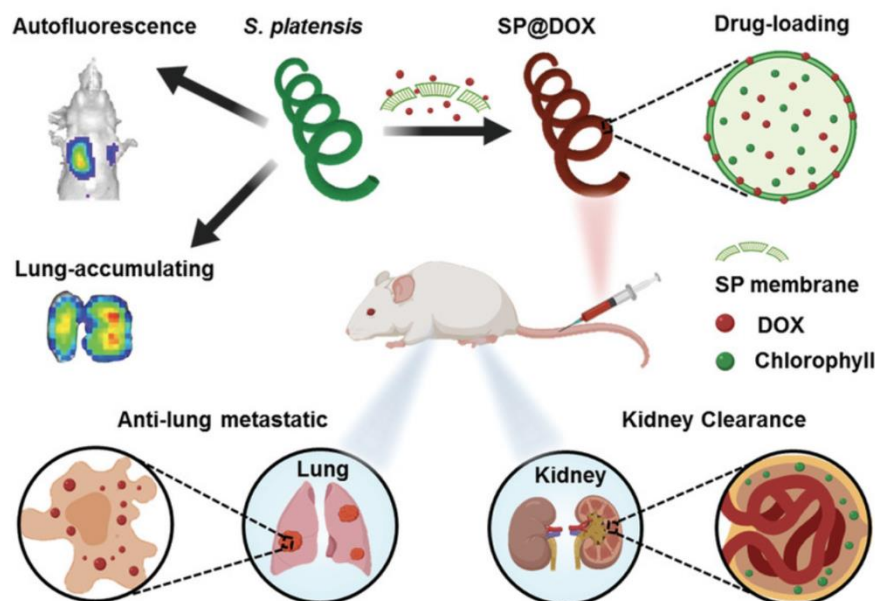


Figure 4. Schematic diagram of SP@DOX-mediated lung-targeted drug delivery and fluorescence imaging-guided chemotherapy for inhibiting breast cancer lung metastasis

3.3 Advantages and disadvantages compared with ECN

Microalgae have high metabolic diversity relative to ECN. For example, several microalgae accumulate high concentrations of carotenoids, and many microalgae contain large amounts of omega-3 fatty acids, which can be used as high-value nutritional supplements.^[19] Microalgae have great potential for targeted drug delivery in vitro and in vivo because they can effectively load drug molecules through their active surfaces. As mentioned above, the helical structure of some microalgae SPs may make them not only more easily captured by intestinal villi, but also adhere to the intestinal wall, thereby prolonging the retention time of drugs in the intestine. The intrinsic fluorescent chlorophyll produced by SPs allows non-invasive fluorescence imaging without the need for chemical modification. All of the above are unique advantages of microalgae relative to ECN.^[27]

Microalgae production is growing rapidly worldwide, and companies are constantly developing new products. Nevertheless, the market success of new microalgae products depends not only on cost-effective production processes, but also on consumer acceptance and demand (especially food applications). In addition, all products must strictly comply with relevant regulations (e.g., food and feed regulations), which may be complex and unsuitable for new microalgae products. For example, in the European Union, only a few microalgae species are currently authorized for the food market, and obtaining commercial authorization remains a bottleneck limiting the potential of microalgae products.^[19]

Reference:

- [1] Su LH, Chiu CH. (2007) Salmonella: clinical importance and evolution of nomenclature. *Chang Gung Med J.* 2007 May-Jun;30(3):210-9.
- [2] Andino A, Hanning I. (2015) Salmonella enterica: survival, colonization, and virulence differences among serovars. *ScientificWorldJournal.* 2015;2015:520179.
- [3] Winter, S., Thiennimitr, P., Winter, M. et al. (2010) Gut inflammation provides a respiratory electron acceptor for Salmonella. *Nature* 467, 426–429
- [4] Toley, B. J. & Forbes, N. S. (2012) Motility is critical for effective distribution and accumulation of bacteria in tumor tissue. *Integr. Biol.* 4, 165–176
- [5] Bhattacharjee, T. & Datta, S. S. (2019) Bacterial hopping and trapping in porous media. *Nat. Commun.* 10, 2075
- [6] Zhou, S., Gravekamp, C., Bermudes, D. & Liu, K. (2018) Tumour-targeting bacteria engineered to fight cancer. *Nat. Rev. Cancer* 18, 727–743
- [7] Wang, W., Xu, H., Ye, Q. et al. (2022) Systemic immune responses to irradiated tumours via the transport of antigens to the tumour periphery by injected flagellate bacteria. *Nat Biomed Eng* 6, 44–53
- [8] Jochen Stritzker, Stephanie Weibel, Philip J. Hill, Tobias A. Oelschlaeger, Werner Goebel, Aladar A. Szalay, (2007) Tumor-specific colonization, tissue distribution, and gene induction by probiotic *Escherichia coli* Nissle 1917 in live mice, *International Journal of Medical Microbiology*, Volume 297, Issue 3, 2007, Pages 151-162
- [9] Janet R. Donaldson, Kamil Hercik, Aswathy N. Rai, Sweetha Reddy, Mark L. Lawrence, Bindu Nanduri, Mariola Edelmann, (2015) Chapter 8 - *Listeria* and -Omics Approaches for Understanding its Biology, Editor(s): Steven C. Ricke, Janet R. Donaldson, Carol A. Phillips, Food Safety, Academic Press, 2015, Pages 135-158
- [10] Paterson Y, Guirnalda PD, Wood LM. (2010) *Listeria* and *Salmonella* bacterial vectors of tumor-associated antigens for cancer immunotherapy[J]. *Semin Immunol*, 2010, 22(3):183-189
- [11] Wood, L.M.; Paterson, Y. (2014) Attenuated *Listeria monocytogenes*: A powerful and versatile vector for the future of tumor immunotherapy. *Front. Cell. Infect. Microbiol.* 2014, 4, 51.
- [12] Coley, W.B. (1893) The treatment of malignant tumors by repeated inoculations of erysipelas: With a report of ten original cases. *Am. J. Med. Sci.* 1893, 105, 487–511.
- [13] Mackaness, G.B. (1962) Cellular resistance to infection. *J. Exp. Med.* 1962, 116, 381–406.
- [14] Schafer, R.; Portnoy, D.A.; Brassell, S.A.; Paterson, Y. (1992) Induction of a cellular immune response to a foreign antigen by a recombinant *Listeria monocytogenes* vaccine. *J. Immunol.* 1992, 149, 53–59.

- [15] Flickinger, John C., Jr., Ulrich Rodeck, and Adam E. Snook. (2018) "Listeria monocytogenes as a Vector for Cancer Immunotherapy: Current Understanding and Progress" *Vaccines* 6, no. 3: 48.
- [16] Brockstedt, D.G.; Giedlin, M.A.; Leong, M.L.; Bahjat, K.S.; Gao, Y.; Luckett, W.; Liu, W.; Cook, D.N.; Portnoy, D.A.; Dubensky, T.W. (2004) Listeria-based cancer vaccines that segregate immunogenicity from toxicity. *Proc. Natl. Acad. Sci. USA* 2004, 101, 13832–13837.
- [17] Ikonomidis, G.; Paterson, Y.; Kos, F.J.; Portnoy, D.A. (1994) Delivery of a viral antigen to the class I processing and presentation pathway by *Listeria monocytogenes*. *J. Exp. Med.* 1994, 180, 2209–2218.
- [18] 汪舒颖,丁承超,马俊飞,等. (2019) 减毒单核细胞增生李斯特氏菌作为疫苗载体在肿瘤治疗中的应用. *微生物学杂志*,2019,39(5):87-97.
- [19] Eli S.J. Thoré, Koenraad Muylaert, Michael G. Bertram, Tomas Brodin, (2023) *Microalgae*, *Current Biology*, Volume 33, Issue 3, 2023, Pages R91-R95
- [20] P. Spolaore, C. Joannis-Cassan, E. Duran, A. Isambert, (2006) Commercial applications of microalgae. *J. Biosci. Bioeng.* 101, 87–96.
- [21] B. Da Silva Vaz, J. B. Moreira, M. G. de Moraes, J. A. V. Costa, (2016) Microalgae as a new source of bioactive compounds in food supplements. *Curr. Opin. Food Sci.* 7, 73–77.
- [22] O. Yasa, P. Erkoc, Y. Alapan, M. Sitti, (2018) Microalga-powered microswimmers toward active cargo delivery. *Adv. Mater.* 30, 1804130.
- [23] X. Yan, J. Xu, Q. Zhou, D. D. Jin, C. I. Vong, Q. Feng, D. H. L. Ng, L. Bian, L. Zhang, (2019) Molecular cargo delivery using multicellular magnetic microswimmers. *Appl. Mater. Today* 15, 242–251.
- [24] D. Zhong, D. Zhang, T. Xie, M. Zhou, (2020) Biodegradable microalgae-based carriers for targeted delivery and imaging-guided therapy toward lung metastasis of breast cancer. *Small* 16, 2000819.
- [25] X. Yan, Q. Zhou, M. Vincent, Y. Deng, J. Yu, J. Xu, T. Xu, T. Tang, L. Bian, Y.-X. J. Wang, K. Kostarelos, L. Zhang, (2017) Multifunctional biohybrid magnetite microrobots for imaging-guided therapy. *Sci. Robot.* 2, eaaq1155.
- [26] D. Zhong, W. Li, Y. Qi, J. He, M. Zhou, (2020) Photosynthetic biohybrid nanoswimmers system to alleviate tumor hypoxia for FL/PA/MR imaging-guided enhanced radio-photodynamic synergetic therapy. *Adv. Func. Mater.* 30, 1910395.
- [27] Danni Zhong et al. , (2021) Orally deliverable strategy based on microalgal biomass for intestinal disease treatment.*Sci. Adv.*7,eabi9265.

10.Policies related to bacterial therapy

Name	Issue time	Institution	Relevant content
《The 13th Five-Year National Innovation Plan》	2016	The State Council	We will develop advanced and efficient biotechnology, accelerate innovation breakthroughs and application development of leading technologies such as synthetic biology, and develop a number of innovative pharmaceutical and biological products
《The 13th Five-Year Plan for biological industry development》	2016	The National Development and Reform Commission	Accelerate the construction of the innovation system of the biological manufacturing industry, solve the problems of the design, synthesis and optimization of artificial biology, accelerate the creation and industrialization of new drugs, and promote the development of translational medicine by relying on advanced technology.
《Guidance Catalogue of Key Products and Services for Strategic Emerging Industries (2016 edition)》	2016	The National Development and Reform Commission	Related technologies such as genetic engineering drugs are mentioned in biotech drugs.
《Catalogue of Key Supporting Industries for Intellectual Property Rights (2018 version)》	2018	State Intellectual Property Office	Synthetic biotechnology and gene editing and other related technologies have been mentioned in the advanced biological industry
《General Theory on Microecological Live bacterial Prod-	2020	The Chinese Pharmacopoeia Commission	The concept of microecological live bacteria products, and the related preparation requirements and detection

ucts (2020 edition)》			methods.
《The 14th Five-Year Special Plan for Health and Health Science and Technology Innovation》	2022	Ministry of Science and Technology of China	Accelerate breakthroughs in cutting-edge technologies, accelerate technological innovation breakthroughs and applied research, and focus on malignant tumors in drug research and development.
《Notice of the 13th Five-Year National Strategic Emerging Industries Development Plan》	2016	The State Council	Grasp the new trend of deep development of life science, extensive application and integration of new biological technology and innovation, and promote the development of medical treatment to precision medicine and personalized medicine.
《Regulations on the Clinical Application and Management of New Biomedical Technology (Draft)》	2019	National Health Commission	The thetic organisms are proposed to be approved by the National Health Commission. The approval of clinical research by the health administrative department includes academic and ethical review, clarifying the responsibilities of all parties and the technical entry threshold
《Guidelines on promoting the healthy development of the pharmaceutical industry》	2016	The State Council	Promote innovation capacity, accelerate the promotion of research and development of biological technologies such as new drug development.
《Fifth-Year Plan for the Development of Biological Economy (2021-2025)》	2021	The National Development and Reform Commission	We will strengthen originality and lead basic research, target frontier fields such as clinical medicine and health management, new drug de-

			velopment, and synthetic biology, and build a national strategic scientific and technological force in the field of biology.
《Reference Relationships between Strategic Emerging Industry Classification and International Patent Classification (2021) (Trial)》	2021	State Intellectual Property Office	Determine the biological drugs, genetic engineering drugs as a strategic industry.
《2023 Outline for Building a Strong Intellectual Property Country and the plan for promoting the implementation of the 14th Five-Year Plan》	2023	State Intellectual Property Office	Financial institutions are encouraged to develop financing and insurance products adapted to the characteristics of intellectual property services.
《Guiding Catalogue for the Adjustment of Industrial Structure (2024 edition)》	2023	National Development and Reform Commission	Breakthrough and application of pharmaceutical core technologies.
Early trials of in vivo biotherapeutic products: chemistry, manufacturing, and control information	2016	FDA	To advise the IND sponsor on early clinical trials using live organisms in the United States.
Policy on quantitative labeling of dietary supplements containing active microorganisms: industry guidelines	2018	FDA	CFU should be added to the quantitative label of dietary supplements containing active microorganisms
Dispersed clinical trials of drugs, biological products, and	2023	FDA	Provide recommendations for clinical trials for manu-

devices			facturers and other stakeholders
《National Biotechnology and Bio-fufacrication Program》	2022	The White House	Improve domestic bio-manufacturing capabilities (i. e. synthetic biology capabilities)
The American Innovation and Competition Law	2021	Congress	Invest funds in basic and advanced technology research, including synthetic biology.
Framework for Coordinating and Cooperation in Biotechnology	2017	The White House	It includes biotechnology products and products and regulations under the jurisdiction of relevant departments, providing a good process and legal reference for product manufacturers.

Note: Although China and the United States have a certain legislative basis for synthetic biology, the current laws on bacterial therapy are not yet complete and sound. However, it can be seen that synthetic biology and biomedicine are areas of common investment for both parties. EcN-related pharmaceutical products manufactured based on synthetic biology technology can take advantage of policy advantages to develop rapidly and seize related markets.

11. Overview of key companies in the ECN industry market

1. Synlogic Therapeutics

Using ECN as a chassis to transform cancer treatment

Introduction: Founded in Boston, USA in 2013, it was listed on the Nasdaq through a reverse merger in 2017. It uses synthetic biology technology to transform microorganisms, and repairs metabolic disorders in patients by synthesizing microorganisms carrying special DNA chains (gene circuits), thereby achieving the effect of treating related diseases.

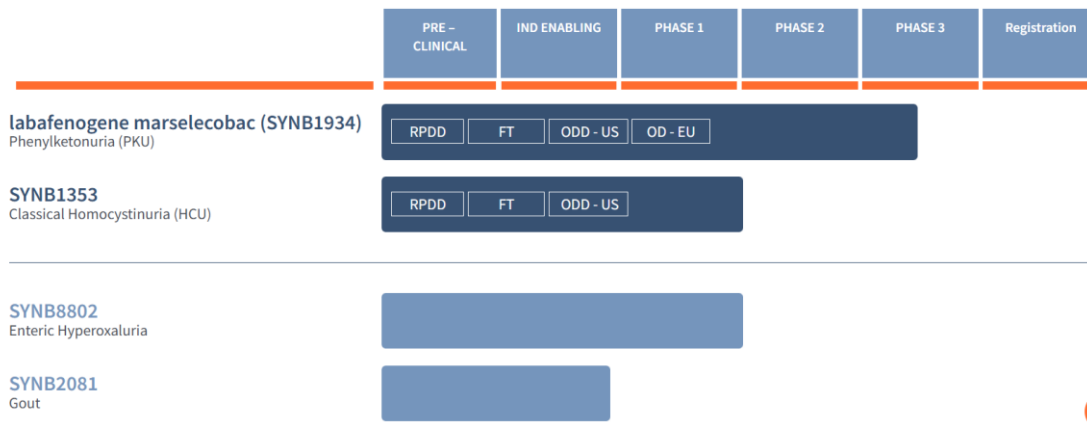


Figure 1. Synlogic Therapeutics product pipeline

2. Ardeypharm

Live bacteria tablets

Introduction: Arderpharm, located in Herdecke, Germany, is a leading pharmaceutical expert in the development and production of probiotic drugs. EcN live bacteria tablets (mutaflor) are sold directly with EcN as the main ingredient.



Figure 2. Secretion-related proteins after EcN modification

3. Angeluo (Shenzhen) Biotechnology Co., Ltd.

Medicinal protein

Introduction: Angeluo (Shenzhen) Biotechnology Co., Ltd. was established on April 10, 2023. In the field of SOD development and application, it is one of the few synthetic biology companies in the world that has molecular acquisition, collision and gene transcription technology, and has achieved mass production on a large-scale industrial scale. The company is devoted to multidisciplinary cross-disciplinary research in biology, chemistry, etc., and provides green biotechnology solutions for applications in medicine, agriculture, food and other fields, as well as the supply of key raw materials.



ANGELO® EcN-SOD药用蛋白

ANGELO® EcN-SOD Medicinal protein

EcN-SOD靶向定植肠道菌群，释放75万U/g SOD酶活，预防治疗肠胃炎。EcN-SOD具有靶向粘附癌细胞的特性，通过联合表达抑癌基因 (TP53、CIP-KIP、INK4、PTEN)，并实现定量释放，维持体内癌细胞平衡。

稳定技术参数：
SOD酶活：75万U/g
EcN菌落数：51万U/g

Figure 3. Angeluo product display

12.Forecast of the development prospects of EcN in China

Wide range of indications:

EcN was originally suitable for the treatment of inflammatory bowel disease (IBD), etc. After transformation, its potential application areas have been further broadened. This includes but is not limited to the treatment of cancer, autoimmune diseases, etc., and as a biosensor to monitor the condition, which brings huge market share potential for EcN.

Policy support:

Governments of various countries are gradually recognizing the importance of EcN and other biotherapeutic technologies, and have introduced relevant policies to support them. For example, the Chinese government has increased its investment in scientific research in the field of biomedicine in the "Eleventh Five-Year Plan", which provides a good policy environment for the development of EcN.

Meet clinical needs:

EcN has the advantages of good oral compliance, lasting efficacy, and precise treatment, which are all urgent needs to be met in current clinical treatment. Especially for patients who need to take medication for a long time, EcN provides a more convenient treatment option.

Capital market attention:

Although there is not much capital market data on EcN at present, with the overall development of the biomedical industry, more and more capital has begun to pay attention to and invest in EcN-related companies and R&D projects. From the perspective of the probiotics market, China's probiotics market reached a scale of RMB 109.38 billion in 2022, increasing from RMB 64.77 billion in 2018 to 2022, with an average annual compound growth rate of 14%. In terms of investment and financing, the investment and financing activity of the probiotics industry was relatively high in 2019, with 9 financing events and a financing amount of more than RMB 840 million. However, since the outbreak of the COVID-19 pandemic in 2020, the financing activity has declined, and the financing amount in 2022 has dropped to RMB 33 million. In terms of the industry competition landscape, the overall market concentration of the probiotics industry is very high, with CR4 of 95.7% and CR7 of 99.8% in 2022. In terms of regional competition, North China and the Yangtze River Delta are areas where the probiotics industry has concentrated production capacity, among which North China has

the largest number of representative companies, including Ketuo Biotechnology and Weilan Biotechnology. Among them, Weikang Probiotics is in the first echelon, with an annual production capacity of more than 500 tons; the second echelon includes Weilan Biotechnology, Ketuo Biotechnology, etc., with an annual production capacity between 100 tons and 500 tons. It is estimated that by 2028, the scale of China's probiotics market will exceed 190 billion yuan, with an estimated annual compound growth rate of 10%.

Application in environmental protection and secretion:

EcN not only has application prospects in the field of disease treatment, but also shows potential in the fields of environmental protection and secretion regulation. For example, the application of EcN in bioremediation can help remove environmental pollution.

Public awareness improvement:

As the public's understanding of probiotics and biotherapy deepens, the acceptance and market demand of EcN (*Escherichia coli* Nissle 1917) are expected to increase. Probiotics are a type of active microorganisms that can promote the ecological balance of the host's intestinal microbial flora and have a positive effect on health. As a probiotic, EcN has multiple biological functions, including antibacterial, anti-inflammatory, and regulation of intestinal flora, which makes it an ideal candidate for disease treatment and diagnosis.

Market data:

According to a report by Fortune Business Insights, the global oncology drug market is expected to grow from \$150 billion in 2021 to \$250 billion in 2028. This growth trend reflects the continued growth in demand and technological advances in the field of oncology treatment. As a potential tumor treatment, the market potential of EcN (*Escherichia coli* Nissle 1917) cannot be ignored.

EcN is a probiotic with multiple biological functions, including antibacterial, anti-inflammatory, and regulation of intestinal flora, which makes it an ideal candidate for disease treatment and diagnosis. Through gene editing technology, EcN can be transformed into a carrier that can accurately deliver drugs to tumor sites, reduce drug damage to normal cells, and improve treatment effects.

The growth of the global oncology drug market is partly due to the development of targeted therapy and immunotherapy. Targeted therapy dominates the global tumor market, accounting for 61.3% of the market share in 2022. The transformation and optimization of EcN makes it expected to become a new member of targeted therapy, providing new possibilities for tumor treatment.

In addition, with the aging of the global population and the advancement of medical technology, the demand for oncology drugs is expected to continue to grow. As a new type of treatment, EcN has great market potential and is expected to occupy a place in the future tumor treatment market.

However, despite the huge market potential, the clinical application of EcN still faces some challenges, including genetic stability, safety and other issues. In order to achieve the widespread clinical application of EcN, further research and development are needed to solve these technical and safety issues.

In summary, with the continuous growth of the global tumor drug market, EcN has a broad market prospect as a potential treatment. Through continuous research and technological innovation, EcN is expected to play an important role in the future field of tumor treatment.

Technological progress:

With the rapid development of biotechnology, especially the continuous advancement of gene editing technology, the transformation and optimization of engineered bacteria such as *Escherichia coli* Nissle 1917 (EcN) has become possible and has shown great potential in clinical applications. As a probiotic, EcN has multiple biological functions, including antibacterial, anti-inflammatory and regulation of intestinal flora, which makes it an ideal candidate for disease treatment and diagnosis.

In terms of clinical application, the engineering transformation of EcN mainly focuses on two aspects: surface modification and genetic engineering. Through these modifications, EcN can be used as a drug delivery system for the treatment of diseases such as cancer and inflammatory bowel disease. For example, through surface modification, the adhesion of EcN to the target can be enhanced, its tolerance in the gastrointestinal environment can be improved, and its biocompatibility can be improved, thereby improving the therapeutic effect.

Genetic engineering involves mutating the endogenous genes of EcN or introducing exogenous genes to achieve specific therapeutic purposes. For example, through genetic engineering, EcN can be transformed into bacteria that can express specific drug molecules or used as biosensors to monitor the disease. These modified EcNs have shown good therapeutic effects in animal models, but to be widely used in clinical practice, genetic stability, safety and other issues need to be addressed.

In addition, EcNs are also used to build smart drug delivery systems that can respond to specific biological signals such as temperature, oxygen concentration or metabolite levels to precisely release drugs at the target site. For example, by modifying EcN to locally release the cytokine GM-CSF and blocking nanoantibodies against PD-L1 and CTLA-4 at the tumor site, it has been shown in mouse models that it can significantly reduce the burden of adenoma.

Although the transformation and optimization of EcN has shown great potential in clinical applications, it still faces some challenges. For example, the genetic instability of bacterial plasmids may lead to uncertainty in therapeutic effects, and the transfer of antibiotic genes may also bring safety issues. In addition, the stability and safety of large-scale industrial production of recombinant strains are also issues that need attention.

In general, the transformation and optimization of EcN is an active research field, and with the continuous advancement of technology, it is expected to be more widely used in clinical practice in the future.

13.Side effects and biosafety prospects of EcN-related therapies

The safety and genetic plasticity of EcN make it an engineered bacterial carrier for the treatment of a variety of diseases. These characteristics of EcN give it a wide range of potential applications in medicine, including as a drug delivery system, anti-tumor therapy, and treatment of specific metabolic diseases. However, EcN still has certain side effects and risks in disease treatment.

1. Immune response

The immune response activated by bacterial therapy may sometimes lead to side effects. For example, excessive immune response may trigger a systemic inflammatory response complex, leading to serious complications such as shock and multiple organ failure. In addition, the activation of the immune system may lead to autoimmune diseases and attack normal body tissues. ^[1]

The immunogenicity of engineered EcN in participating therapies mainly involves the immune response that it may induce in vivo as a living drug. Although EcN itself has a high safety as a probiotic, the immunogenicity of EcN after engineering modification will be affected by many factors. Engineered EcN may express exogenous proteins or drugs, which may cause an immune response in the host. For example, EcN is currently used for CRISPR/Cas9 targeted delivery for gene therapy research on diseases such as tumors. The Cas9 protein carried by EcN is derived from two bacteria that often infect humans - *Streptococcus pyogenes* and *Staphylococcus aureus*. A study by Stanford University showed that antibodies against SaCas9 were detected in 78% of donors, while antibodies against SpCas9 were detected in 58% of donors. This indicates that there is a pre-existing adaptive immune response in the human body, and the immune response that may be caused by this is considered in clinical applications. ^[2]

2. Targeting

In the related therapies for colonization, detection and treatment of tumor cells, *Escherichia coli* Nissle 1917 (EcN) will only form a long-term niche in the intestine in the presence of adenoma tissue in genetically engineered mouse models. And although EcN will selectively and significantly enrich in tumors in the absence of antibiotic pretreatment in mouse models compared with adjacent tissues and organs, it will also exist in trace amounts in adjacent healthy tissues and peripheral organs, which cannot be ignored. ^[4,5]

3. Gene transfer risk

Generally, if the antibiotic resistance gene of bacteria belongs to nucleoplasmic mutation, it is relatively stable; on the contrary, if the resistance gene is located in a mobile gene element, horizontal gene transfer may occur, becoming the antibiotic resistance library of pathogens in the gastrointestinal tract^[2]. Engineered EcN that can express exogenous proteins or drugs for disease treatment. Its antibiotic gene is usually located in the introduced exogenous plasmid, and the transfer of plasmid DNA between bacteria can be achieved through conjugation of pili. The transfer of antibiotic genes may confer resistance to pathogens in its environment, thereby leading to the generation of drug resistance, and further increasing the difficulty of treating bacterial infectious diseases caused by it.

4. Risk of unlimited proliferation in vivo

EcN is currently on the market as a drug ingredient and participates in the treatment of IBD. The reasonable use of probiotics will benefit patients, but if used in patients with severe damage to the intestinal mucosal barrier or severe acute IBD, probiotics may be transferred from the intestine to the blood, leading to probiotic bacteremia and fungal diseases. Therefore, for patients with severe damage to the intestinal mucosal barrier or severe acute IBD, be vigilant about bacteremia when using probiotics. Attention should be paid to monitoring vital signs and infection indicators. If there are any abnormalities, the probiotics should be discontinued in time and blood cultures should be performed to identify the pathogens. At the same time, anti-infection and symptomatic supportive treatment should be given. ^[2]

5. Genotoxicity

Although EcN has probiotic properties, the genotoxin colibactin encoded by the pks gene island it carries is genotoxic and can cause DNA double-strand breaks in host cells, causing cell cycle arrest and ultimately cell death. It may be a carcinogenic compound. At present, studies have been conducted on engineering EcN to specifically inactivate ClbP peptidase to uncouple EcN antagonists from gene activity, thereby providing safety guarantees for the use of EcN. ^[3]

6. Prospects for the biosafety of EcN in the therapeutic field

Overall, as an engineered bacterium, EcN has a promising application prospect in disease treatment and diagnostic strategies. With the continuous advancement of synthetic biology and genetic engineering, the transformation and optimization of EcN will continue to improve its safety and therapeutic potential, bringing more effective and safer treatment options to patients.

Reference:

- [1]史恩宇,王宁,赵溪悦,陈博,曹鸣芯,李长义.细菌介导的肿瘤治疗研究进展[J].中国生物工程杂志,2024(Z1):112-123.DOI:10.13523/j.cb.2307016.
- [2]杨荟平, 王志青, 刘乐,等.炎症性肠病中应用益生菌的临床疗效与潜在风险[J] . 中华炎症肠病杂志, 2022, 06(2) :155-160 .
- [3]Clémence M. Priscilla B.et al.Deciphering the interplay between the genotoxic and probiotic activities of *Escherichia coli* Nissle 1917.PLoS Pathog.2019 Sep; 15(9): e1008029.
- [4]Gurbatri CR, Radford GA, Vrbanac L, Im J.et al. Engineering tumor-colonizing *E. coli* Nissle 1917 for detection and treatment of colorectal neoplasia. Nat Commun. 2024 Jan 20;15(1):646. PMID: 38245513; PMCID: PMC10799955.
- [5]Luke JJ, Piha-Paul SA, Medina T.et al.Phase I Study of SYNBI891, an Engineered *E. coli* Nissle Strain Expressing STING Agonist, with and without Atezolizumab in Advanced Malignancies. Clin Cancer Res. 2023 Jul 5;29(13):2435-2444. PMID: 37227176; PMCID: PMC11225568.
- [6]Charlesworth, C. T. et al. Nat. Med. 25, 249–254 (2019).

14. Introduction to the EcN project used by each team

PekingHSC

curmino: Based on the VIII secretion system of E.coli Nissle 1917, a bacterial platform for cancer treatment

It is generally believed that Gram-negative bacteria do not have secretion ability. But we found that E. coli has T8SS, which forms Curli fiber on the surface of E. coli. The corresponding gene sequence is called Curli specific gene (Csg), which contains seven proteins csgABCDEFG. Among them, CsgG is a channel that mediates the secretion of CsgA and CsgB. Curli fiber is composed of csgA and CsgB. CsgA is the main component of Curli fiber, and CsgB plays the role of condensation nucleus. i

iGEMers from PKUHSC-China brainstormed a variety of ways to transform T8SS.

1 CsgA-mediated intratumoral protein drug delivery system: CsgA is an amyloid protein with 5 repeats. There is evidence that csgG can only mediate the secretion of amyloid protein. ii Based on this, we use a linker that can be broken by MMP to connect CsgA or truncated CsgA with the protein of interest (POI). Taking advantage of the natural targeting of EcN to tumors, EcN secretes CsgA-fused POI inside the tumor, and uses the highly expressed MMP enzyme in the tumor microenvironment to cut off the linker between CsgA and POI, thereby achieving continuous release of POI in situ inside the tumor and achieving highly specific tumor killing.

2 Applying Curli in E.coli Nissle 1917 as ectopic nucleation display system (ENDS) for immune boosting: CsgB plays a role similar to condensation nuclei in the formation of Curli fiber, and through the interaction between CsgB and CsgF on the bacterial surface, Curli fiber grows on the bacterial surface. We knocked out the CsgF gene of EcN, fused a protein targeting the surface of tumor cells at the end of CsgB, and fused a neoantigen at the end of CsgA, so that Curli fiber carrying a large amount of neoantigen grows on the surface of tumor cells, realizing ectopic nucleation for antigen display. Immune cells are recruited by antigens displayed on the tumor surface, achieving immune activation, and thus achieving highly specific tumor killing.

3 ncAA incorporated application of csgA for prodrug click-to-release: We used codon expansion and non-natural amino acid site-directed modification technology to insert the non-natural amino acid Tet2.0 with a tetrazine group into csgA. By utilizing the natural targeting of bacteria to tumors, we can achieve pre-targeting of tumors. Then, trans-cyclooctene TCO caged prodrug is added, and TCO reacts with tetrazine to release active drug molecules, thereby achieving highly tumor-specific drug release and killing.

Spidey Microbe: Parkinson's disease treatment drug delivery platform using probiotic adhesion and interaction network

Probiotics have great potential in treating and alleviating gastrointestinal and gut-brain axis-related diseases, but their weak intestinal colonization ability and complex microbial interactions limit the application of probiotics in the therapeutic field. To overcome these limitations, we designed and constructed a superhero, Spidey Microbe, which includes *Escherichia coli* Nissle 1917 and *Lactococcus lactis* F44. Through the intestinal adhesion system based on biotin-avidin binding and the interaction system based on quorum sensing, probiotics can "weave a web" and colonize in the intestine like Spider-Man. We hope to design gene circuits to control the expression of mucosal binding proteins and the mutual adhesion of probiotics in *Escherichia coli* Nissle 1917 to achieve controllable colonization of the intestine by the "spider silk" of Spidey Microbe. In addition, we plan to construct a butyrate overexpression module in EcN and construct a cell factory in *Lactococcus lactis* F44 to express serotonin and L-dopa to give Spiderman bacteria more superpowers to fight Parkinson's disease. Our transformation of EcN can be summarized as follows.

1 Intestinal colonization ability: We designed a self-assembly system based on the combination of streptavidin and biotin, and used N-hydroxysuccinimide biotin, a reagent that can be conjugated through primary amines (such as the N-terminus and side chains of lysine residues in proteins and peptides), to label the EcN cell surface. Inducible expression of streptavidin and intestinal mucus protein binding protein fusion protein-"spider silk" was constructed in *Escherichia coli* Nissle1917, and the signal peptide was used to locate this fusion protein to the extracellular space. Under induced conditions, controlled colonization of the intestine can be achieved.

2 Butyrate production element: Butyrate, as a short-chain fatty acid, can reduce inflammatory response by improving intestinal mucosal permeability and downregulating the expression of inflammatory factors. We enhance the carbon metabolic flow of acetyl-CoA to butyrate by expressing acetyl-CoA acetyltransferase (ACAT). Thereby improving the butyrate synthesis capacity of EcN.

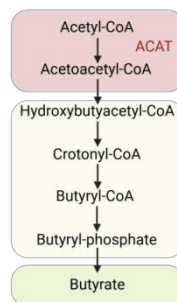


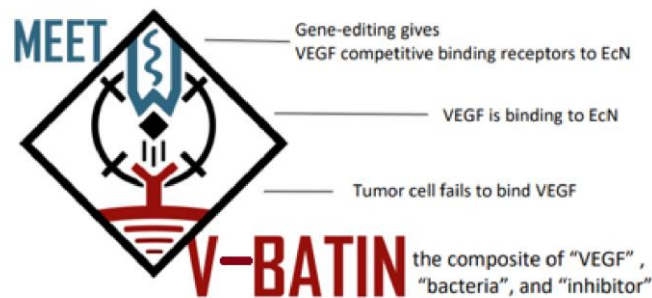
Figure 1. Butyrate production metabolic pathway (from iGEM TJUSX)

USTC

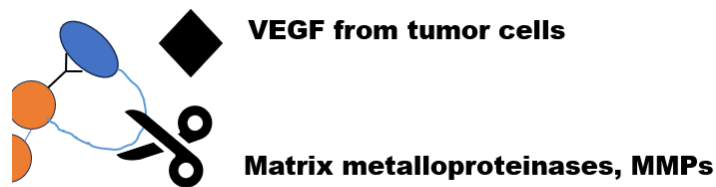
V-batin constructs tumor-suppressing engineered bacteria that compete for vascular endothelial growth factor

A key reason why solid tumors are difficult to treat is that the tumor microenvironment around them inhibits the killing of the immune system, and how to relieve this inhibition is a difficult problem. At present, the strategy of targeting the tumor microenvironment to inhibit or treat solid tumors has attracted widespread attention. However, most existing therapies still face challenges such as off-target.

Our team noticed that the harmless *Escherichia coli* Nissle 1917 strain can automatically colonize in the tumor microenvironment, showing excellent targeting. Therefore, we plan to use the mechanism of vascular endothelial growth factor (VEGF) in the human body to promote angiogenesis, so that EcN can express VEGF under specific conditions in the tumor microenvironment to compete for binding receptors and present them on the surface of bacteria. By consuming VEGF in the tumor microenvironment through adsorption, its concentration is reduced, thereby effectively inhibiting angiogenesis in the tumor area and achieving the therapeutic purpose. This strategy provides a new idea for the treatment of solid tumors and demonstrates the potential of engineered bacteria in the treatment of solid tumors.



First, we screened several different VEGF receptors and presented them on the surface of *E. coli* through a surface display system; secondly, we also designed a series of intrinsic environmental sensing switches such as hypoxia switches. When and only when the bacteria are in the tumor microenvironment, the switch can be turned on and a series of designs will begin to express. Third, different from the off-target effect problem faced by traditional systemic drug delivery methods, in order not to affect VEGF acting in the normal blood circulation system, we designed a masking system. The VEGFR displayed on the surface of *E. coli* is connected to a masking peptide with a similar structure to VEGF. The connection structure can be cut under the tumor-specific high-expression MMP-14, so that the masking structure can be specifically removed in the tumor microenvironment.



GXU-China

This project uses a combination of engineered bacteria and nanoparticles to achieve precise positioning and attack on tumor cells, providing a feasible solution for the treatment of triple-negative breast cancer.

EcN is used as the chassis strain of engineered bacteria in this project, and its transformation is mainly divided into two parts: genetic engineering and chemical modification of nanoparticles.

This project uses the cd47-targeted SIRP α gene to genetically engineer EcN to achieve the purpose of identifying TNBC. The team used the Golden Gate cloning system to transfer the target gene SIRP α into the plasmid vector pET-28a+, and also used the SpyTag/SpyCatcher structure. This year, the team also focused on the surface display of EcN-targeted tumor SIRP α . In order to correctly display tumor-targeting proteins on the cell surface, the target protein (guest protein) needs to be genetically fused to a carrier protein that can promote export through the cell envelope and anchor the guest protein to the bacterial cell surface through appropriate surface exposure. The SC/ST surface display (SpyCatcher/SpyTag Surface Display system, S3D) system was utilized in the project, in which the membrane anchor and the passenger are expressed separately and assembled by post-translational coupling. We selected AIDA-I adhesin as a carrier for tumor-targeted SIRP α . It contains a C-terminal autotransporter structure that forms a porin-like β -barrel channel on the outer membrane to transport the N-terminal guest domain to the cell surface.

This project also further chemically modified EcN with peroxidase-like manganese dioxide nanoparticles to alleviate the problem of hypoxia in the tumor microenvironment by reacting with endogenous hydrogen peroxide and generating oxygen in situ. The engineered bacteria were surface-sulfurized and bound to gold nanoparticles deposited on the surface of manganese dioxide nanoparticles. These nanoparticles can react with GSH in tumor cells to produce GSSH. The modified engineered bacteria can survive on lactate feedstock and compete with tumor cells for glucose availability, preventing lactic acidosis and ultimately promoting the transformation of tumor-associated macrophages (TAMs) from a non-inflammatory phenotype to an inflammatory phenotype.

Intestide

Short peptides have many superior properties, such as higher stability and lower immunogenicity, which make them conducive to stable and safe absorption in the human intestine. However, the existing methods of delivering short peptides to the intestine still have many limitations. In order to establish a safe and efficient short peptide delivery system, we designed two different engineered bacteria: short peptide producers and regulators. Our design includes three modules-secretion module, adhesion module and QS (quorum sensing) module. The secretion module only exists in the short peptide producer, and the recombinant transmembrane protein Lpp'OmpA is fused with the short peptide, and an enterokinase cleavage site is added in the middle. The engineered bacteria colonize the intestine and express the fusion protein. The fusion protein will be located in the outer membrane of the engineered bacteria and cut by enterokinase in the intestine to release the short peptide and complete the delivery. Adhesion modules exist in two types of engineered bacteria. Producers and regulators express Hsp60 and LAP adhesins, respectively. Natural Hsp60 proteins exist on the surface of the intestine. Engineered bacteria expressing LAP adhere to the intestine, and bacteria expressing Hsp60 adhere to LAP engineered bacteria, and so on, forming a bacterial layer that adheres stably to the intestine. QS safety modules exist in short peptide producers and regulators. Based on the principle of population regulation, the population size of producers and regulators is adjusted to obtain the optimal short peptide production ratio.

Based on our purpose - to establish a safe and efficient short peptide delivery system in the intestine, we selected E.coli Nissel 1917 as our chassis strain. It is a recognized effective probiotic for the treatment and prevention of various intestinal diseases or intestinal inflammation. It has high safety, wide application, in-depth research, and a good synthetic biology background. We introduced the plasmid we designed into EcN and conducted a series of verification experiments such as Western Blot and immunofluorescence, in order to successfully achieve the functions of secretion, adhesion and population regulation.

[2] Balagaddé FK, Song H, Ozaki J, Collins CH, Barnet M, Arnold FH, Quake SR, You L. (2008) A synthetic *Escherichia coli* predator-prey ecosystem. *Mol Syst Biol.* 2008;4:187. doi: 10.1038/msb.2008.24. Epub 2008 Apr 15.

NJTech-China-A

Today, the prevalence of diabetes is increasing year by year. How to treat the growing number of diabetic patients has become a global problem. D-tagatose, as a functional sugar, is a potential therapeutic molecule for type II diabetes. However, there are few reports on the naturally occurring 4-tagatose-epimerase that catalyzes the synthesis of tagatose from fructose. The fructose-catalyzed production of tagatose still has problems such as a small source of enzymes and a low conversion rate. Therefore, this project aims to screen out new enzymes with potential tagatose-4-epimerase activity by gene mining. The project combines protein engineering, molecular docking, and molecular simulation. By means of iterative mutation, the conversion rate of tagatose-4-epimerase is further improved to further meet the needs of industrial production. This project has increased the maximum conversion rate of the new tagatose-4-epimerase discovered to 37%, and the maximum yield has reached 24 g/(L·h). At the same time, three strategies are proposed: fusion-promoting tags, molecular chaperones, and fermentation optimization to solve the inclusion body problem in the expression of new enzymes. In addition, we envisioned a high-throughput screening method based on tagatose-specific green fluorescent biosensors to screen mutants with the best performance to improve screening efficiency.

NJTech-China

Inflammatory bowel disease (IBD) is a chronic nonspecific intestinal inflammation with unknown etiology and incurable. The pathogenesis of IBD is complex, and the traditional treatment effect is not ideal. Patients often face problems such as recurrence of inflammation and increased economic burden. This project uses the probiotic *Escherichia coli* Nissle 1917 (EcN) as the chassis cell, and uses synthetic biology thinking to design five modules with different functions. Among them, the sensing module SoxR/SoxS responsive promoter will regulate the opening of the entire gene circuit. By sensing the excessive NO in the IBD intestinal environment, it ensures that our gene circuit is only turned on in the IBD environment, ensuring a certain degree of biosafety and system specificity. The main therapeutic modules include the immune module programmed cell death ligand 1 (PD-L1), the adhesion module mussel foot silk protein (Mfp), and the antioxidant module superoxide dismutase (SOD), which are used to inhibit immune overreaction, repair damaged intestinal parts, and eliminate reactive oxygen. The last safety lysis module PhiX174E is the ultimate guarantee of biosafety. When the number of bacteria reaches a certain level, the bacteria can be automatically lysed. This not only ensures that our engineered bacteria will not overproliferate in the intestine, but also better releases our therapeutic modules into the intestinal environment to achieve better therapeutic effects.

with the antibody Fc region to display commercial monoclonal antibodies. The OMVs produced by the above engineering transformation can achieve targeted delivery to different organs, tissues, and cells.

Finally, we will use the constructed Cas9 RNP outer membrane vesicle delivery system to conduct targeted gene editing experiments in human cells, evaluate the safety of the vector, and the gene editing efficiency, and lay the foundation for the clinical application of this gene editing delivery platform.

15.Contributors

Cai Kailing Li Lin Liu Heyu Zhou Yufan Sun Yuhao Gu Qianhui

Hu Meihuan Zhu Hongyu Yan Xujie Lu Chenxi Xing Yunqi Lei Murong

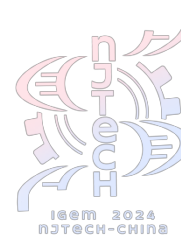
Zhao Yanzong Shu Shaoyin Li Qicheng Yu Shenhan Sun Guzihan Shen Jiawen

Zhou Xinquan Yang Xinyi Chen Xinjie Zhou Suhui Yang Hongdan Zhu Chuning

Wang Tao Chu Huixin Li Shushu Li Qike Xu Kehui Yu Yingying

Ma Jiaca Zhou Dekun Gao Jingyuan Chen Xinnan Gao Jiaxin Chen Jingyan

Hu Chenhao Chen Yuyanyan Zhang Wenrui Yan Yifei Huang Furong Zhu Luping



(In no particular order)

16. Legal Statement

The copyright of this report belongs to all iGEM teams involved in the preparation. Without written permission, no organization or individual may reprint, reproduce, publish or quote in any form. If the iGEM team involved in the preparation agrees to quote or publish, it must be used within the permitted scope and the source must be indicated. No quotation, deletion or modification contrary to the original intention of this report shall be made.

The author of this report has professional research capabilities and guarantees that the data in the report are from legal and compliant channels. The output of opinions and data analysis are based on the analyst's objective understanding of the industry. This report is not authorized or influenced by any third party.

The opinions and information involved in this report are for reference only and do not constitute any investment or academic advice. This report is issued only with the permission of relevant laws and is issued only for information purposes.

The information in this report comes from public information. The iGEM team involved in the preparation has the final right of interpretation of the accuracy, completeness and reliability of the information. The information, opinions and speculations contained in this article only reflect the judgment of the author and the date of the release of this report. The descriptions in the past reports should not be used as a basis for future performance. The Alliance does not guarantee that the information contained in this report is kept up to date. At different times, all iGEM teams involved in the compilation may issue reports and articles that are inconsistent with the information, opinions and speculations contained in this article. At the same time, all iGEM teams involved in the compilation may make modifications to the information contained in this report without notice. Readers should pay attention to the corresponding updates or modifications. Any organization or individual is responsible for all activities carried out using the entire content of this report and bears all consequences.